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Modeling and Analysis of Compositional Data

– Exercises and Solutions –
–Errata–

October 30, 2021

Contents

2	Compositional data and their sample space	3
3	The Aitchison geometry	9
4	Coordinate representation	23
5	Exploratory data analysis	43
6	Random compositions	73
7	Statistical inference	87
8	Linear models	99
9	Compositional processes	129
A	Errata	153

Foreword

This booklet contains the solutions to the exercises in the book *Modelling and Analysis of Compositional Data*. Many of them have long explanations, as we consider them useful for a better understanding of the theory presented in the book.

The software used to solve the exercises and elaborating the figures was:

- Package *compositions* for R.¹²
- CoDaPack³
- Microsoft, Excel 2003, Excel 2010.⁴

Figures and tables are numbered following the same scheme as in the book, i.e. first the chapter number, followed by a sequential number corresponding to the order of the figure within that chapter, either in the book or in this booklet, as if it were at the end of the book. The same holds for equations, although this affects only Chapter 6. **Random compositions** and Chapter 9. **Compositional processes**.

While preparing the booklet, some errata have been found in the book. They have been annotated at the end.

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¹ van den Boogaart, K.G. and Tolosana-Delgado, R. (2008) Compositions: a unified R package to analyze compositional data. *Computers and Geosciences*, 34 (4), 320–338.

² R Development Core Team (2004). R: A language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing. ISBN 3-900051-00-3.

³ CoDaPack - Version 2.01.15, developed by Grup d'Estadística i Anàlisi de Dades, Department of Computer Sciences, Applied Mathematics, and Statistics. University of Girona, Spain.

⁴ Microsoft. Microsoft Excel. Redmond, Washington: Microsoft, 2003, 2012. Computer Software.

Exercise 1

If data are measured in ppm, what is the value of the constant κ in Definition 2.4?

Answer of exercise 1

The units ppm mean *parts per million*, thus $\kappa = 10^6$, a million.

Exercise 2

Plot a ternary diagram using different values for the constant sum κ .

Answer of exercise 2

This exercise can be solved with the material of Appendix A.1. In the case of $\kappa = 1$, the three vertices of the triangle have as coordinates $(0, 0)$, $(\cos(60), \sin(60))$ and $(1, 0)$, just scaled up by κ otherwise. Considering $\kappa = 1$ (proportions), $\kappa = 100$ (percentages), $\kappa = 10^6$ (ppm) or $\kappa = 25$ (an arbitrary constant), the 4 diagrams in Figure 2.3 are obtained. Note that they differ in the scale of the axis of representation.

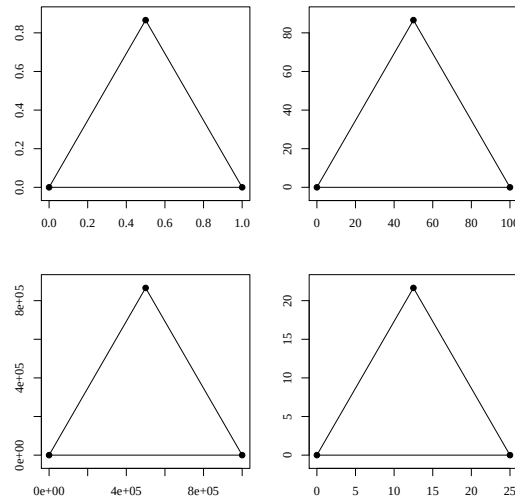


Fig. 2.3. Representation of a ternary diagram using $\kappa = 1$ (proportions), $\kappa = 100$ (percentages), $\kappa = 10^6$ (ppm) and $\kappa = 25$.

Exercise 3

Verify that data in Table 2.1 satisfy the conditions for being compositional. Plot them in a ternary diagram.

Table 2.1. Simulated data set (3 parts, 20 samples).

	1	2	3	4	5	6	7	8	9	10
x_1	79.07	31.74	18.61	49.51	29.22	21.99	11.74	24.47	5.14	15.54
x_2	12.83	56.69	72.05	15.11	52.36	59.91	65.04	52.53	38.39	57.34
x_3	8.10	11.57	9.34	35.38	18.42	18.10	23.22	23.00	56.47	27.11

	11	12	13	14	15	16	17	18	19	20
x_1	57.17	52.25	77.40	10.54	46.14	16.29	32.27	40.73	49.29	61.49
x_2	3.81	23.73	9.13	20.34	15.97	69.18	36.20	47.41	42.74	7.63
x_3	39.02	24.02	13.47	69.12	37.89	14.53	31.53	11.86	7.97	30.88

Answer of exercise 3

The data in table 2.1 are strictly positive and, for each sample, the sum is 100. Thus, the data satisfy the conditions of being compositional. Nevertheless, the main criterion, namely that they carry only relative information, cannot be verified with simulated data. This is a subjective decision of the researcher in practical applications, who has to decide, according to his/her knowledge of the problem analyzed. Figure 2.4 represents the data set in a ternary diagram.

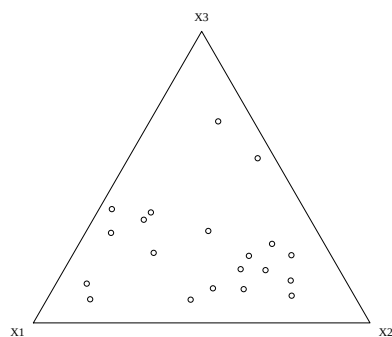


Fig. 2.4. Representation of the data in table 2.1.

Exercise 4

Compute the Euclidean distance between the first two vectors of Table 2.1. Imagine originally a fourth part x_4 was measured, constant for all samples and equal to 5%. Take the first two vectors, close them to sum up to 95%, add the fourth part to them (so that they sum up to 100%), and compute the Euclidean distance between the closed vectors. If the Euclidean distance is subcompositionally dominant, the distance between the four-part compositions must be greater than or equal to the distance between the three-part compositions.

Answer of exercise 4

The two vectors to be compared are

$$\mathbf{a} = [79.07, 12.83, 8.10] \quad \text{and} \quad \mathbf{b} = [31.74, 56.69, 11.57] .$$

Their modified versions, adding the fourth part of 5% and rescaling the first three to 95%, are:

$$\begin{aligned} \mathbf{a}' &= \left[\frac{95}{100} 79.07, \frac{95}{100} 12.83, \frac{95}{100} 8.10, 5 \right] = [75.1165, 12.1885, 7.6950, 5.0000] , \\ \mathbf{b}' &= \left[\frac{95}{100} 31.74, \frac{95}{100} 56.69, \frac{95}{100} 11.57, 5 \right] = [30.1530, 53.8555, 10.9915, 5.0000] . \end{aligned}$$

The corresponding (square) Euclidean distances are

$$\begin{aligned} d^2(\mathbf{a}, \mathbf{b}) &= (79.07 - 31.74)^2 + (12.83 - 56.69)^2 + (8.10 - 11.57)^2 \\ &= 4175.869, \\ d^2(\mathbf{a}', \mathbf{b}') &= (75.1165 - 30.153)^2 + (12.1885 - 53.8555)^2 + \\ &\quad + (7.695 - 10.9915)^2 + (5 - 5)^2 \\ &= 3768.722, \end{aligned}$$

which implies that the Euclidean distance is *not* subcompositionally dominant, because the distance measured with four parts, $\sqrt{3768.722}$, is smaller than the distance measured with three parts, $\sqrt{4175.869}$.

Exercise 5

Consider the two vectors $[0.7, 0.5, 0.8]$ and $[0.25, 0.75, 0.5]$. Perturb one vector by the other with and without previous closure. Is there any difference?

Answer of exercise 5

First we perturb the two vectors as given:

$$\begin{aligned}
 & [0.7, 0.5, 0.8] \oplus [0.25, 0.75, 0.5] \\
 &= \mathcal{C} [0.7 \cdot 0.25, 0.5 \cdot 0.75, 0.8 \cdot 0.5] \\
 &= \frac{1}{0.175 + 0.325 + 0.4} [0.175, 0.325, 0.4] \\
 &= \left[\frac{0.175}{0.9}, \frac{0.325}{0.9}, \frac{0.4}{0.9} \right] \\
 &= [0.1842, 0.3947, 0.4210] .
 \end{aligned}$$

Then, we repeat this, but closing the two compositions first,

$$\begin{aligned}
 \mathcal{C} [0.7, 0.5, 0.8] &= [0.7, 0.5, 0.8] / 2 = [0.35, 0.25, 0.4] \\
 \mathcal{C} [0.25, 0.75, 0.5] &= [0.25, 0.75, 0.5] / 1.5 = [1/6, 1/2, 1/3] .
 \end{aligned}$$

Perturbing these two vectors,

$$\begin{aligned}
 & [0.35, 0.25, 0.4] \oplus [1/6, 1/2, 1/3] \\
 &= \mathcal{C} [0.35/6, 0.25/2, 0.4/3] \\
 &= \frac{1}{0.0583 + 0.1250 + 0.1333} [0.0583, 0.1250, 0.1333] \\
 &= \left[\frac{0.0583}{0.3167}, \frac{0.1250}{0.3167}, \frac{0.1333}{0.3167} \right] \\
 &= [0.1842, 0.3947, 0.4210] .
 \end{aligned}$$

Thus, applying the closure before perturbation has no effect. Actually, in a chain of operations of powering and perturbation, we may skip the closure until the last step.

Exercise 6

Perturb each sample of the data set given in Table 2.1 with $\mathbf{s} = \mathcal{C}[0.1, 0.1, 0.8]$ and plot the initial and the resulting perturbed data set. What do you observe?

Answer of exercise 6

The plot of the original and the perturbed data are represented in Figure 3.5. Perturbing by $\mathbf{s}$ moves the data towards the apex corresponding to the

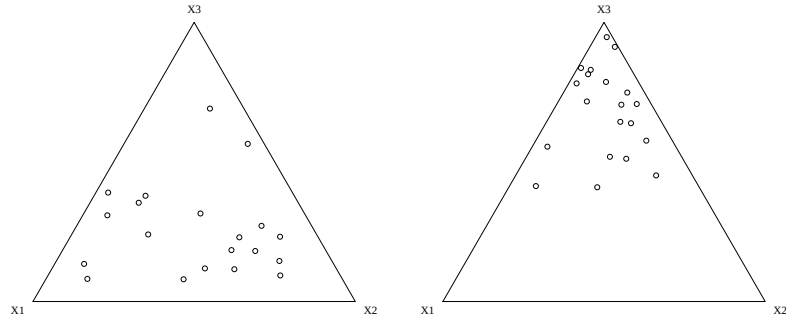


Fig. 3.5. Original (left) and perturbed data of Table 2.1.

component with the largest value in $\mathbf{s}$, the third part. Recall that perturbation is a shifting that does not alter the geometric relationships between samples.

Exercise 7

Apply powering with α ranging from -8 to $+8$ in steps of 1 unit to $\mathbf{t} = \mathcal{C}[0.7, 0.5, 0.8]$ and plot the resulting set of compositions. Join them by a line. What do you observe?

Answer of exercise 7

The 17 points are represented in Figure 3.6. They form a kind of arch, actually a compositional line as defined in Section 3.4, passing through the barycenter $\mathbf{n} = [1, 1, 1]/3$ (for $\alpha = 0$) and with leading vector $\mathbf{t}$.

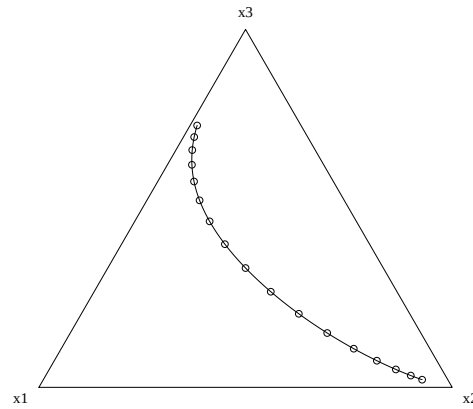


Fig. 3.6. Representation of the point $\mathbf{t} = \mathcal{C}[0.7, 0.5, 0.8]$ powered by α ranging from -8 to $+8$ in steps of 1 unit.

Exercise 8

Perturb the compositions obtained in Exercise 7 by $\mathbf{r} = \mathcal{C}[0.25, 0.75, 0.5]$. What is the result?

Answer of exercise 8

Figure 3.7 presents the result: another compositional line with leading vector $\mathbf{t}$, but now passing through $\mathbf{r}$. The figure includes also the compositional line of Exercise 7 as a dashed line, to show that the two lines are parallel and cut at infinity (the border of the simplex), as expected for two lines with the same leading vector.

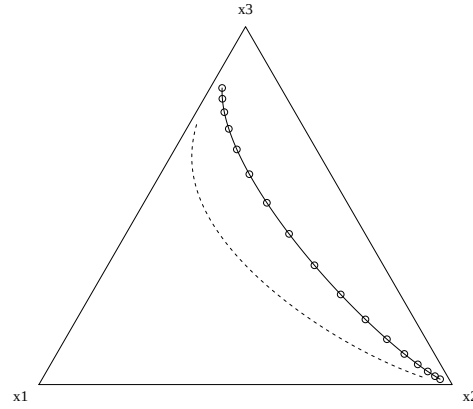


Fig. 3.7. Representation of the compositions obtained in Exercise 7 perturbed by $\mathbf{r} = \mathcal{C}[0.25, 0.75, 0.5]$.

Exercise 9

Compute the Aitchison inner product of $\mathbf{x} = \mathcal{C} [0.7, 0.4, 0.8]$ and $\mathbf{y} = \mathcal{C} [2, 8, 1]$. Are they orthogonal?

Answer of exercise 9

According to Definition 3.5, the scalar product sought can be computed in two ways: using all pairwise logratios (Eq. 3.2), or using the geometric means of the components of each vector (Eq. 3.3 or Eq. 3.4). We follow both approaches here to illustrate their equivalence. First, all pairwise logratios of the components of each vector have to be computed. They are summarized in Table 3.1. Note that both tables are antisymmetric, because $\ln(a/b) =$

Table 3.1. Pairwise logratios between the two vectors of Exercise 9.

$\ln(\cdot/\cdot)$	x_1	x_2	x_3		y_1	y_2	y_3
x_1	0.0000	0.5596	-0.1335	y_1	0.0000	-1.3863	0.6931
x_2	-0.5596	0.0000	-0.6931	y_2	1.3863	0.0000	2.0794
x_3	0.1335	0.6931	0.0000	y_3	-0.6931	-2.0794	0.0000

$-\ln(b/a)$. To do the scalar products of these vectors of pairwise logratios, it is enough to use either the lower triangle or the upper triangle of these matrices, as $\ln(a/b) \cdot \ln(c/d) = \ln(b/a) \cdot \ln(d/c)$. Using the lower triangle, the scalar product sought is

$$\begin{aligned} \langle \mathbf{x}, \mathbf{y} \rangle_a &= \frac{1}{3} (-0.5596 \times 1.3863 + 0.1335 \times (-0.6931) + 0.6931 \times (-2.0794)) \\ &= -0.7699. \end{aligned}$$

The second way to compute this scalar product requires the following steps:

(a) Computation of the geometric means

$$g_m(\mathbf{x}) = \sqrt[3]{0.7 \cdot 0.4 \cdot 0.8} = 0.607, \quad g_m(\mathbf{y}) = \sqrt[3]{2 \cdot 8 \cdot 1} = 2.520.$$

(b) Computation of the logratio of each component divided by the geometric mean, shown in Table 3.2.

Table 3.2. Logratios of the two vectors of Exercise 9.

$\ln(\frac{x_1}{g_m(\mathbf{x})})$	$\ln(\frac{x_2}{g_m(\mathbf{x})})$	$\ln(\frac{x_3}{g_m(\mathbf{x})})$		$\ln(\frac{y_1}{g_m(\mathbf{y})})$	$\ln(\frac{y_2}{g_m(\mathbf{y})})$	$\ln(\frac{y_3}{g_m(\mathbf{y})})$
0.1420	-0.4176	0.2756		-0.2310	1.1552	-0.9242

- (c) Computation of the products of each $\ln(\frac{x_i}{g_m(\mathbf{x})})$ with the corresponding $\ln(\frac{y_i}{g_m(\mathbf{y})})$ and adding them to obtain the sought result:

$$\begin{aligned}\langle \mathbf{x}, \mathbf{y} \rangle_a &= (0.1420 \times -0.2310) + (-0.4176 \times 1.1552) + (0.2756 \times -0.9242) \\ &= (-0.0328) + (-0.4824) + (-0.2547) \\ &= -0.7699.\end{aligned}$$

As expected, the same result is obtained.

Exercise 10

Compute the Aitchison norm of $\mathbf{x}_1 = \mathcal{C}[0.7, 0.4, 0.8]$ and call it a . Compute $\alpha \odot \mathbf{x}_1$ with $\alpha = 1/a$. Compute the Aitchison norm of the resulting composition. How do you interpret the result?

Answer of exercise 10

According to Definition 3.6, the squared Aitchison norm is

$$\|\mathbf{x}\|_a^2 = \frac{1}{2D} \sum_{i=1}^D \sum_{j=1}^D \left(\ln \frac{x_i}{x_j} \right)^2.$$

Applying this formula to the vector $\mathbf{x}_1 = \mathcal{C}[0.7, 0.4, 0.8]$, and taking into account that $\ln \frac{x_i}{x_i} = 0$ for all i , we obtain

$$\begin{aligned} a^2 &= \frac{1}{2 \cdot 3} \left(\left(\ln \frac{0.7}{0.4} \right)^2 + \left(\ln \frac{0.7}{0.8} \right)^2 + \left(\ln \frac{0.4}{0.7} \right)^2 \right. \\ &\quad \left. + \left(\ln \frac{0.4}{0.8} \right)^2 + \left(\ln \frac{0.8}{0.7} \right)^2 + \left(\ln \frac{0.8}{0.4} \right)^2 \right) \\ &= \frac{1}{6} (0.3132^2 + 0.0178^2 + 0.3132^2 + 0.4805^2 + 0.0178^2 + 0.4805^2) \\ &= 0.2705, \end{aligned}$$

and $a = 0.5201$. Thus, $\alpha \odot \mathbf{x}_1 = (1/a) \odot \mathbf{x}_1 = \mathcal{C}[0.5037, 0.1717, 0.6511]$ and $\|\alpha \odot \mathbf{x}_1\|_a = 1$. Consequently, the operation of powering by the inverse of the norm is a normalization to unit length.

Exercise 11

Redo Exercise 4, but using the Aitchison distance given in Definition 3.7. Is it subcompositionally dominant?

Answer of exercise 11

The two vectors to be compared are

$$\mathbf{a} = [79.07, 12.83, 8.10] \quad \text{and} \quad \mathbf{b} = [31.74, 56.69, 11.57] .$$

As shown in Exercise 4, their modified versions, adding the fourth part of 5% and rescaling the first three to 95%, are:

$$\mathbf{a}' = [75.1165, 12.1885, 7.6950, 5.00] , \quad \mathbf{b}' = [30.1530, 53.8555, 10.9915, 5.00] .$$

The squared Aitchison distance given in Definition 3.7 is in this case (omitting zeros)

$$\begin{aligned} d_a^2(\mathbf{a}, \mathbf{b}) &= \|\mathbf{a} \ominus \mathbf{b}\|_a^2 = \frac{1}{2 \cdot 3} \sum_{i=1}^D \sum_{j=1}^D \left(\ln \frac{a_i}{a_j} - \ln \frac{b_i}{b_j} \right)^2 \\ &= \frac{1}{6} (2.3986^2 + 1.2693^2 + 1.1293^2 + 2.3986^2 + 1.2693^2 + 1.1293^2) \\ &= \frac{1}{6} (5.7531 + 1.6111 + 1.2752 + 5.7531 + 1.6111 + 1.2752) \\ &= \frac{17.2790}{6} = 2.8798 , \end{aligned}$$

and taking the square root, we obtain $d_a(\mathbf{a}, \mathbf{b}) = 1.6970$.

For vectors $\mathbf{a}'$ and $\mathbf{b}'$, the analogous computations give $d_a(\mathbf{a}', \mathbf{b}') = 1.7181$, showing that the Aitchison distance is subcompositionally dominant.

Exercise 12

In a two-part composition $\mathbf{x} = [x_1, x_2]$, simplify the formula for the Aitchison distance, taking $x_2 = 1 - x_1$ (using $\kappa = 1$). Use it to plot seven equally spaced points on the segment $(0, 1) = \mathcal{S}^2$, from $x_1 = 0.014$ to $x_1 = 0.986$.

Answer of exercise 12

For a 2-part composition $\mathbf{x} = [x_1, x_2]$, taking $x_2 = 1 - x_1$ and using $\kappa = 1$, the squared Aitchison distance reduces to

$$d_a^2(\mathbf{x}, \mathbf{y}) = \frac{1}{2} \left(\ln \frac{x_1}{1 - x_1} - \ln \frac{y_1}{1 - y_1} \right)^2.$$

The Aitchison distance between the two endpoints

$$\mathbf{x} = [0.014, 1 - 0.014] \quad \text{and} \quad \mathbf{y} = [0.986, 1 - 0.986]$$

is then $d_a(\mathbf{x}, \mathbf{y}) = 6.0169$. To obtain seven equally spaced points, this distance has to be divided by 6, which is approximately 1.0028. The easiest way to determine the points is to use the only logratio coordinate corresponding to a 2-part composition. The coordinate for the starting point $\mathbf{x}_0$ is -3.0085 , and adding 1.0028 we obtain for $\mathbf{x}_1$ the coordinate -2.0056 . Repeating this last step six times, the remaining five points are obtained as given in Table 3.3. The plot can be done either on a segment, as in Figure 3.8, or as a scatterplot, as in Figure 3.9. Note that a distance $d_a(\mathbf{x}, \mathbf{y}) = 6$ and a starting point $\mathbf{x}_0$ with coordinate -3 would give nearly identical results.

Table 3.3. Seven equally space points: logratio coordinate and parts in $\mathcal{S}^2$

point $\mathbf{x}_i$	$\frac{1}{\sqrt{2}} \ln \frac{x_i}{1-x_i}$	x_i	$1 - x_i$
$\mathbf{x}_0$	-3.0085	0.014	0.986
$\mathbf{x}_1$	-2.0056	0.055	0.945
$\mathbf{x}_2$	-1.0028	0.195	0.805
$\mathbf{x}_3$	0.0000	0.500	0.500
$\mathbf{x}_4$	1.0028	0.805	0.195
$\mathbf{x}_5$	2.0056	0.945	0.055
$\mathbf{x}_6$	3.0085	0.986	0.014



Fig. 3.8. Plot of seven equally spaced points in $\mathcal{S}^2$ as a segment.

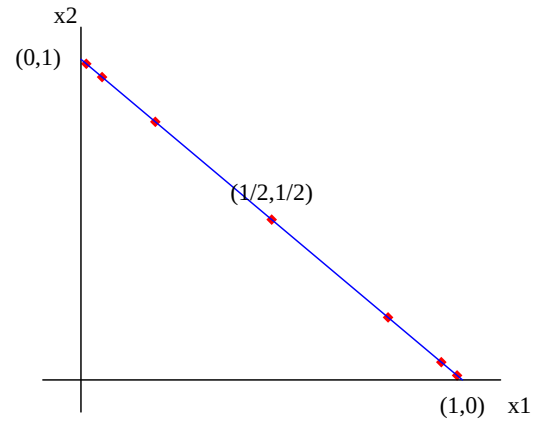


Fig. 3.9. Plot of seven equally spaced points in $\mathcal{S}^2$ as a scatterplot in $\mathbb{R}^2$. The blue line is the two-part simplex $\mathcal{S}^2$.

Exercise 13

In a rock sample, several radioactive isotopes have been measured, obtaining $[^{238}\text{U}, ^{232}\text{Th}, ^{40}\text{K}] = [150, 30, 110]$ ppm. Which will be the composition after $\Delta t = 10^9$ years? And after another Δt years? Which was the composition Δt years ago? And Δt years before that? Close these five compositions and represent them in a ternary diagram. What do you see? Could you write the evolution as an equation? (Half-life disintegration periods: $[^{238}\text{U}, ^{232}\text{Th}, ^{40}\text{K}] = [4.468; 14.05; 1.277] \cdot 10^9$ years).

Answer of exercise 13

The rate at which a radioactive isotope decays is measured in half-life. It is the time taken for the activity of a given amount of a radioactive substance to decay to half of its initial value. The assumption that the mass $N(t)$ of a given radioactive isotope decays exponentially at the disintegration rate, or decay rate, λ , leads to

$$N(t) = N(0) \cdot e^{-\lambda t},$$

where $N(0)$ is the mass of the isotope at the time origin. The relation between the half-life $t_{1/2}$ and the disintegration rate is then obtained as

$$\frac{N(0)}{2} = N(0) \cdot e^{-\lambda t_{1/2}} \quad , \quad \lambda = \frac{\ln 2}{t_{1/2}}.$$

Particular isotopes decay at different rates, as each has its own decay constant λ . With three radioactive components, the three equations can be grouped in a vector form as

$$\mathbf{N}(t) = \mathbf{N}(0) \exp(-\boldsymbol{\lambda}t),$$

where $\mathbf{N}(t) = [N_1(t), N_2(t), N_3(t)]$ are, at time t , the masses of the isotopes $[^{238}\text{U}, ^{232}\text{Th}, ^{40}\text{K}]$. Note that $\mathbf{N}(t)$ (ppm) are not absolute masses but proportions of mass with respect to the total mass of the radioactive sample, which may include other radioactive isotopes and inert materials. This means that $[\mathbf{N}(t), F(t)]$ is a composition, closed to $\kappa = 10^6$, where $F(t)$ represents the fill-up value $F(t) = 10^6 - N_1(t) - N_2(t) - N_3(t)$. As interest is focussed in the proportions of the isotopes in time, within their own subcomposition, the previous equation can be closed to proportions (closure constant $\kappa = 1$) to obtain

$$\mathbf{x}(t) = \mathcal{C}\mathbf{N}(t) = \mathbf{x}(0) \oplus [t \odot \exp(-\boldsymbol{\lambda})],$$

which shows the linear structure of the evolution model. The initial values for the masses (ppm) of the sample are $\mathbf{N}(0) = [150, 30, 110]$ ppm and the half-life periods are $[4.468, 14.05, 1.277] \cdot 10^9$ years. The initial composition is $\mathbf{x}(0) = \mathcal{C}\mathbf{N}(0) = [0.5172414, 0.1034483, 0.3793103]$. For each isotope the disintegration rate is $\lambda = \ln 2 / t_{1/2}$ (year⁻¹), therefore, the vector of disintegration rates is

$$\begin{aligned} \boldsymbol{\lambda} &= \left[\frac{\ln 2}{4.468 \cdot 10^9}, \frac{\ln 2}{14.05 \cdot 10^9}, \frac{\ln 2}{1.277 \cdot 10^9} \right] \\ &= [1.551359, 0.4933432, 5.427934] \cdot 10^{-10}. \end{aligned}$$

The composition $\mathbf{x}(t)$ for times $t = -2\Delta t, -\Delta t, 0, \Delta t, 2\Delta t$, with $\Delta t = 10^9$ (years), can be computed either from $\mathbf{x}(t)$ or from $\mathbf{N}(t)$, taking closure afterwards. Using the second strategy,

$$\mathbf{N}(t) = [150, 30, 110] \oplus [\exp(-4.468 \cdot t), \exp(-14.05 \cdot t), \exp(-1.277 \cdot t)] ,$$

and then $\mathbf{x}(t) = \mathcal{C}\mathbf{N}(t)$. This way allows us to compare $\mathbf{N}(t)$ and $\mathbf{x}(t)$. $\mathbf{N}(t)$ is the vector of the first three parts of the four-part composition $[\mathbf{N}(t), F(t)]$ (closed to $\kappa = 10^6$), whereas $\mathbf{x}(t)$ is a subcomposition closed to $\kappa = 1$. Figure 3.10 shows the evolution in time of the subcomposition in a time interval $(-15, 15) \cdot 10^9$ (years) (red line). Times $t = (-2, -1, 0, 1, 2) \cdot 10^9$ (years) are marked with circles. Figure 3.11 shows the evolution of the sample in time in

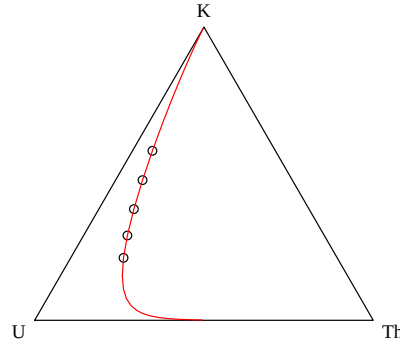


Fig. 3.10. Proportions of the isotopes ^{238}U , ^{232}Th , ^{40}K evolving in time due to radioactive disintegration represented in a ternary diagram. Times $t = (-2, -1, 0, 1, 2) \cdot 10^9$ (years) are marked with circles. The line represent the evolution of proportions of the three isotopes in the time-interval $(-15, 15) \cdot 10^9$ (years).

the same time interval. Left panel shows the proportions in the subcomposition $\mathbf{x}(t)$ and the right panel the same expressed in ppm's (the fill-up component $F(t)$ is not shown). The differences between the left and right panel are very apparent, but the ratios between the three components are exactly the same, thus recalling that vectors of proportional positive components are compositionally equivalent. Moreover, Figure 3.11, left panel, shows the typical behavior of proportions following a straight-line in the simplex (shown in Figure 3.10). The numerical values at times $t = (-2, -1, 0, 1, 2) \cdot 10^9$ (years) are shown in Table 3.4, both in proportions of the subcomposition and in ppm.

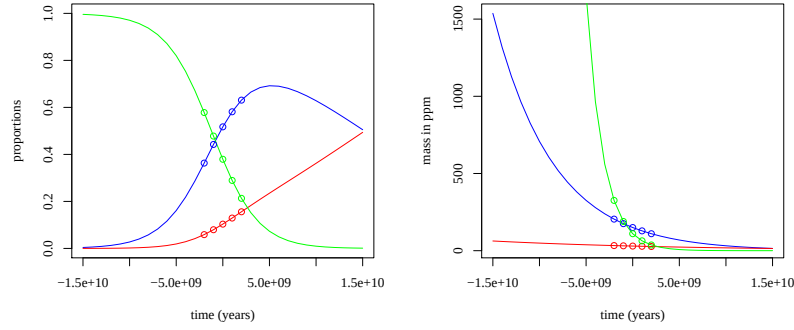


Fig. 3.11. Proportions of the isotopes ^{238}U , ^{232}Th , ^{40}K evolving in time due to radioactive disintegration. Times $t = (-2, -1, 0, 1, 2) \cdot 10^9$ (years) are marked with circles. Lines represent the evolution of proportions of the three isotopes in the time-interval $(-15, 15) \cdot 10^9$ (years). Left panel, evolution of proportions in the subcomposition: ^{238}U (blue), ^{232}Th (red), ^{40}K (green). Right panel, the same evolution maintaining the ppm units on the total mass.

Table 3.4. Proportions of ^{238}U , ^{232}Th , ^{40}K at times $t = (-2, -1, 0, 1, 2) \cdot 10^9$, given as proportions in $\mathbf{x}(t)$ and $\mathbf{N}(t)$ in ppm.

t (years)	^{238}U	^{232}Th	^{40}K	^{238}U (ppm)	^{232}Th (ppm)	^{40}K (ppm)
	proportions in $\mathbf{x}(t)$					
$-2 \cdot 10^9$	0.3630916	0.05876897	0.5781395	204.6	33.1	325.7
$-1 \cdot 10^9$	0.4423789	0.07959309	0.4780280	175.2	31.5	189.3
0	0.5172414	0.10344828	0.3793103	150.0	30.0	110.0
$+1 \cdot 10^9$	0.5813977	0.12925649	0.2893458	128.4	28.6	63.9
$+2 \cdot 10^9$	0.6309648	0.15593126	0.2131040	110.0	27.2	37.1

Exercise 14

Consider the data set given in Table 2.1. Compute the clr coefficients defined in Equation (4.3). Verify that the sum of the transformed components equals zero.

Answer of exercise 14

The computation of clr coefficients can be carried out in two different ways. The first one, given in Equation (4.3), is to compute the geometric mean of each composition, $g_m(\mathbf{x}_i)$, $i = 1, 2, \dots, 20$; then to divide each part x_{ij} , $j = 1, 2, 3$ by $g_m(\mathbf{x}_i)$ and, finally, to take logarithms. A computationally more efficient alternative is to take first logarithms on the whole data matrix, $\ln(x_{ij})$; then to compute the average of the logarithms for each composition and, finally, to subtract it from each component of the log-composition, i.e.

$$\text{clr}(\mathbf{x}_i) = \ln[x_{i1}, x_{i2}, x_{i3}] - \left[\frac{1}{3} \sum_{j=1}^3 \ln x_{ij}, \frac{1}{3} \sum_{j=1}^3 \ln x_{ij}, \frac{1}{3} \sum_{j=1}^3 \ln x_{ij} \right],$$

for $\ln g_m(\mathbf{x}_i) = (1/3) \sum_j \ln x_{ij}$. Theoretically, the sum of the clr coefficients is zero. However, depending on the way of computing clr-coefficients and the precision of the calculator, their numerical sums can be e.g. $4.440892 \cdot 10^{-16}$, $0.000000 \cdot 10^{+00}$, or $3.784524 \cdot 10^{-40}$, due to rounding errors.

Exercise 15

Using the sign matrix of Table 4.1 and Equation (4.10), compute the coefficients of the corresponding contrast matrix, a 5×6 matrix. Which are the balancing elements in the corresponding basis?

Answer of exercise 15

Table 4.5, in its upper part, shows the contrast matrix Ψ corresponding to the SBP code of signs obtained using Equation (4.10). The rows of the Ψ matrix are the clr transformations of the balancing elements. Taking clr^{-1} , the balancing elements (compositions in $\mathcal{S}^6$) are obtained. They are shown in the lower part of Table 4.5. The rows of Ψ are the clr of the balancing

Table 4.5. Upper table: entries of the contrast matrix Ψ . Lower table: balancing elements in the corresponding basis of $\mathcal{S}^6$.

	$\Psi_{.1}$	$\Psi_{.2}$	$\Psi_{.3}$	$\Psi_{.4}$	$\Psi_{.5}$	$\Psi_{.6}$
$\Psi_{1.}$	0.2886751	0.2886751	-0.5773503	-0.5773503	0.2886751	0.2886751
$\Psi_{2.}$	0.8660254	-0.2886751	0.0000000	0.0000000	-0.2886751	-0.2886751
$\Psi_{3.}$	0.0000000	0.8164966	0.0000000	0.0000000	-0.4082483	-0.4082483
$\Psi_{4.}$	0.0000000	0.0000000	0.0000000	0.0000000	0.7071068	-0.7071068
$\Psi_{5.}$	0.0000000	0.0000000	0.7071068	-0.7071068	0.0000000	0.0000000
Balancing elements						
$\mathbf{e}_1$	0.2065586	0.2065586	0.0868827	0.0868827	0.2065586	0.2065586
$\mathbf{e}_2$	0.3588479	0.1130916	0.1509386	0.1509386	0.1130916	0.1130916
$\mathbf{e}_3$	0.1516947	0.3432183	0.1516947	0.1516947	0.1008487	0.1008487
$\mathbf{e}_4$	0.1533464	0.1533464	0.1533464	0.1533463	0.3110041	0.0756103
$\mathbf{e}_5$	0.1533464	0.1533464	0.3110041	0.0756103	0.1533464	0.1533463

elements $\mathbf{e}_i$, $i = 1, 2, \dots, 5$. Therefore, the rows of Ψ add to zero, and the sum of the squared entries add up to 1 (check it!). The balancing elements are closed compositions; their parts add to 1. If $\Psi_{ij} = 0$, then the j -th part of $\mathbf{e}_i$ is the geometric mean of the D components in $\mathbf{e}_i$ (check it!).

Exercise 16

The two rows of the following matrix are six-part compositions in percentage:

$$\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} 3.74 & 9.35 & 16.82 & 18.69 & 23.36 & 28.04 \\ 9.35 & 28.04 & 16.82 & 3.74 & 18.69 & 23.36 \end{pmatrix}.$$

Using the binary partition of Table 4.1 and Equation (4.9), compute their five balances. Compute their Aitchison squared-norms and inner product. Which is the angle between the two compositions?

Answer of exercise 16

The rows of the following matrix are the balances associated with the SBP in Table 4.1:

$$\begin{pmatrix} \mathbf{x}^* \\ \mathbf{y}^* \end{pmatrix} = \begin{pmatrix} -0.4220417 & -1.3748952 & -0.8221731 & -0.1291226 & -0.07454317 \\ 0.9713092 & -0.7813041 & 0.2401522 & -0.1577106 & 1.06312306 \end{pmatrix}.$$

Note that, although the two compositions have their parts permuted, i.e. they have identical parts but in a different order, the corresponding balances do not appear permuted, but different. The Aitchison square-norm can be obtained by computing the ordinary Euclidean norm of the balances:

$$\|\mathbf{x}\|_a^2 = \|\mathbf{x}^*\|^2 = 2.766654.$$

Moreover, $\|\mathbf{x}\|_a^2 = \|\mathbf{y}\|_a^2$, as the Aitchison norm is permutation invariant. This is so, despite the fact that the balances $\mathbf{x}^*$ are not a permutation of the balances $\mathbf{y}^*$. The Aitchison inner product is

$$\langle \mathbf{x}, \mathbf{y} \rangle_a = \langle \mathbf{x}^*, \mathbf{y}^* \rangle = 0.4079471.$$

The Aitchison inner product, like all inner products, can be expressed as a function of the cosine of the angle between the two vectors, i.e.

$$\langle \mathbf{x}, \mathbf{y} \rangle_a = \|\mathbf{x}\|_a \|\mathbf{y}\|_a \cos \alpha,$$

where α is the angle between $\mathbf{x}$ and $\mathbf{y}$ in radians. Thus, $\alpha = 1.422805$ radians or, in degrees, $\alpha = 81.52^\circ$.

Exercise 17

Consider the logratios $c_1 = \ln x_1/x_3$ and $c_2 = \ln x_2/x_3$ in a simplex $\mathcal{S}^3$. They are coordinates when using the alr transformation. Find two unitary vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ such that $\langle \mathbf{x}, \mathbf{e}_i \rangle_a = c_i$, $i = 1, 2$. Compute the inner product $\langle \mathbf{e}_1, \mathbf{e}_2 \rangle_a$, and determine the angle between them. Does the result change if the considered simplex is $\mathcal{S}^7$?

Answer of exercise 17

We are looking for vectors $\mathbf{e}_i$, $i = 1, 2$, such that $\langle \mathbf{x}, \mathbf{e}_i \rangle_a = c_i$. The Aitchison inner product can be computed using *Property 4.2 (b)*, $\langle \mathbf{x}, \mathbf{e}_i \rangle_a = \langle \text{clr}(\mathbf{x}), \text{clr}(\mathbf{e}_i) \rangle$, where $\langle \cdot, \cdot \rangle$ denotes the ordinary inner product in $\mathbb{R}^2$. Logratio c_1 satisfies

$$c_1 = [\ln x_1, \ln x_2, \ln x_3][1, 0, -1]^\top = \text{clr}[\mathbf{x}][1, 0, -1]^\top = \langle \text{clr}(\mathbf{x}), [1, 0, -1] \rangle,$$

where matrix notation using row vectors has been used. The last two members are obtained taking into account that the k^{th} component of $\text{clr}(\mathbf{x})$ can be expressed as a difference of logarithms, $\text{clr}_k(\mathbf{x}) = \ln x_k - \ln g_m(\mathbf{x})$, and that the components of $[1, 0, -1]$ add to zero. Therefore, subtracting $\ln g_m(\mathbf{x})$ from $\ln x_k$ does not change c_1 . Consequently, $[1, 0, -1]$ is the clr of $\mathbf{e}_1$. A similar reasoning leads to $[0, 1, -1] = \text{clr}(\mathbf{e}_2)$. However, these two clr-vectors are not unitary as required by the statement. In fact, the norm of both is $\sqrt{2}$. Strictly speaking the problem has no solution, as $\mathbf{e}_1$ and $\mathbf{e}_2$ are not unitary. Alternatively, two unitary clr vectors can be considered

$$\text{clr}(\mathbf{e}_1) = \left[\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}} \right] \quad , \quad \text{clr}(\mathbf{e}_2) = \left[0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right] ,$$

which correspond to the compositions

$$\mathbf{e}_1 = \mathcal{C} \left[\exp \left(\frac{1}{\sqrt{2}} \right), 1, \exp \left(-\frac{1}{\sqrt{2}} \right) \right] , \mathbf{e}_2 = \mathcal{C} \left[1, \exp \left(\frac{1}{\sqrt{2}} \right), \exp \left(-\frac{1}{\sqrt{2}} \right) \right] ,$$

such that $\langle \mathbf{x}, \mathbf{e}_i \rangle_a = \sqrt{2} c_i$, $i = 1, 2$.

To compute the angle α between $\mathbf{e}_1$ and $\mathbf{e}_2$ (unitary versions), the inner product between their clr's can be used:

$$\cos \alpha = \langle \mathbf{e}_1, \mathbf{e}_2 \rangle_a = \langle \text{clr}(\mathbf{e}_1), \text{clr}(\mathbf{e}_2) \rangle = 0 + 0 + \frac{1}{2} = \frac{1}{2} .$$

This means that $\alpha = \pi/3$ or $\alpha = 2\pi - \pi/3$ or if preferred $\pm 60^\circ$. This result is generalized to $\mathcal{S}^D$, particularly for $D = 7$. The angle is always $\alpha = \pi/3$ as the computation of the inner product $\langle \text{clr}(\mathbf{e}_i), \text{clr}(\mathbf{e}_j) \rangle = 0 + 0 + \dots + 1/2 = 1/2$. Note that this stands in agreement with the fact that the alr leads to coordinates of an oblique basis. For more details, see Egozcue et al. (2011).¹

¹ J. J. Egozcue, C. Barceló-Vidal, J. A. Martín-Fernández, E. Jarauta-Bragulat, J. L. Díaz-Barrero and G. Mateu-Figueras (2011) Elements of simplicial linear algebra and geometry, in Pawlowsky-Glahn, V. and Buccianti A. (Eds.) (2011) *Compositional Data Analysis: Theory and Applications*, John Wiley & Sons, Ltd. 378p.

Exercise 18

When computing the clr of a composition $\mathbf{x} \in \mathcal{S}^D$, a clr coefficient is $\xi_i = \ln(x_i/g_m(\mathbf{x}))$. It can be considered as a balance between two groups of parts. Which are the parts and which is the corresponding balancing element?

Answer of exercise 18

The coefficient $\xi_i = \ln(x_i/g_m(\mathbf{x}))$ is a balance up to a multiplicative constant. It is computed as

$$\begin{aligned}\xi_i &= \ln \frac{x_i}{\prod_{k=1}^D x_k^{1/D}} = \ln \frac{x_i^{(1-1/D)}}{\prod_{k \neq i}^D x_k^{1/D}} = \frac{D-1}{D} \ln \frac{x_i}{\prod_{k \neq i}^D x_k^{1/(D-1)}} \\ &= \sqrt{\frac{D-1}{D}} \left(\sqrt{\frac{D-1}{D}} \ln \frac{x_i}{\prod_{k \neq i}^D x_k^{1/(D-1)}} \right).\end{aligned}$$

The last parenthesis includes a balance of the part x_i over the geometric mean of the $D-1$ remaining parts; therefore, the multiplicative constant is $\sqrt{(D-1)/D}$. This result was presented in Egozcue, and Pawlowsky-Glahn (2006).² In order to get the balancing element corresponding to the balance

$$b = \sqrt{\frac{D-1}{D}} \ln \frac{x_i}{\prod_{k \neq i}^D x_k^{1/(D-1)}},$$

Definition 4.8 can be used. The sign code of this balance is

x_1	x_2	$\dots$	x_{i-1}	x_i	x_{i+1}	$\dots$	x_D	r	s
-1	-1	$\dots$	-1	+1	-1	$\dots$	-1	1	$D-1$

where the only +1 is placed at the part x_i . The clr of the corresponding balancing element is

$$\text{clr}(\mathbf{e}_i) = \left[-\sqrt{\frac{1}{(D-1)D}}, \dots, +\sqrt{\frac{D-1}{D}}, \dots, -\sqrt{\frac{1}{(D-1)D}} \right],$$

where the only positive square-root is in the i th-component. Taking inverse clr, the balancing element is

$$\mathbf{e}_i = \exp \left[-\sqrt{\frac{1}{(D-1)D}}, \dots, +\sqrt{\frac{D-1}{D}}, \dots, -\sqrt{\frac{1}{(D-1)D}} \right].$$

Note that the balancing element is unitary, as the squares of the components of $\text{clr}(\mathbf{e}_i)$ add to 1. The result can be checked computing

$$\langle \mathbf{x}, \mathbf{e}_i \rangle_a = \langle \text{clr}(\mathbf{x}), \text{clr}(\mathbf{e}_i) \rangle = b.$$

² Egozcue, J. J. and V. Pawlowsky-Glahn (2006) Simplicial geometry for compositional data. In: Buccianti, A., Mateu-Figueras, G. and Pawlowsky-Glahn, V., (eds) Compositional Data Analysis in the Geosciences: From Theory to Practice. Geological Society, London, Special Publications, 264, 67-77, 2006.

Exercise 19

Six parties have contested an election. In five districts, they have obtained the votes in Table 4.4. Parties are divided into left (L) and right (R) wing parties. Is there some relationship between the L-R balance and the relative votes of R1-R2? Select an adequate SBP to analyze this question and obtain the corresponding balance coordinates. Find the correlation matrix of the balances and give an interpretation to the maximum correlated two balances. Compute the distances between the five districts. Which are the two districts with the maximum and minimum interdistance? Are you able to distinguish some cluster of districts?

Table 4.4. Votes obtained by six parties in five districts.

district	L1	L2	R1	R2	L3	L4
d1	10	223	534	23	154	161
d2	43	154	338	43	120	123
d3	3	78	29	702	265	110
d4	5	107	58	598	123	92
d5	17	91	112	487	80	90

Answer of exercise 19

To analyze the stated question, it is convenient to include the balance between left and right wing parties and the balance between the two right wing parties in the sequential binary partition (SBP). This is easily obtained starting a SBP separating L and R parties. The partitions between L parties are arbitrary as no question is stated within the L-subcomposition. For instance, such a partition can be coded with the sign Table 4.6. Note that, with this SBP,

Table 4.6. Sign table coding the SBP.

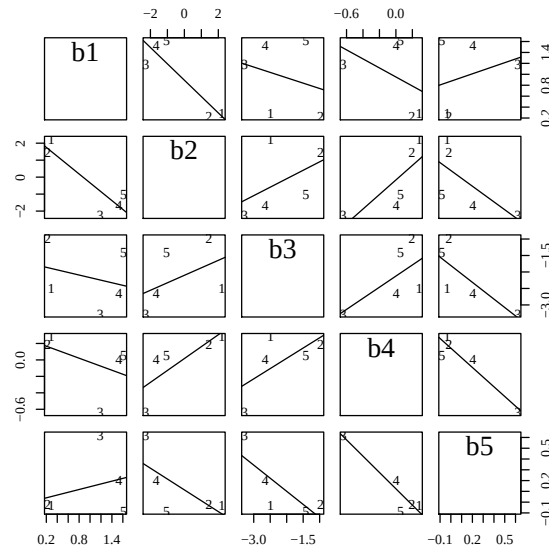
balance	L1	L2	R1	R2	L3	L4
b_1	-1	-1	+1	+1	-1	-1
b_2	0	0	+1	-1	0	0
b_3	+1	-1	0	0	-1	-1
b_4	0	+1	0	0	-1	-1
b_5	0	0	0	0	+1	-1

balances b_1 and b_2 are those required in the statement. The computed balances are presented in Table 4.7.

Table 4.7. Computed balances.

district	b_1	b_2	b_3	b_4	b_5
d1	0.2897335	2.223781	-2.4877369	0.284135330	-0.03143214
d2	0.2224713	1.457945	-0.9679359	0.193603222	-0.01746031
d3	1.1855678	-2.253293	-3.2738884	-0.639640375	0.62172326
d4	1.5276191	-1.649785	-2.6496013	0.004770274	0.20534082
d5	1.6130060	-1.039281	-1.4125035	0.057106876	-0.08328518

Scatter plots of balances are useful to examine any relationship between them. Figure 4.6 shows all scatter plots between balances and regression lines fitted; the numbers in the plot correspond to districts.

**Fig. 4.6.** Scatter plots and regression lines; numbers correspond to districts.

Focussing on the relationship between b_1 and b_2 , two groups of districts are distinguished: (d1,d2) and (d3,d4,d5). Within each group the values of b_1 and b_2 seem similar. The correlation coefficient between b_1 and b_2 is -0.88 , but the linear relationship is dominated by leverage of the two groups of districts. Other high correlations can be found, e.g. -0.94 between b_4 and b_5 , but also leverage effects invalidate any further linear relationship. In this case, any attempt of statistical modelling is irrelevant. Inter-distances are pos-

sibly the only result that may clarify similarities between districts. Note that (Euclidean) distances computed from balances are the Aitchison distances when votes are assumed compositional. Table 4.8 shows the inter-Aitchison distances between the proportions of votes in the five districts.

Table 4.8. Inter-Aitchison distances between proportions of votes in five districts.

	d1	d2	d3	d4
d2	1.705642			
d3	4.769143	4.595771		
d4	4.086218	3.778186	1.208137	
d5	3.689035	2.896607	2.470582	1.412962

These inter distances confirm that d1 is relatively close to d2. Also the pairs (d3,d4) and (d4,d5) are similarly close, however d3 and d5 are more distant. It is in agreement with the two groups of districts visualized in the scatter plot of balances b_1 and b_2 .

Exercise 20

Consider the data set given in Table 2.1. Check the data for zeros. Apply the alr transformation to compositions with no zeros. Plot the transformed data in $\mathbb{R}^2$.

Answer of exercise 20

The data set shown in Table 2.1 contains no zeros. However, the statement insists on checking for zeros or even negative values, because for large data sets this is not so obvious. If zeros were found, the analyst should then ask for the nature of zeroes and, in general, for the whole data set. Are the data truly compositional? Are the zeroes under detection limit values? Or are they recording errors? If the data set is made of counting observations, the presence of zeroes may correspond to the nature of the sampling. Another case is the case of essential zeroes, what should be concluded from the nature of the data and the sampling procedures. In the present exercise, the data set is simulated. If it was sampled from a given distribution on the simplex, the presence of a zero would imply a wrong technique of simulation of compositional data. For example, a simulation of a 3-dimensional normal distribution truncated for negative values could give zeroes.

Data in Table 2.1 are shown in a ternary diagram in Figure 4.7 (left panel). This data set has been alr-transformed so that

$$\text{alr}_1(\mathbf{x}) = \ln \frac{x_1}{x_3} \quad , \quad \text{alr}_2(\mathbf{x}) = \ln \frac{x_2}{x_3} \quad .$$

Results are shown in Figure 4.7 (right panel, black circles) representing the alr as being Cartesian.

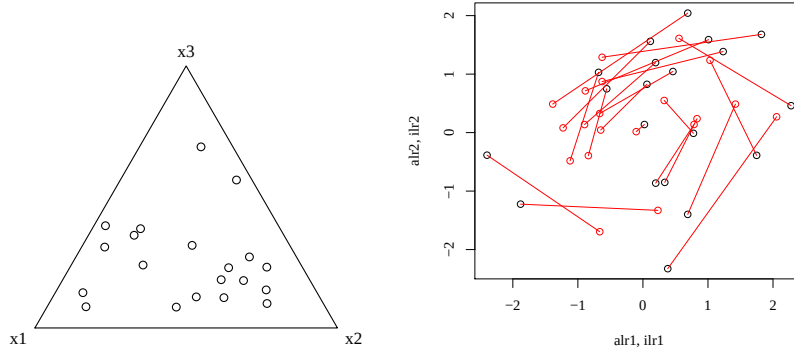


Fig. 4.7. Data of Table 2.1. Left: ternary diagram. Right: alr-transformed data (black circles); balances (red circles).

In order to compare the alr-coordinates with ilr coordinates, the following balances have been computed

$$b_1 = \frac{1}{\sqrt{2}} \ln \frac{x_1}{x_3} \quad , \quad \sqrt{\frac{2}{3}} \ln \frac{(x_1 x_3)^{1/2}}{x_2} \quad ,$$

and then plotted in Figure 4.7 (right panel, red circles). Red and black circles are linked by lines to visualize the two representations of the same data point. Note that alr_1 and b_1 differ only by a factor $\sqrt{2}$, which disperses red points more than black points in the x-axis direction. On the other hand, the red links give an impression of a kind of rotation. It is due to the fact that alr produces coordinates in an oblique basis of the simplex and the ilr -coordinates are Cartesian orthonormal coordinates.

Exercise 21

Consider the data set given in Table 2.1 and take the components in a different order. Apply the alr transformation to compositions with no zeros. Plot the transformed data in $\mathbb{R}^2$. Compare the result with those obtained in Exercise 20.

Answer of exercise 21

Permuting the parts of the composition leads to take a different denominator. For instance, take the following alr coordinates

$$\text{alr}_1(\mathbf{x}) = \ln \frac{x_1}{x_2} \quad , \quad \text{alr}_2(\mathbf{x}) = \ln \frac{x_3}{x_2} \quad .$$

Figure 4.8 shows the alr-coordinates obtained in Exercise 20 (black circles) compared with those obtained with the latter ones (red circles), linked by lines. Note that the second alr-coordinate of Exercise 20 and the present one differ only in the sign; the plot shows clearly this feature.

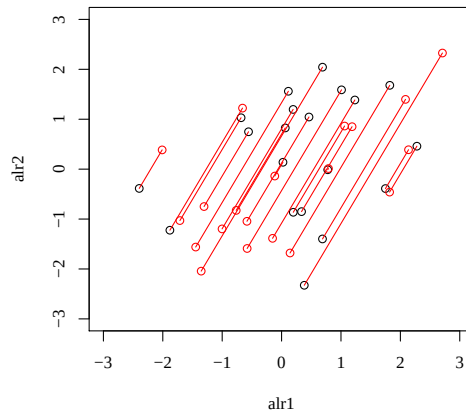


Fig. 4.8. Cartesian representation of alr transformed data.

Exercise 22

Consider the data set given in Table 2.1. Apply the ilr transformation to compositions with no zeros. Plot the transformed data in $\mathbb{R}^2$. Compare the result with the scatterplot obtained in Exercises 20 and 21 using the alr transformation.

Answer of exercise 22

Figure 4.7 (right panel) compares the alr coordinates and ilr-coordinates given by the two balances b_1 and b_2 . See Exercise 20.

Exercise 23

Compute the alr and ilr coordinates, as well as the clr coefficients of the six-part composition

$$[x_1, x_2, x_3, x_4, x_5, x_6] = [3.74, 9.35, 16.82, 18.69, 23.36, 28.04] \%$$

Answer of exercise 23

Computation of alr coordinates requires the decision on which part is taken as denominator in the logratios. According to Definition 4.10, the last part x_6 is chosen as reference part. Then the logratios to be computed are

$$\text{alr}_i(\mathbf{x}) = \ln x_i - \ln x_6, \quad i = 1, 2, \dots, 5.$$

Note that previous closure of the vector is not required, as any multiplicative constants cancel out when computing logratios.

There are infinitely many ilr coordinates depending on the chosen basis of the simplex. A simple option is to use the sequential binary partition in Table 4.9. The bottom part of the table is the contrast matrix Ψ corresponding to the SBP. It is computed using Definition 4.3 and Equation (4.10) for the construction of the entries of Ψ . Note also that Example 4.1 shows a very similar construction. The corresponding ilr-coordinates (balances) are found

Table 4.9. Sequential Binary Partition of a 6-part composition.

balance	x_1	x_2	x_3	x_4	x_5	x_6
b_1	+1	-1	-1	-1	-1	-1
b_2	0	+1	-1	-1	-1	-1
b_3	0	0	+1	-1	-1	-1
b_4	0	0	0	+1	-1	-1
b_5	0	0	0	0	+1	-1
contrast matrix Ψ						
	$\frac{1}{5}\sqrt{\frac{1.5}{1+5}}$	$-\frac{1}{1}\sqrt{\frac{1.5}{1+5}}$	$-\frac{1}{1}\sqrt{\frac{1.5}{1+5}}$	$-\frac{1}{1}\sqrt{\frac{1.5}{1+5}}$	$-\frac{1}{1}\sqrt{\frac{1.5}{1+5}}$	$-\frac{1}{1}\sqrt{\frac{1.5}{1+5}}$
	0	$\frac{1}{4}\sqrt{\frac{1.4}{1+4}}$	$-\frac{1}{1}\sqrt{\frac{1.4}{1+4}}$	$-\frac{1}{1}\sqrt{\frac{1.4}{1+4}}$	$-\frac{1}{1}\sqrt{\frac{1.4}{1+4}}$	$-\frac{1}{1}\sqrt{\frac{1.4}{1+4}}$
	0	0	$\frac{1}{3}\sqrt{\frac{1.3}{1+3}}$	$-\frac{1}{1}\sqrt{\frac{1.3}{1+3}}$	$-\frac{1}{1}\sqrt{\frac{1.3}{1+3}}$	$-\frac{1}{1}\sqrt{\frac{1.3}{1+3}}$
	0	0	0	$\frac{1}{2}\sqrt{\frac{1.2}{1+2}}$	$-\frac{1}{1}\sqrt{\frac{1.2}{1+2}}$	$-\frac{1}{1}\sqrt{\frac{1.2}{1+2}}$
	0	0	0	0	$\frac{1}{1}\sqrt{\frac{1.1}{1+1}}$	$-\frac{1}{1}\sqrt{\frac{1.1}{1+1}}$

using the matrix equation (4.7).

The clr coefficients can be computed from Definition 4.1. However, if they are computed manually, the following equivalent expression is more efficient

$$\text{clr}(\mathbf{x}) = \ln x_i - \frac{1}{6} \sum_{j=1}^6 \ln x_j ,$$

where the last term is the natural logarithm of the geometric mean of the parts. Note that in order to compute alr, clr and ilr logratios, closure of the vector $\mathbf{x}$ is not required, as any multiplicative constant cancels out when computing logratios which are log-contrasts.

Table 4.10 shows the alr, clr and ilr coordinates for $\mathbf{x}$. The label $\text{ilr}^1(\mathbf{x})$ denotes the coordinates proposed in this exercise. The last row, labelled $\text{ilr}^2(\mathbf{x})$, corresponds to the ilr coordinates (balances) associated with the SBP in Table 4.1, as proposed in the following Exercise 24. Comparing the two ilr's, it can be observed that the fifth balance of $\text{ilr}^1(\mathbf{x})$ is equal to the forth balance of $\text{ilr}^2(\mathbf{x})$, as can be checked comparing the code of both SBP's. Moreover, although other values in both ilr's seem to be unrelated, the norm of $\mathbf{x}$ computed from the clr and ilr coordinates is

$$\|\mathbf{x}\|_a = \sqrt{\sum_{i=1}^6 [\text{clr}_i(\mathbf{x})]^2} = \sqrt{\sum_{j=1}^5 [\text{ilr}_j^k(\mathbf{x})]^2} = 1.663326 ,$$

for $k = 1$ (this Exercise) and for $k = 2$ (Exercise 24).

Table 4.10. alr, clr and ilr coordinates for $\mathbf{x}$.

alr($\mathbf{x}$)	-2.0145465	-1.0982557	-0.5110634	-0.4056434	-0.1826069	
clr($\mathbf{x}$)	-1.3125271	-0.3962364	0.1909559	0.2963759	0.5194124	0.7020193
ilr ¹ ($\mathbf{x}$)	-1.4378010	-0.7364957	-0.2727806	-0.2566575	-0.1291226	
ilr ² ($\mathbf{x}$)	-0.4220417	-1.3748950	-0.8221731	-0.1291226	-0.0745432	

Exercise 24

Consider the six-part composition of the preceding exercise. Using the binary partition of Table 4.1 and Equation (4.9), compute its five balances. Compare with the results of the preceding exercise.

Answer of exercise 24

See solution of Exercise 23.

Exercise 25

Show that the vectors of an alr basis of a four-part composition are not orthogonal.

Answer of exercise 25

Let $\mathbf{x} = [x_1, x_2, x_3, x_4]$ be a composition in $\mathcal{S}^4$. Taking x_4 as reference part the alr-coordinates are

$$\zeta_i = \ln \frac{x_i}{x_4}, \quad i = 1, 2, 3,$$

and the composition is recovered from ζ_i as

$$\mathbf{x} = \zeta_1 \odot \mathcal{C}[e, 1, 1, 1] \oplus \zeta_2 \odot \mathcal{C}[1, e, 1, 1] \oplus \zeta_3 \odot \mathcal{C}[1, 1, e, 1],$$

as noted in the first formula of Section 4.7. This means that $\mathbf{e}_1 = \mathcal{C}[e, 1, 1, 1]$, $\mathbf{e}_2 = \mathcal{C}[1, e, 1, 1]$, $\mathbf{e}_3 = \mathcal{C}[1, 1, e, 1]$ are the three compositions of the basis used in the alr transformation. Their respective clr coefficients are

$$\text{clr}(\mathbf{e}_1) = \left[\frac{3}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4} \right], \quad \text{clr}(\mathbf{e}_2) = \left[-\frac{1}{4}, \frac{3}{4}, -\frac{1}{4}, -\frac{1}{4} \right],$$

$$\text{clr}(\mathbf{e}_3) = \left[-\frac{1}{4}, -\frac{1}{4}, \frac{3}{4}, -\frac{1}{4} \right],$$

To check that these compositions are not orthogonal, the Aitchison inner product of each pair is computed,

$$\langle \mathbf{e}_i, \mathbf{e}_j \rangle_a = \langle \text{clr}(\mathbf{e}_i), \text{clr}(\mathbf{e}_j) \rangle = \frac{1}{2} \neq 0,$$

whenever $i \neq j$ and, therefore, they are not orthogonal compositions. In this case, the angle they form is such that $\cos \alpha = 1/2$ which corresponds to $\alpha = \pm 60^\circ$. On the other hand, $\langle \mathbf{e}_i, \mathbf{e}_i \rangle_a = \|\mathbf{e}_i\|_a^2 = 3/4$, which shows that the vectors of the basis are not unitary.

Exercise 26

Consider two $(D-1, D)$ -contrast matrices Ψ_0, Ψ_1 defining ilr-transformations, $\text{ilr}_0(\cdot), \text{ilr}_1(\cdot)$. Prove:

- (a) if $\mathbf{x} \in \mathcal{S}^D$, then $\text{ilr}_i(\mathbf{x})\Psi_i = \text{clr}(\mathbf{x}), i = 1, 2$.
- (b) $[\Psi_1\Psi_2^\top]^{-1} = \Psi_2\Psi_1^\top$.

Answer to exercise 26

Recall the main properties of $(D-1, D)$ -contrast matrices (Equation 4.5):

$$\Psi_i\Psi_i^\top = \mathbf{I}_{D-1}, \quad \Psi_i^\top\Psi_i = \mathbf{I}_D - (1/D)\mathbf{1}_D^\top\mathbf{1}_D,$$

where $\mathbf{1}_D$ is a row-vector with entries equal to 1. Then, for $\mathbf{x} \in \mathcal{S}^D$,

$$\text{ilr}_i(\mathbf{x})\Psi_i = \ln \mathbf{x} \Psi_i^\top\Psi_i = \ln \mathbf{x} (\mathbf{I}_D - (1/D)\mathbf{1}_D^\top\mathbf{1}_D) = \ln \mathbf{x} - \ln \mathbf{g}(\mathbf{x}),$$

where $\mathbf{g}(\mathbf{x})$ is a D -row vector with all entries equal to the geometric mean of entries of $\mathbf{x}$. This proves statement (a).

To prove statement (b), consider the equation

$$\mathbf{x}^*\Psi_1\Psi_2^\top = \mathbf{z}^*, \quad \mathbf{x}^*, \mathbf{z}^* \in \mathbb{R}^{D-1},$$

and isolate $\mathbf{x}^*$:

$$\mathbf{x}^*\Psi_1\Psi_2^\top\Psi_2 = \mathbf{x}^*\Psi_1 - (1/D)\mathbf{x}^*\Psi_1\mathbf{1}_D^\top\mathbf{1}_D = \mathbf{z}^*\Psi_2.$$

Taking into account that $\mathbf{x}^*\Psi_1\mathbf{1}_D^\top$ is null due to the clr-coefficient properties stated in Definition 4.1, it yields

$$\mathbf{x}^*\Psi_1 = \mathbf{z}^*\Psi_2, \quad \text{and thus} \quad \mathbf{x}^* = \mathbf{z}^*\Psi_2\Psi_1^\top,$$

what proves statement (b).

Exercise 27

Consider a composition $\mathbf{x} = [x_1, x_2, x_3]$ in $\mathcal{S}^3$ and the two coordinates $w_1 = \ln(x_1/(x_2 + x_3))$, $w_2 = \ln(x_2/x_3)$, which constitute a transformation of $\mathcal{S}^3$ onto $\mathbb{R}^2$, known as the *multiplicative logratio* transformation (Aitchison 1982; 1986).

- (a) Check that this transformation is one to one by finding the inverse transformation, that is, expressing $\mathbf{x} = [x_1, x_2, x_3]$ as a function of $[w_1, w_2]$.
- (b) Assume that $\mathbf{x}, \mathbf{y}$ are compositions in $\mathcal{S}^3$ and their coordinates are $[w_1, w_2]$ and $[z_1, z_2]$, respectively. Find the composition $\mathbf{x} \oplus_m \mathbf{y}$ which coordinates are $[w_1 + z_1, w_2 + z_2]$. Is the perturbation $\oplus_m$ easily interpretable?
- (c) Consider the straight lines in $\mathbb{R}^2$ given by $w_1 = -2$, $w_1 = 0$, $w_1 = 2$, $w_2 = -2$, $w_2 = 0$, $w_2 = 2$, and the circle $w_1^2 + w_2^2 = 2$. Plot them in a ternary diagram and in ilr-coordinates.

Answer of exercise 27

- (a) The multiplicative logratio transformation is scale invariant, as any positive constant multiplying each part cancels out in the ratios. However, it is not permutation invariant, as it lacks symmetry. To obtain the inverse transformation, take exponentials of w_1 and w_2 , and express x_1 and x_2 as a function of x_3 . It yields

$$\mathbf{x} = \mathcal{C}[x_3 \exp(w_1)(1 + \exp(w_2)), x_3 \exp(w_2), x_3].$$

Common factors in the parts are removed by the closure operation, for instance, the common factor x_3 . After performing the closure operation, the desired inverse transformation is ($K = (1 + \exp(w_1))(1 + \exp(w_2))$)

$$\mathbf{x} = \left[\frac{\exp(w_1)(1 + \exp(w_2))}{K}, \frac{\exp(w_2)}{K}, \frac{1}{K} \right].$$

Since both the initial transformation and its inverse are well defined for $\mathbf{x} \in \mathcal{S}^3$ and for any $[w_1, w_2] \in \mathbb{R}^2$, it shows that the transformation is one-to-one.

- (b) The operation in $\mathcal{S}^3$ which corresponds to $[w_1 + z_1, w_2 + z_2]$ is

$$\mathbf{x} \oplus_m \mathbf{y} = \mathcal{C}[\exp(w_1 + z_1)(1 + \exp(w_2 + z_2)), \exp(w_2 + z_2), 1].$$

Substituting the w 's and z 's by their respective expressions in the transformation,

$$\mathbf{x} \oplus_m \mathbf{y} = \mathcal{C} \left[\frac{x_1 y_1}{(x_2 + x_3)(y_2 + y_3)} \left(1 + \frac{x_2 y_2}{x_3 y_3} \right), \frac{x_2 y_2}{x_3 y_3}, 1 \right].$$

This expression can be simplified in several ways. One of them gives

$$\mathbf{x} \oplus_m \mathbf{y} =$$

$$\mathcal{C}[x_1 y_1 (x_2 y_2 + x_3 y_3), (x_2 + x_3)(y_2 + y_3)(x_2 y_2 + x_3 y_3), x_3 y_3 (x_2 + x_3)(y_2 + y_3)],$$

which interpretation as operation in the simplex seems extremely challenging. Other equivalent expressions share similar difficulties. Despite of the involved form of $\mathbf{x} \oplus_m \mathbf{y}$, it defines an additive group in the simplex, obviously different to the ordinary perturbation.

- (c) Figure 4.9 shows the proposed lines and circle in the ternary diagram (left). An important feature is that the given lines, indeed coordinate lines, appear in the ternary diagram as straight lines. The lines converging to the vertex x_1 are straight lines in the sense of the Aitchison geometry of the simplex; parallel lines to the side x_2, x_3 , are not straight lines in the sense of the Aitchison geometry. All these lines and circle are transformed to ilr-coordinates (right). The straight lines in the simplex are now easily identified. Also the asymmetric distortion and non-orthogonality of coordinate lines is clear. This shows that the (Euclidean) geometry induced by the multiplicative logratio transformation is interpreted with difficulties.

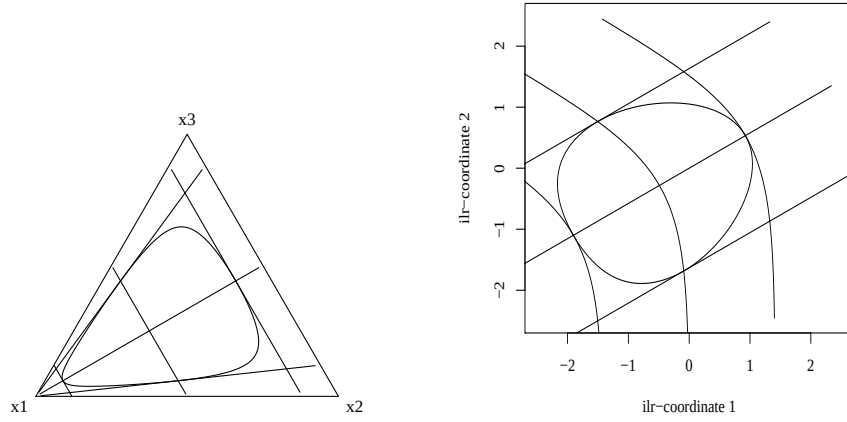


Fig. 4.9. Three horizontal and three vertical coordinate lines plus a circle, centered at $(0, 0)$ and with radius 2 in multiplicative logratio coordinates, represented in the ternary diagram (left) and in ilr-coordinates (right).

Exercise 28

This exercise aims at illustrating the problems of classical statistics applied to compositional data. Using the data given in Table 5.1, compute the classical correlation coefficients between the following pairs of parts: (MnO vs. CaO), (FeO vs. Na₂O), (MgO vs. FeO), and (MgO vs. Fe₂O₃). Now ignore the structural oxides Al₂O₃ and SiO₂ from the data set, close the remaining components to percentages within the subcomposition, and compute the same correlation coefficients as above. Compare the results. Compare the correlation matrix between the feldspar-constituent parts (CaO,Na₂O,K₂O), as obtained from the original data set, and after closing this three-part subcomposition.

Answer of exercise 28

For the data presented in Table 5.1, the correlation coefficients between the four pairs are reported in the first column of Table 5.9 below. Taking the subcomposition without Al₂O₃ and SiO₂ and closing the other 10 components to sum up to 100%, the correlations of the second column are obtained.

parts	composition	subcomposition
MnO vs. CaO	0.328	0.953
FeO vs. Na ₂ O	-0.308	0.699
MgO vs. FeO	0.207	-0.787
MgO vs. Fe ₂ O ₃	0.215	-0.001

Table 5.9. Correlation coefficients in a 12-part composition and in a 10-part subcomposition

Now select the three main feldspar-forming cations (CaO,Na₂O,K₂O), and compute their correlation coefficients with the original data, and after closing this 3-part subcomposition. Results are reported in Table 5.10. As can be

parts	composition	subcomposition
CaO vs. Na ₂ O	0.814	-0.702
CaO vs. K ₂ O	0.661	-0.346
Na ₂ O vs. K ₂ O	0.415	-0.425

Table 5.10. Correlation coefficients in a 12-part composition and in a 3-part subcomposition

seen, not a single correlation keeps its sign and/or its value. Thus, simple questions like *Are FeO and Na₂O associated?* cannot be answered with a simple correlation coefficient.

Exercise 29

For the data given in Table 2.1, compute and plot the center with the samples in a ternary diagram. Compute the total variance and the variation matrix.

Answer of exercise 29

Center and variation matrix are reported in Table 5.11, and the required ternary diagram in Figure 5.18.

Table 5.11. Center (in %) and variation matrix of the data of Table 2.1

	X1	X2	X3
center	0.36.9000	36.4408	26.6593
variation	X1	X2	X3
X1	0.000	2.004	1.272
X2	2.004	0.000	1.453
X3	1.272	1.453	0.000

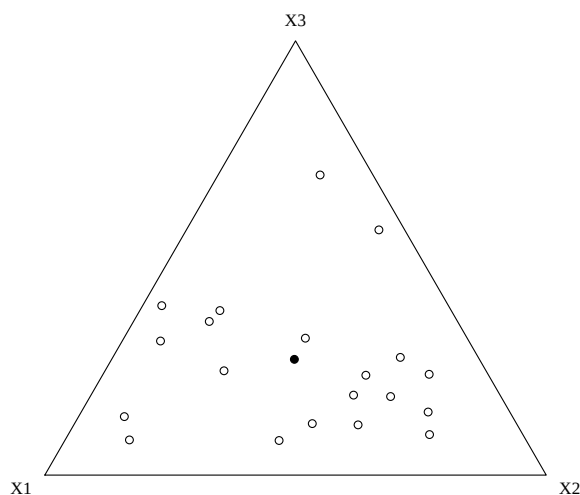


Fig. 5.18. Ternary diagram of the data of Table 2.1

Exercise 30

Perturb the data given in Table 2.1 with the inverse of the center. Compute the center of the perturbed data set and plot it with the samples in a ternary diagram. Compute the total variance and the variation matrix. Compare your results numerically and graphically with those obtained in Exercise 29.

Answer of exercise 30

Center and variation matrix are reported in Table 5.12, and the required ternary diagram in Figure 5.19. Note that the variation matrix did not change at all, as expected.

Table 5.12. Center (in %) and variation matrix of the data of Table 2.1 after centering

	X1	X2	X3
center	33.333	33.333	33.333
variation	X1	X2	X3
X1	0.000	2.004	1.272
X2	2.004	0.000	1.453
X3	1.272	1.453	0.000

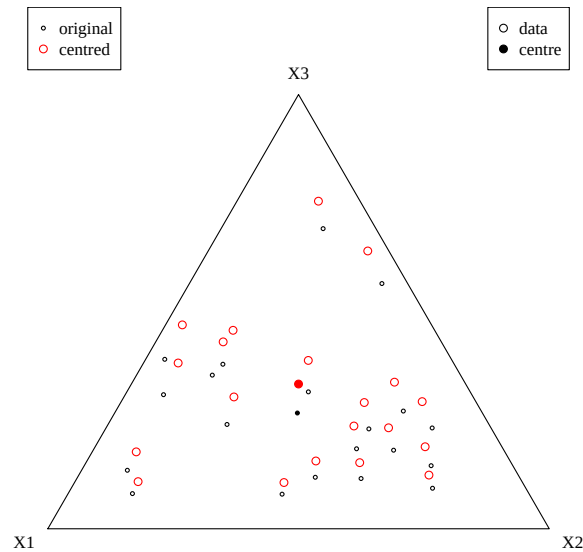


Fig. 5.19. Ternary diagram of the data of Table 2.1, before and after centering

Exercise 31

Make a biplot of the data given in Table 2.1 and give an interpretation.

Answer of exercise 31

The biplot is represented in Figure 5.20. A biplot of three parts has seldom

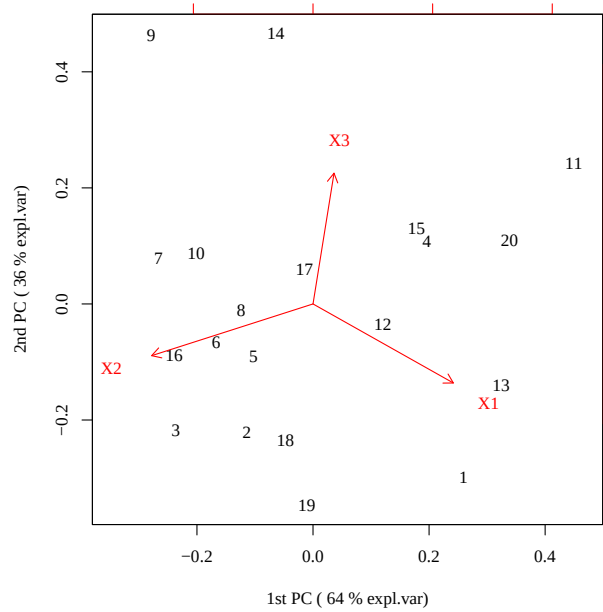


Fig. 5.20. Biplot of the data of Table 2.1. Explained variability: 100%.

an interpretation, except in the case that the three parts come from closing a vector of three masses which are totally independent and with the same variance (which forms a perfect triangle) or the case where they are linked by a one dimensional process (which is detected because the three parts fall quasi aligned to the first principal component). This is none of the cases here.

Exercise 32

Figure 5.5 shows the biplot of the data given in Table 5.1. How would you interpret the different patterns that can be observed?

Answer of exercise 32

A compositional biplot should be interpreted looking for three kinds of patterns: groups of clr-parts near to each other with short links between them; groups of clr-parts aligned on a link; and two groups of clr-parts whose links are orthogonal. We can see two clusters of clr-parts: $\{\text{SiO}_2, \text{MnO}, \text{FeO}\}$ and $\{\text{Na}_2\text{O}, \text{TiO}_2, \text{P}_2\text{O}_5, \text{K}_2\text{O}, \text{Al}_2\text{O}_3\}$, which should have a low variability (recall that a biplot is a projection, and the variability might be in another dimension). The total variation of these two subcompositions is 0.0022578 and 0.0098705, respectively. With respect to the total variation 0.3163323 of the whole composition (discarding CO_2) they represent contributions of 0.7% and 3%, respectively.

Several groups of three or more aligned clr-parts can be spotted:

$$\{\text{K}_2\text{O}, \text{FeO}, \text{MgO}\}, \quad \{\text{Fe}_2\text{O}_3, \text{MnO}, \text{FeO}\}, \quad \{\text{K}_2\text{O}, \text{CaO}, \text{FeO}\}$$

are some examples. They probably plot in a one-dimensional pattern on a ternary diagram. Of these, two appear as quasi-orthogonal: $\{\text{K}_2\text{O}, \text{FeO}, \text{MgO}\}$ and $\{\text{Fe}_2\text{O}_3, \text{MnO}, \text{FeO}\}$: variations within these subcompositions should be tested to see if they are uncorrelated. This is explored in the next exercises.

Exercise 33

Select three-part subcompositions that behave in a particular way in Figure 5.5 and plot them in a ternary diagram. Do they reproduce properties mentioned in the previous description?

Answer of exercise 33

Figure 5.21 shows ternary diagrams of the subcompositions $\{K_2O, FeO, MgO\}$, $\{Fe_2O_3, MnO, FeO\}$ and $\{K_2O, CaO, FeO\}$, with a compositional straight line along their first principal component. The agreement between the dots and the curve is remarkable in all cases.

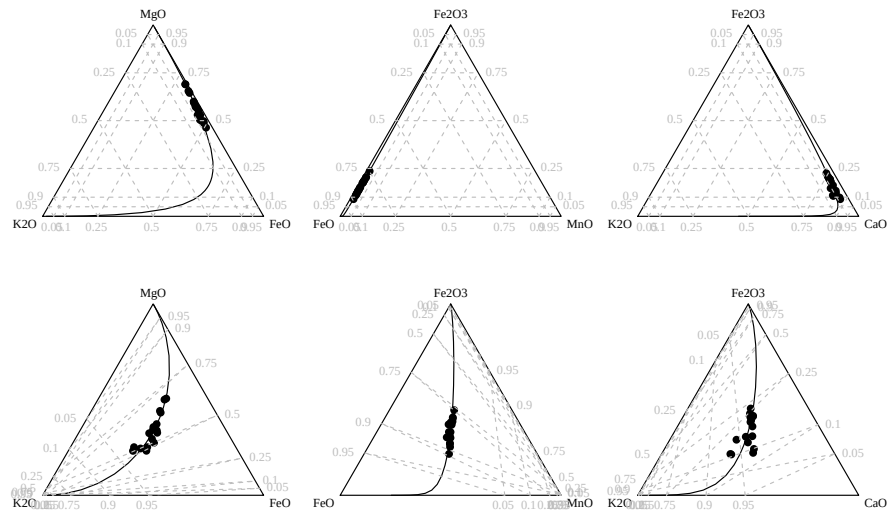


Fig. 5.21. Ternary diagrams of one-dimensional subcompositions, in original units (above) and centered (below).

Exercise 34

Do a scatter plot of the logratios

$$\frac{1}{\sqrt{2}} \ln \frac{K_2O}{MgO} \quad \text{against} \quad \frac{1}{\sqrt{2}} \ln \frac{Fe_2O_3}{FeO},$$

identifying each point. Compare with the biplot. Compute the total variance of the subcomposition (K_2O , MgO , Fe_2O_3 , FeO) and compare it with the total variance of the full data set.

Answer of exercise 34

Figure 5.22 represents the desired scatterplot, together with the biplot of Fig-

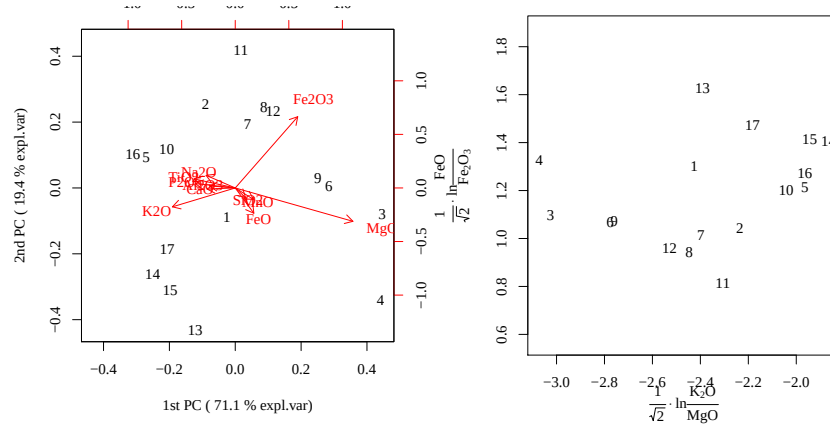


Fig. 5.22. Biplot of the whole composition (left) and scatterplot of two selected ilr coordinates (right) of the Kilauea data set. Note the similar—although rotated—distribution of points (numbered). Explained variability: 90.5%.

ure 5.5 for ease reference. Observing the pattern of distribution of the individual data (marked with numbers), one can see that they are almost the same between the two points (with slight differences on locations of the data 14, 15 and 17) after a 180° rotation. The sum of the variances of these two coordinates is 0.1846, the total variance of the subcomposition (K_2O , MgO , Fe_2O_3 , FeO) is 0.2083, and the total variance of the whole composition is 0.3163. The conclusion is that the scatterplot of those two very simple logratios captures a good deal of the relevant structure and variability of this 11-part data set. This is often the case for logratios constructed from opposed clr-parts in the biplot which form wide angles, the more similar to 90° the better. Moreover, it was said in exercise 32 that the subcompositions of these elements should show uncorrelated variations, thus these two logratios should show no to low correlation: the Pearson correlation between them is $\rho = 0.2856$, a non-significant

value (a t-test of null correlation delivered a p -value of 0.2665, clearly above the critical value of 0.05).

Exercise 35

How would you recast the data in Table 5.1 from mass proportion of oxides (as they are) to molar proportions? You may need the following molar weights. Any idea of how to do that with a perturbation?

SiO ₂	TiO ₂	Al ₂ O ₃	Fe ₂ O ₃	FeO	MnO
60.085	79.899	101.961	159.692	71.846	70.937
MgO	CaO	Na ₂ O	K ₂ O	P ₂ O ₅	
40.304	56.079	61.979	94.195	141.945	

Answer of exercise 35

The conversion of a mass quantity m_i of species i into its molar quantity n_i is obtained by division with the molar weight of that species M_i , i.e. $n_i = m_i/M_i$. If we have a vector $\mathbf{m} = [m_1, m_2, \dots, m_D]$ of *absolute masses* of D different species, then the vector of *absolute moles* $\mathbf{n}$ is obtained component wise as $n_i = m_i/M_i$. However, we typically have a vector of *mass proportions* $\mathbf{x} = [x_1, x_2, \dots, x_D]$, i.e. $\mathbf{x}$ is proportional to $\mathbf{m}$, $\mathbf{x} \propto \mathbf{m}$. Now, applying an inverse perturbation with the vector of molar masses $\mathbf{M} = [M_1, M_2, \dots, M_D]$ at both sides, we find $\mathbf{x} \ominus \mathbf{M} \propto \mathbf{m} \ominus \mathbf{M} = \mathbf{n}$. But the vector of *molar proportions* (also known as *molar fractions*) $\mathbf{X} = \mathcal{C}[\mathbf{n}] \propto \mathbf{n}$ by definition. Plugging these two relations,

$$\mathbf{x} \ominus \mathbf{M} \propto \mathbf{m} \ominus \mathbf{M} = \mathbf{n} \propto \mathbf{X},$$

which implies that the molar fractions can be computed from the mass composition as $\mathbf{X} = \mathbf{x} \ominus \mathbf{M}$. Table 5.13 contains all the data in molar fractions (in %). Readers are invited to check that it doesn't matter whether we close any of the involved vectors or not,

$$\mathbf{X} = \mathbf{x} \ominus \mathbf{M} = \mathcal{C}[\mathbf{x}] \ominus \mathbf{M} = \mathbf{x} \ominus \mathcal{C}[\mathbf{M}] = \mathcal{C}[\mathbf{x}] \ominus \mathcal{C}[\mathbf{M}],$$

all results are the same.

Table 5.13. Kilauea data set (except CO₂) expressed as molar percentages

	SiO ₂	TiO ₂	Al ₂ O ₃	Fe ₂ O ₃	FeO	MnO	MgO	CaO	Na ₂ O	K ₂ O	P ₂ O ₅
1	48.795	1.771	6.836	0.605	8.476	0.154	20.457	10.664	1.861	0.284	0.098
2	50.297	1.913	7.515	0.833	8.106	0.148	17.014	11.743	2.017	0.309	0.105
3	43.631	1.223	4.696	0.763	8.016	0.138	32.886	7.154	1.233	0.195	0.065
4	43.205	1.100	4.572	0.572	8.291	0.137	33.790	6.908	1.178	0.188	0.060
5	50.994	2.568	7.380	0.689	8.560	0.149	16.139	10.701	2.258	0.429	0.131
6	45.577	1.466	5.478	0.796	8.020	0.149	28.154	8.585	1.462	0.237	0.075
7	48.767	1.783	6.826	0.862	8.018	0.155	20.623	10.717	1.857	0.297	0.094
8	48.395	1.762	6.652	0.935	7.904	0.154	21.570	10.429	1.821	0.290	0.090
9	46.163	1.486	5.735	0.788	7.991	0.150	26.833	9.037	1.506	0.232	0.079
10	50.889	2.125	7.650	0.713	8.674	0.158	15.509	11.701	2.097	0.370	0.114
11	49.535	1.901	7.115	1.082	7.625	0.156	19.099	11.161	1.915	0.313	0.100
12	48.258	1.697	6.632	0.914	7.886	0.154	21.988	10.341	1.778	0.263	0.090
13	48.897	1.784	6.923	0.395	8.752	0.154	19.905	10.954	1.849	0.290	0.098
14	50.140	1.909	7.269	0.535	8.707	0.156	17.062	11.878	1.717	0.522	0.104
15	49.520	1.864	6.945	0.528	8.650	0.155	18.706	11.376	1.645	0.512	0.099
16	51.733	2.376	7.933	0.628	8.442	0.150	13.743	12.245	2.264	0.366	0.119
17	50.561	1.932	7.461	0.487	8.706	0.157	16.102	12.167	2.012	0.315	0.100

Exercise 36

Re-do all the descriptive analysis (and the related exercises) with the Kilauea data set expressed in molar proportions. Compare the results.

Answer of exercise 36

An advantage to see the conversion between mass percentages and molar fractions as a perturbation is that we can immediately foresee which compositional statistical results will yield exactly the same results in both data sets, and which not. Even more, we can already say which will be the relationships between the results for those cases in which they are not different:

- those descriptive statistics that describe the typical values (center) will be perturbed as the data set, i.e. $\text{cen}(\mathbf{X}) = \text{cen}(\mathbf{x}) \ominus \mathbf{M}$; the logratio means will be translated equivalently: $\bar{\mathbf{X}}^* = \bar{\mathbf{x}}^* - \text{ilr}(\mathbf{M})$;
- those descriptive statistics that describe the spread, variability and codependence (variation matrix, total variance, logratio covariance or correlation) will not change in any sense;
- all statistical methods that rely on those variability measures (PCA, LDA, regression, centered data in ternary diagrams, lack of correlation of two subcompositions, one-dimensional patterns in subcompositions, etc) will remain unchanged as well, of course.

For these reasons, we will not repeat all those analysis, as results are exactly the same as shown elsewhere. Just for illustration, we report the centers of molar proportions and of mass proportions (Table 5.14) and the ternary dia-

Table 5.14.

	mass center	molar center
SiO ₂	48.570	49.018
TiO ₂	2.354	1.787
Al ₂ O ₃	11.228	6.677
Fe ₂ O ₃	1.838	0.698
FeO	9.913	8.367
MnO	0.179	0.153
MgO	13.744	20.679
CaO	9.649	10.434
Na ₂ O	1.823	1.784
K ₂ O	0.480	0.309
P ₂ O ₅	0.221	0.095

grams from exercise 33 (Figure 5.23), for the sake of comparison. Note that the molar fraction center is only notably different from the mass compositional center in those parts that have either a very large M_i or a very small one. Note as well how the ternary diagrams in original scale differ slightly between

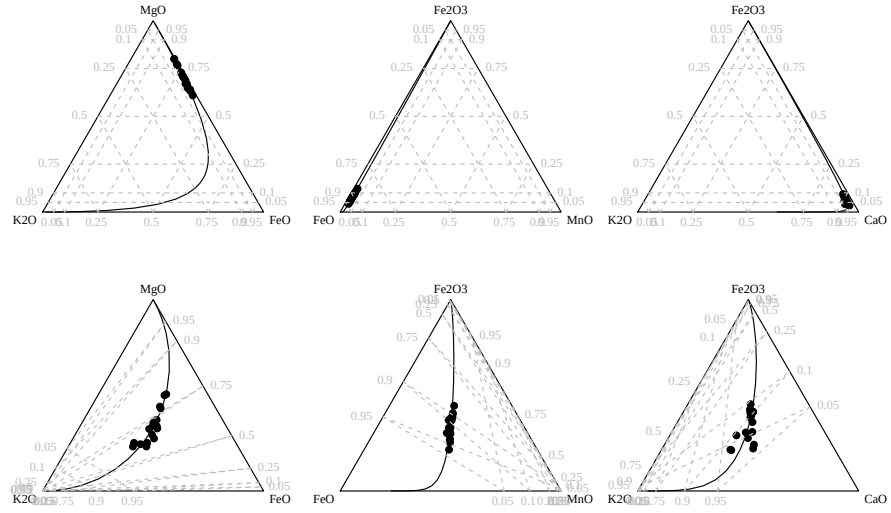


Fig. 5.23. Ternary diagrams of one-dimensional subcompositions expressed in molar fractions, in original units (above) and centered (below).

mass composition (5.21) and molar fraction (5.23). Note that the centered diagrams of both sets of ternary diagrams do not change, as expected.

Exercise 37

Compute the vector of arithmetic means of the ilr transformed data from Table 2.1. Apply the ilr^{-1} backtransformation and compare it with the center.

Answer of exercise 37

Table 5.15 reports the three different ways to compute an ilr as balances between three components. For instance, the first ilr is obtained by using the contrast matrix

$$\Psi = \begin{pmatrix} \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}.$$

Taking the mean of the first set of transformed data, $\bar{\mathbf{x}} = [0.138, 0.221]$, and transforming it back,

$$\text{ilr}^{-1}(\bar{\mathbf{x}}) = \mathcal{C}[\exp(\bar{\mathbf{x}} \cdot \Psi)] = [0.3689997, 0.3644077, 0.2665926],$$

the result given in Table 5.11 is obtained, as expected. The same will happen with the other two sets of ilr transformations.

Table 5.15. Three different results of applying an ilr transformation to data from Table 2.1, with their centers.

	$\frac{1}{\sqrt{6}} \ln \frac{X_1^2}{X_2 X_3}$	$\frac{1}{\sqrt{2}} \ln \frac{X_2}{X_3}$	$\frac{1}{\sqrt{6}} \ln \frac{X_2^2}{X_1 X_3}$	$\frac{1}{\sqrt{2}} \ln \frac{X_3}{X_1}$	$\frac{1}{\sqrt{6}} \ln \frac{X_3^2}{X_1 X_2}$	$\frac{1}{\sqrt{2}} \ln \frac{X_1}{X_2}$
1	1.673	0.325	-0.555	-1.611	-1.118	1.286
2	0.175	1.124	0.885	-0.714	-1.061	-0.410
3	-0.271	1.445	1.387	-0.488	-1.116	-0.957
4	0.622	-0.602	-0.832	-0.237	0.210	0.839
5	-0.050	0.739	0.665	-0.326	-0.615	-0.412
6	-0.330	0.847	0.898	-0.138	-0.568	-0.709
7	-0.977	0.728	1.119	0.482	-0.142	-1.210
8	-0.286	0.584	0.649	-0.044	-0.362	-0.540
9	-1.800	-0.273	0.664	1.695	1.136	-1.422
10	-0.760	0.530	0.839	0.393	-0.079	-0.923
11	1.262	-1.645	-2.056	-0.270	0.794	1.915
12	0.640	-0.009	-0.327	-0.549	-0.312	0.558
13	1.587	-0.276	-1.032	-1.236	-0.555	1.512
14	-1.036	-0.865	-0.231	1.330	1.267	-0.465
15	0.514	-0.611	-0.786	-0.139	0.272	0.750
16	-0.544	1.103	1.227	-0.081	-0.684	-1.023
17	-0.037	0.098	0.103	-0.016	-0.066	-0.081
18	0.442	0.980	0.628	-0.872	-1.069	-0.107
19	0.802	1.187	0.627	-1.288	-1.429	0.101
20	1.133	-0.988	-1.422	-0.487	0.289	1.475
center	0.138	0.221	0.122	-0.230	-0.260	0.009

Exercise 38

Take the parts of the compositions in Table 2.1 in a different order. Compute the vector of arithmetic means of the ilr transformed sample. Apply the ilr^{-1} backtransformation. Compare the result with the previous one.

Answer of exercise 38

Using the centers given in Table 5.15, i.e. $\bar{\mathbf{x}}^{(2)} = [0.122, -0.230]$ and $\bar{\mathbf{x}}^{(2)} = [-0.260, 0.009]$, together with the two corresponding matrices of logcontrasts,

$$\boldsymbol{\Psi}^{(2)} = \begin{pmatrix} -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$$

and

$$\boldsymbol{\Psi}^{(3)} = \begin{pmatrix} -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{pmatrix},$$

readers can check that they obtain exactly the same result as in the preceding exercise. Note that a change in the signs of the rows of any of these logcontrast matrices is allowed: the corresponding coordinate will also show the opposite sign.

Exercise 39

Center the data set of Table 2.1. Compute the vector of arithmetic means of the ilr transformed data. What do you obtain?

Answer of exercise 39

You will obtain the vector $[0, 0]$, no matter which ilr basis you use.

Exercise 40

Compute the covariance matrix of the ilr transformed data set of Table 2.1 before and after perturbation with the inverse of the center. Compare both matrices.

Answer of exercise 40

Both the original and the centered data should give exactly the same matrices, depending on which ilr coordinates you use. These results will even be the same if you switch the sign of one of the coordinates:

$$\mathbf{S}^{(1)} = \begin{pmatrix} 0.8498274 & -0.2113527 \\ -0.2113527 & 0.7267026 \end{pmatrix}$$

$$\mathbf{S}^{(2)} = \begin{pmatrix} 0.9405207 & 0.1589909 \\ 0.1589909 & 0.6360094 \end{pmatrix}$$

$$\mathbf{S}^{(3)} = \begin{pmatrix} 0.57444702 & 0.05236177 \\ 0.05236177 & 1.00208303 \end{pmatrix}$$

Exercise 41

Take the Catalan elections data set, and select the data for 2006 (e.g., filtering it in a spreadsheet software). Compute center, total variance, and variation matrix of the subcomposition (PSC, CiU, PP, ICV, ERC). Compute these statistics for the whole composition of parties (including C's with all the preceding ones). Compare the two variation matrices. Take the mean of the whole composition, remove the value for C's, reclose the remaining five means to 100% and compare with the mean originally computed for the subcomposition.

Answer to exercise 41

Once selected the data for the two target subcompositions, i.e. (PSC, CiU, PP, ICV, ERC) and (PSC, CiU, PP, ICV, ERC, C's). Table 5.16 shows the centers (means in the simplex), variation matrices and total variances for both subcompositions. In the first two rows of Table 5.16, the center of both com-

Table 5.16. Centers, variation matrix and total variances corresponding to the whole composition (PSC, CiU, PP, ICV, ERC, C's) and the subcomposition (PSC, CiU, PP, ICV, ERC).

center (mean in the simplex)						
	CiU	PSC	PP	ICV	ERC	C's
	0.4024	0.2453	0.0806	0.0759	0.1871	0.0087
	0.4059	0.2475	0.0813	0.0766	0.1887	
variation matrix						
	CiU	PSC	PP	ICV	ERC	C's
CiU	0	0.1291	0.2195	0.1118	0.0663	1.16969
PSC	0.1291	0	0.0904	0.0755	0.2357	0.6729
PP	0.2195	0.0904	0	0.1442	0.3366	0.4885
ICV	0.1118	0.0755	0.1442	0	0.1635	0.7157
ERC	0.0663	0.2357	0.3366	0.1635	0	1.3134
C's	1.1697	0.6729	0.4885	0.7157	1.3134	0
total variances						
	full comp.: 0.9888			comp. no C's: 0.3146		

position and subcomposition without C's can be compared; they are slightly different in the common parts. It can be checked that taking the subcomposition (PSC, CiU, PP, ICV, ERC) in the center of the full composition and closing to 1, exactly the same center of the subcomposition is obtained. Note that the proportion of C's in the center is relatively small and this causes that the rest of proportions in the subcomposition are similar.

Only one variation matrix, that corresponding to the full composition is shown in Table 5.16. The reason for this is that the variation matrix of the

subcomposition without C's is equal to the submatrix after deletion of the column and the row corresponding to C's. The main features to be observed in the variation matrix are: (a) the higher variances of log ratios are those which include C's, thus pointing out that C's introduces the most variability in the data set. This can be checked comparing total variances: suppressing C's the total variance is reduced by a factor larger than three. (b) the variances of the log ratios ERC over CiU (0.0663), PSC over ICV (0.0755), or even PP over PP over PSC (0.0904) are quite small compared with other variances. This means that traditional catalan nationalist parties (ERC, CiU), and parties corresponding to traditional parties in Spain (PP, PSC, ICV) maintain their proportions across the districts in the sample, apparently despite of their left-right wing characteristics. Note that a low variance of the log ratio of two parties suggests proportionality in the number of votes across the studied sample.

Exercise 42

Take the Catalan elections dataset, and select the data prior to 2006 (e.g., filtering it in a spreadsheet software). Compute center and total variance of the subcomposition (PSC, CiU, PP, ICV, ERC). Compare with the equivalent statistics obtained in the preceding exercise. Which was the perturbation occurring in an average within this subcomposition as a consequence of the appearance of C's? Think that no change would mean a perturbation of $[1, 1, 1, 1, 1]/5$. What happened with the variability? Compare by quotient.

Answer of exercise 42

As a first step, the data set must be checked. Zeros appeared in two different cases. In the 1999 elections, IC and PSC contested in coalition. Under the label ICV (or at that time IC), there are zero votes in most districts. Amalgamation of votes of both parties can be considered, but here the option of a redistribution of votes between the two parties has been taken. This has been done according to the approach explained in Section 5.9 (page 98). Other zero vote cases appear in the 1988 elections, when PP was not contesting in many districts. This zeros are essential, as nobody could vote PP in those districts. Consequently, 1988 data has been removed from the data set. Table 5.17 shows the centers (mean in the simplex) of the subcomposition (PSC, CiU, PP, ICV, ERC) for each of the available elections (2006, 2003, 1999, 1995, 1992), and also the center assuming that elections before 2006 constitute a single sample. The Aitchison norm of these centers, and the total variance of the samples, have been also computed and are shown in Table 5.17. Results in Table 5.17

Table 5.17. Centers and their norms, and total variance of the samples by elections for the subcomposition (CiU, PSC, PP, ICV, ERC). The results by elections are completed with "before 2006", where 1992, 1995, 1999 and 2003 election data have been assumed as a single sample.

			center				
	CiU	PSC	PP	ICV	ERC	A-norm	tot var
2006	0.4059	0.2475	0.0813	0.0766	0.1887	1.4433	0.3146
2003	0.4033	0.2511	0.0872	0.0504	0.2080	1.6809	0.3410
1999	0.4941	0.2774	0.0741	0.0364	0.1180	2.0734	1.0307
1995	0.5066	0.2213	0.1013	0.0475	0.1233	1.7734	0.4850
1992	0.5691	0.2349	0.0530	0.0378	0.1051	2.2108	0.6164
before 2006	0.4956	0.2483	0.0777	0.0431	0.1352	1.9132	0.7366

deserve some comments. The sample center, which plays the role of the mean for real samples, is the average across a sample of 41 districts. Accordingly, a district with more than one million electors has equal weight in the sample center weight as a district with ten thousand electors and these centers cannot

be interpreted as a result of political elections, it is simply an average of the proportions in different districts. The Aitchison norm of the centers can be interpreted as an inequality index between parties: the larger the Aitchison norm, the greater the difference between one or more parties compared with others. For instance, the 1992 elections show the large Aitchison norm and, consequently, the greater dominance of some parties (in this case one party, CiU) over the other ones in the subcomposition in the districts. Contrarily, the most balanced elections in the sample were those of 2006, since the Aitchison norm of the center is the smallest.

The total variance informs about the dispersion of the results across the 41 districts (4 times 41 in the case before 2006). The fact that the last two elections considered (2003,2006) have the lower total variances suggests a possible social homogenization between districts. The reasons for this require a deeper social and political analysis. Comparing the 2006 results with previous elections, it is clear that 2006 was the most balanced and homogeneous result across districts. Whether this is due to the appearance of C's is not clear from these results.

In order to study change of proportions of votes from election to election, the perturbation difference between consecutive elections is useful. If $\mathbf{x}_i$ is the center for the i -th election, the perturbation difference is $\mathbf{x}_{i+1} \ominus \mathbf{x}_i$. Note that this perturbation difference is essentially a ratio between the proportions of votes in those elections. After carrying out these differences, they describe the change of the studied composition from election to election. Table 5.18, in its upper part, shows these results closed to proportions, followed by the Aitchison norm of the perturbation difference. The middle sub-table shows the same perturbation after multiplying by $1/5 = 0.2$, i.e. compared to the neutral element $[0.2, 0.2, 0.2, 0.2]$. Note that this does not produce any effect in the perturbation-difference from the compositional point of view, as the closed perturbation and this new normalization preserve the ratios between parts and they are compositionally equivalent. One can realize that interpretation of the closed perturbation differences are harder to interpret than the non-closed version. In fact, the normalized difference-perturbations parts which are larger than one point out increments and, conversely, values smaller than one mean decrements. The bottom sub-table represents the same perturbation differences in the common form, seen e.g. in newspapers, of percent of increment/decrement. They are obtained from the middle sub-table subtracting one and multiplying by 100. Note that this last representation of the perturbation-differences is not a compositional representation as ratios, or even differences, of these signed percents are just meaningless.

Looking at the Aitchison norms in Table 5.18, one can conclude that the 2003 and 1995 elections represented larger changes than those in years 2006 and 1999, as the Aitchison norms of the perturbation differences were ~ 0.6 and ~ 0.4 , respectively. For instance the 2003 elections are characterized, with respect to the 1999 elections, by a substantial decrease of the traditional

Table 5.18. Centers of perturbation-differences between consecutive elections and for elections before 2006 and 2006 elections referred to the subcomposition (CiU, PSC, PP, ICV, ERC). Top sub-table shows the perturbation-difference closed to 1. Middle sub-table shows the same perturbation-differences after dividing by $1/5 = 0.2$. Bottom sub-table shows the same perturbations expressed in percent of increment or decrement (according the sign). Aitchison norm of the center perturbation-difference are shown in the upper sub-table.

	perturbation to next election					
	CiU	PSC	PP	IC	ERC	A-norm
2003 to 2006	0.1881	0.1842	0.1741	0.2839	0.1696	0.4217
1999 to 2003	0.1350	0.1497	0.1945	0.2292	0.2915	0.6260
1995 to 1999	0.2082	0.2676	0.1563	0.1635	0.2043	0.4318
1992 to 1995	0.1443	0.1527	0.3097	0.2033	0.1900	0.6052
before 2006	0.1358	0.1652	0.1734	0.2943	0.2314	0.6086
	normalized 1/5					
	CiU	PSC	PP	IC	ERC	
2003 to 2006	0.941	0.921	0.871	1.420	0.848	
1999 to 2003	0.675	0.749	0.973	1.146	1.458	
1995 to 1999	1.041	1.338	0.782	0.818	1.022	
1992 to 1995	0.721	0.763	1.549	1.016	0.950	
before 2006	0.679	0.826	0.867	1.472	1.157	
	as % increment/decrement					
	CiU	PSC	PP	IC	ERC	
2003 to 2006	-5.94%	-7.90%	-12.93%	41.97%	-15.21%	
1999 to 2003	-32.50%	-25.14%	-2.74%	14.62%	45.76%	
1995 to 1999	4.10%	33.82%	-21.83%	-18.25%	2.17%	
1992 to 1995	-27.86%	-23.67%	54.87%	1.64%	-4.99%	
before 2006	-32.12%	-17.41%	-13.32%	47.15%	15.69%	

parties PSC and CiU compared to an increase of IC and the relevant one of ERC.

Both Tables 5.17 and 5.18 can be visualized in a plot where the proportions evolve in time. However, these plots are not very informative and the mentioned tables allow a better interpretation. This suggests finding another way of showing the differences of the centers from election to election. The compositional dendrogram can be useful to this end. Table 5.19 shows two sign codes for the SBP's. The first one (left part of Table 5.19) starts separating CiU and ERC, clear self-declared catalan nationalist parties, from other parties; the second one (right part of Table 5.19), takes the first partition separating self-declared left wing parties (PSC, IC, ERC) from right wing parties (CiU, PP). Using these two SBP's the corresponding compositional dendrogram have been built up and are shown in Figure 5.24. In Figure

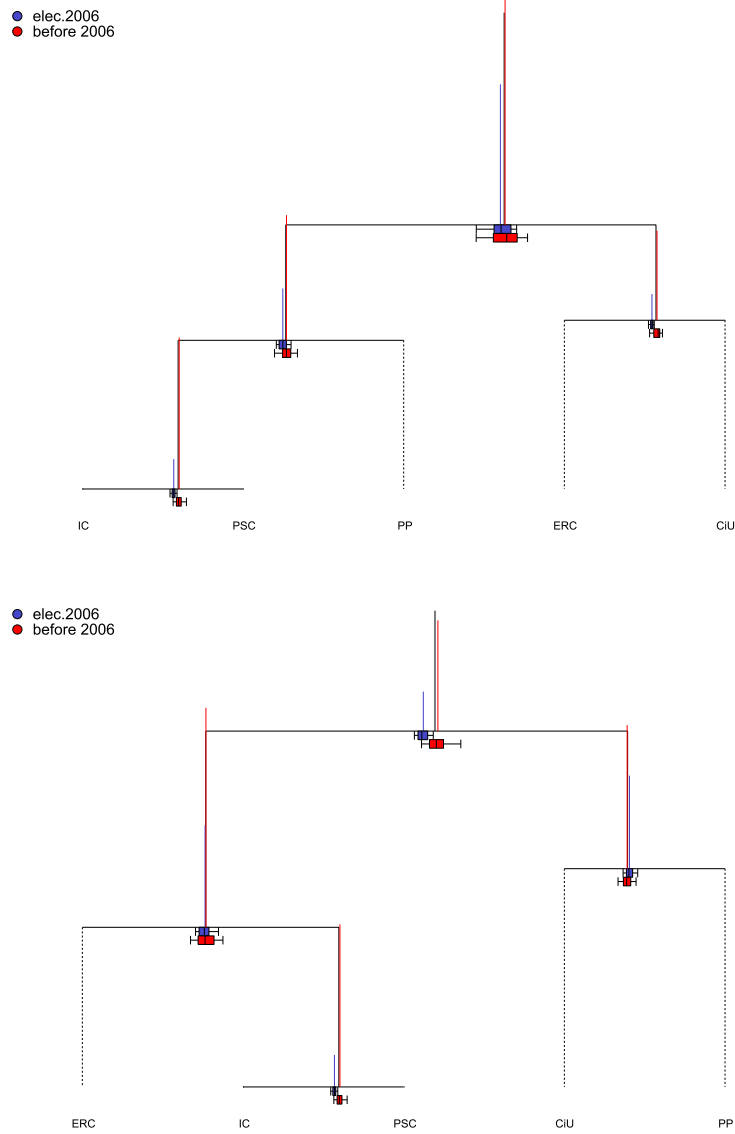


Fig. 5.24. Two balance dendrogram comparing means and variances of the 2006 elections with the 2003, 1999, 1995, and 1992 elections taken as a single sample. The SBP's correspond to the sign codes in Table 5.19. Black vertical lines: dendrogram of all elections considered together as one sample. Blue lines: 2006 elections. Red lines: elections before 2006.

Table 5.19. Two sign code SBP's. The left one separates self-declared catalan nationalist parties from other parties in the first partition. The right sign code separates self-declared left wing parties from right wing parties in the first partition.

CiU	PSC	PP	IC	ERC	CiU	PSC	PP	IC	ERC
+1	-1	-1	-1	+1	+1	-1	+1	-1	-1
+1	0	0	0	-1	+1	0	-1	0	0
0	-1	+1	-1	0	0	-1	0	-1	+1
0	+1	0	-1	0	0	+1	0	-1	0

5.24, separation between blue lines (2006 elections) and red lines (before 2006 elections) represent the differences between the means of the corresponding balances; the same feature can be seen comparing blue and red box-plots. In the upper dendrogram all blue bars are shift to the left with respect to the red ones thus pointing out an increase of the proportion of votes of the branch placed at the left. For instance, the balance between IC and PSC indicates a larger proportion of votes to IC, with respect to PSC, in the 2006 elections than before the 2006 elections. Also the balance ERC versus CiU favors ERC in the 2006 elections. Note that comparison of box-plots is an exploratory tool.

Similar features are observed in the bottom dendrogram of Figure 5.24. In that case, the balance left-right parties, (PSC, IC, ERC) versus (CiU, PP), experiences a significant change in the 2006 elections in favor of the formers. Also an important change is visualized in the balance IC over PSC, where the right position of the blue bar indicates an increment of the proportion of votes for IC compared with PSC.

Exercise 43

As in the preceding exercise, compute the two matrices of variation of the subcomposition (PSC, CiU, PP, ICV, ERC) before and after 2006. Compare each entry of the variation matrices by quotient. Which pairs notably increased or decreased their contribution to the total variability? Which seem to have an equivalent contribution, that is, which have a similar ratio $t_{ij}^{after}/t_{ij}^{before}$ as the total variance ratio calculated before?

Answer of exercise 43

Table 5.20 shows the two variation matrices asked for in the statement (top and middle sub-tables). The bottom sub-table is the componentwise ratio of the two first variation matrices. The first observation is that all the entries of the variation matrix corresponding to elections before 2006 are greater than those from the 2006 elections (all ratios are smaller than 1). There is a clear reason for this fact. The 2006 election data are collected in 41 districts. Variances of log-ratios, the entries of the variation matrix, thus represent variability across districts. The data from elections before 2006 include the same district four times for different elections, totalizing $4 \times 41 = 164$ data points. The entries of this second variation matrix reflect both variability across the 41 districts and those intra-district along different elections. Although this fact does not imply larger variances than for 2006 election, the result is not unexpected. Looking at the ratios of variances, the value 0.80 for the ratio of variances of the logratio PP over ERC indicates that this logratio increased substantially its value. Also values ~ 0.60 are considerable; those are coming from the logratios CiU-PSC, CiU-PP, PSC-ERC. More details on the change of variability between the two samples can be observed in the compositional dendrogram shown in Figure 5.24, comparing the vertical blue and red lines for each balance. There, the more apparent change of variance corresponds to the balance PSC-IC. Other increments of variance are not reflected, as they correspond to simple logratios in the SBP's used in Exercise 42.

Table 5.20. Variation matrices for the elections 2006 and for elections before 2006 (2003,1999,1995,1992). Componentwise ratio of those matrices is shown in the bottom sub-table.

variation matrix for 2006 elections					
	CiU	PSC	PP	IC	ERC
CiU	0	0.1291	0.2195	0.1118	0.0663
PSC	0.1291	0	0.0904	0.0755	0.2357
PP	0.2195	0.0904	0	0.1442	0.3366
IC	0.1118	0.0755	0.1442	0	0.1635
ERC	0.0663	0.2357	0.3366	0.1635	0
variation matrix for before 2006 elections					
	CiU	PSC	PP	IC	ERC
CiU	0.	0.2036	0.3324	0.5212	0.2230
PSC	0.2036	0	0.2405	0.3763	0.3839
PP	0.3324	0.2405	0	0.4142	0.4216
IC	0.5212	0.3763	0.4142	0	0.5665
ERC	0.2230	0.3839	0.4216	0.5665	0
ratio of matrices (2006 over before)					
	CiU	PSC	PP	IC	ERC
CiU		0.63	0.66	0.21	0.30
PSC	0.63		0.38	0.20	0.61
PP	0.66	0.38		0.35	0.80
IC	0.21	0.20	0.35		0.29
ERC	0.30	0.61	0.80	0.29	

Exercise 44

Draw a ternary diagram of the subcomposition (PP, PSC, ICV) using only data before 2006. Include the first PC. How much variability is captured by this first PC? Compare with the proportion of variance explained by the first component in the biplot of Figure 5.16a. Repeat the exercise with the subcomposition (PP, CiU, ERC), and compare with the second PC of the same biplot.

Answer of exercise 44

To examine the data corresponding to the subcomposition (PSC, PP, ICV) it is convenient to check first for possible outliers with great influence within this subcomposition. The data point corresponding to the district *Segrià* in the 1995 elections causes a large distortion when carrying out the computation of principal components. Remember that the 1999 elections were contested by a coalition of PSC and IC in many districts, and for this exercise the votes were redistributed according to the results of the 2003 and 1999 elections (see Example 5.9). The district *Segrià* was an exception, and IC collected nine votes separately, not in coalition. Figure 5.25 shows the ternary diagram of the data set, and the first principal component using the complete data set before (left panel) and after (right panel) removal of the outlying data point. There is a large change in the principal component. The total variance explained by the first principal component is 65.3% for the complete data set, and 60.0% after removal of the outlier. This is a considerable difference when taking into account that the two principal components explain 100% of variability. The outlying point can be identified as a point placed almost on

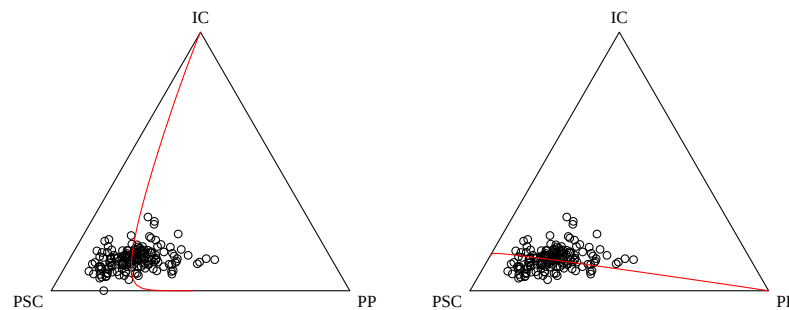


Fig. 5.25. Ternary diagram for elections before 2006, subcomposition PSC, PP, IC. The first principal component is shown as a red line. Right panel, complete data set; Left panel, after removal of one outlier.

the border of the ternary diagram with a low proportion of IC votes. More details can be visualized by representing the data in a form biplot. Recall that in a form biplot the data points are obtained as the two first coordinates

in the principal directions; in this case they are the only two coordinates and they are, in fact, ilr coordinates. Figure 5.26 shows the biplot, again for the complete data set (left panel) and after outlier removal (right panel). Elections are colored differently. Comparing both panels, the change of the principal component is clearly revealed: a rotation of almost 60 degrees due to the leverage of the outlier (the green point separated from the rest in the direction `clr.IC`). The rounded shape of the whole data set is the reason for

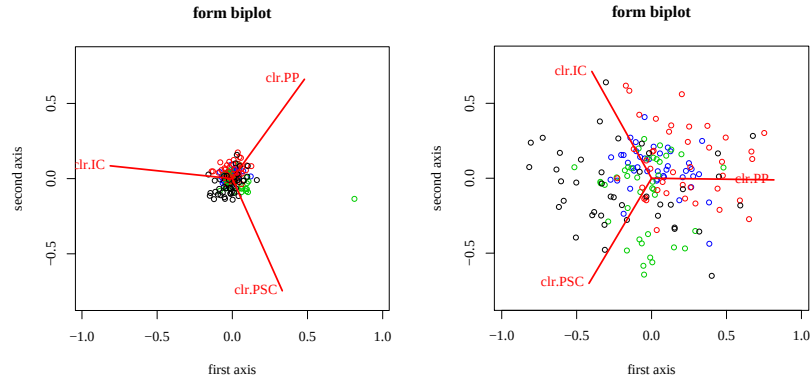


Fig. 5.26. Form biplot for elections before 2006, subcomposition PSC, PP, IC. Colors of data points indicate year of election: black, 2003; blue, 1999; green, 1995; red, 1992. Right panel, complete data set; Left panel, after removal of one outlier. Explained variability: 100%.

the first principal component explaining a poor 60% of total variability.

The exercise can be repeated for the subcomposition (PP, CiU, ERC). Figure 5.27 shows the ternary diagram (left panel) with the first principal component (red line). The first principal component explains 67.6% of the total variance. The right panel shows the form biplot. In this case, the effect of outliers is not dramatic and no data point has been removed. It is interesting to see that the different elections appear quite separated in the form biplot, thus pointing out the evolution in time of the proportions for these parties across the districts.

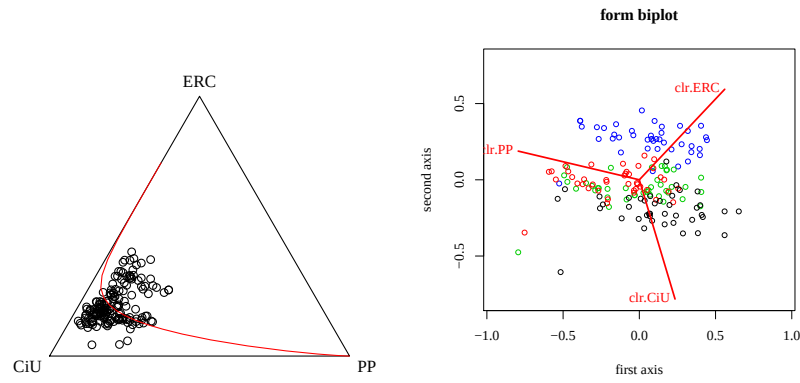


Fig. 5.27. Ternary plot (left panel) and form biplot (right panel) for elections before 2006, subcomposition CiU, PP, ERC. Colors of data points indicate year of election: black, 2003; blue, 1999; green, 1995; red, 1992. Explained variability: 100%.

Exercise 45

[*Sample space of ipsative data*] An individual is asked to order three items, x_1, x_2, x_3 , following his/her preference. The answer is arranged in an array in which the order of preference, 1, 2, 3, is placed in the position labeled x_i . For instance, $[3, 1, 2,]$ means that x_2 is the most preferred item and x_1 is the third one preferred. Nonresponse is excluded. The random variable $\mathbf{Z} = [Z_1, Z_2, Z_3]$, containing the orders of preference, is then considered.

- (a) Determine the minimal sample space, $\mathcal{M}$, for $\mathbf{Z}$, made of all possible triplets $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_m$.
- (b) The analyst decided that the possible values of $\mathbf{Z}$ can be reasonably represented in the simplex $\mathcal{S}^3$ and the Aitchison geometry is valid. Give reasons supporting these ideas.
- (c) Accepting statement (b), compute the Aitchison interdistances between the compositions $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_m$.
- (d) The probabilities of getting the answer $\mathbf{z}_i$ are given. Consider, for instance, two cases. The first is $p_i = 1/m, i = 1, 2, \dots, m$. The second is

$$q_i = \binom{m-1}{i-1} q^{i-1} (1-q)^{m-i} \quad , \quad q = 0.9 \quad , \quad i = 1, 2, \dots, m.$$

Compute the center and total variance of $\mathbf{Z}$, when the sample space $\mathcal{M}$. Repeat the computation when $\mathcal{M}$ is assumed embedded in $\mathcal{S}^3$.

Answer of exercise 45

- (a) The sample space must contain triplets containing permutations of the order indices 1, 2, 3. Therefore, there are $m = 3! = 6$ permutations. They can be ordered as $\mathbf{z}_1 = [1, 2, 3]$, $\mathbf{z}_2 = [1, 3, 2]$, $\mathbf{z}_3 = [2, 1, 3]$, $\mathbf{z}_4 = [2, 3, 1]$, $\mathbf{z}_5 = [3, 1, 2]$, $\mathbf{z}_6 = [3, 2, 1]$. They are the elements of $\mathcal{M}$, the minimal sample space of $\mathbf{Z}$. These points are shown in Figure 6.7 (left panel).
- (b) A reason for considering $\mathcal{M} \subset \mathcal{S}^3$ is that the number of elements of $\mathbf{z}_i$ is always 6. However, this is not conclusive; principles of compositional analysis should hold (Section 2.2). Scale invariance is in this case accomplished since labelling the order indices as 2, 4, 6 or as 0.1, 0.2, 0.3 should give equivalent results in the analysis. However, this is somewhat arbitrary, as the orders could be expressed in other numbers, e.g. 1, 3, 5. If the analyst wants to study only the preferences for items x_1 and x_2 using the same sample of preferences, he/she needs to guarantee that distances in $\mathcal{S}^2$ are smaller than or equal distances computed in $\mathcal{S}^3$. For instance, the distance between $\mathbf{z}_1 = [1, 2, 3]$ and $\mathbf{z}_2 = [1, 3, 2]$, should be greater than or equal to the distance between $\mathbf{z}_1^s = [1, 2]$ and $\mathbf{z}_2^s = [1, 3]$. This is granted using Aitchison distances, while other distances, like the Euclidean distance, fail at this point. Note that the representation in $\mathcal{S}^3$

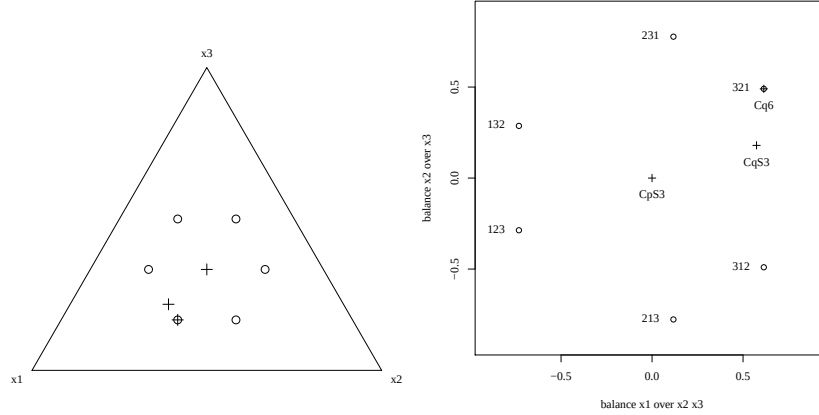


Fig. 6.7. Left panel: points $\mathbf{z}_i \in \mathcal{M}$ representing preferences: circles. Center of $\mathbf{Z}$ when the sample space is $\mathcal{M}$ with probabilities q_i : cross over circle. Centers of $\mathbf{Z}$ when the sample space is $\mathcal{S}^3$: crosses. If probabilities are the p_i 's the center is placed at the barycenter. Right panel: same points shown in the left panel in ilr-coordinates of the simplex. Aitchison distances between points are now visualized. Labels: permutation for each sample point; Cq6, center when sample space is $\mathcal{M}$ and probabilities are the q_i 's; CpS3, CqS3, centers when the sample space is $\mathcal{S}^3$ and probabilities are the p_i 's and q_i 's, respectively.

of the two cases $\mathbf{z}_1 = [1, 2, 3]$ and $\mathbf{z}_5 = [3, 1, 2]$, is given by their closures, $\mathcal{C}[1, 2, 3] = [1/6, 2/6, 3/6]$, $\mathcal{C}[3, 1, 2] = [2/6, 3/6, 1/6]$, and of their subcompositions in $\mathcal{S}^2$ by $\mathcal{C}[1, 2] = [1/3, 2/3]$ and $\mathcal{C}[3, 1] = [3/4, 1/4]$, respectively. These changes of representatives may affect the computation of distances when they are not scale invariant. For instance, if the ordinary Euclidean distance is used on the representatives, the square-distance between $\mathcal{C}[1, 2, 3]$ and $\mathcal{C}[3, 1, 2]$ is 0.167, whereas for $\mathcal{C}[1, 2]$ and $\mathcal{C}[3, 1]$ it is 0.347, a violation of the subcompositional coherence principle, as for two parts it is larger than for three parts.

- (c) The square-Aitchison distances between the $\mathbf{z}_i$ compositions are shown in Table 6.3, together with the two sets of probabilities $p_i = 1/6$ and q_i , as given in the statement. The Aitchison inter-distances between the $\mathbf{z}_i$, shown in Table 6.3, can be easily visualized in Figure 6.7 (right panel). It shows that Aitchison distances between points produced by a permutation of preferences 1 and 3 are larger than those caused by permutation of ordinates (1,2) and (2,3). This can be a reason to adopt the Aitchison geometry of the simplex. However, this is a subjective choice of the analyst.

The associated variability is shown in the right-hand-side of Table 6.3 for the two alternative probabilities. For instance, the variability associated with $\mathbf{z}_6$ with probabilities q_i is $\sum_k d_a^2(\mathbf{z}_6, \mathbf{z}_k) = 0.355$. In order to determine the center and total variance, the minimum in the columns

Table 6.3. Square Aitchison distances between compositions in $\mathcal{M} \subset \mathcal{S}^3$, assigned probabilities $(\mathbf{p}_i, \mathbf{q}_i)$, and associated variability (var_p, var_q)

	$\mathbf{z}_1$	$\mathbf{z}_2$	$\mathbf{z}_3$	$\mathbf{z}_4$	$\mathbf{z}_5$	$\mathbf{z}_6$	p_i	q_i	var_p	var_q
$\mathbf{z}_1$	0.000	0.329	0.961	1.852	1.852	2.414	0.167	$1.0 \cdot 10^{-5}$	1.235	2.176
$\mathbf{z}_2$	0.329	0.000	1.852	0.961	2.414	1.852	0.167	$4.5 \cdot 10^{-4}$	1.235	1.970
$\mathbf{z}_3$	0.961	1.852	0.000	2.414	0.329	1.852	0.167	$8.1 \cdot 10^{-3}$	1.235	1.378
$\mathbf{z}_4$	1.852	0.961	2.414	0.000	1.852	0.329	0.167	$7.3 \cdot 10^{-2}$	1.235	0.822
$\mathbf{z}_5$	1.852	2.414	0.329	1.852	0.000	0.961	0.167	0.328	1.235	0.706
$\mathbf{z}_6$	2.414	1.852	1.852	0.329	0.961	0.000	0.167	0.590	1.235	0.355

var_p, var_q is the corresponding total variance, and the point $\mathbf{z}_i$ where the minimum occurs is the center when the sample space is $\mathcal{M}$. Note the minimum is computed over the points in the sample space $\mathcal{M}$ and not over the whole $\mathcal{S}^3$. Accordingly, the total variance for probabilities p_i is 1.235 but the center is indistinguishable among all z_i 's. For probabilities q_i , the total variance is 0.355 and the center is $\mathbf{z}_6$ (Fig. 6.7). Whenever $\mathcal{M}$ is taken as the sample space, the center is one or more points in $\mathcal{M}$ and there is no possibility of finding a center outside $\mathcal{M}$.

When the sample space is the whole $\mathcal{S}^3$, variability has to be computed for all points in $\mathcal{S}^3$. Fortunately, the center can be computed as $\text{clr}^{-1}\text{E}[\text{clr}(\mathbf{Z})]$ taking into account that expectation is taken with respect to the discrete probabilities p_i or q_i . For probabilities $p_i = 1/6$, the center is the neutral element of the simplex $[1/3, 1/3, 1/3]$, labelled CpS3 in Figure 6.7 (right panel). For probabilities q_i , the center is $[0.500, 0.281, 0.218]$ (Fig. 6.7, SqS3 in right panel). In both cases, these centers are not in $\mathcal{M}$, i.e. they are impossible results of the random variable $\mathbf{Z}$.

Exercise 46

Plot the density of a normal on $\mathcal{S}^2$. Do it for several logistic means and variances. For instance, $\mu = 0$, $\mu = -1$ and logistic variances $\sigma^2 = 0.5^2$, $\sigma^2 = 2^2$. These densities can be plotted taking the Lebesgue measure as reference measure in $[0, 1]$ and, alternatively, taking the Aitchison measure on $\mathcal{S}^2$ identified with $[0, 1]$.

Answer of exercise 46

The random composition $\mathbf{X} = [X_1, X_2]$ has logistic-normal distribution or normal on the simplex $\mathcal{S}^2$ if its single random coordinate follows a normal distribution,

$$B = (1/\sqrt{2}) \ln(X_1/X_2) \sim \mathcal{N}(\mu, \sigma^2).$$

The required plots are densities of X_1 with respect to the Lebesgue measure and the Aitchison measure in $[0, 1]$. The densities for X_2 are equal up to symmetry with respect to the value 0.5. The transformation from X_1 to B and its inverse can be written as

$$B = \frac{1}{\sqrt{2}} \ln \frac{X_1}{1 - X_1}, \quad X_1 = \frac{\exp(\sqrt{2}B)}{1 + \exp(\sqrt{2}B)}.$$

They are the logit transformation and its inverse up to the constant $1/\sqrt{2}$. The Jacobian of the transformation is $J = dB/dx_1 = 1/[\sqrt{2}x_1(1 - x_1)]$. Therefore, the density of X_1 with respect to the Lebesgue measure is

$$f_L(x_1) = \frac{J}{[\sqrt{2\pi\sigma^2}]} \exp\left(-\frac{1}{2} \frac{((1/\sqrt{2})(\ln x_1 - \ln(1 - x_1)) - \mu)^2}{\sigma^2}\right).$$

The density of X_1 with respect to the Aitchison measure is obtained suppressing the derivative $dB/dx_1 = 1/[\sqrt{2}x_1(1 - x_1)]$ (Jacobian) from f_L , i.e.

$$f(x_1) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \frac{((1/\sqrt{2})(\ln x_1 - \ln(1 - x_1)) - \mu)^2}{\sigma^2}\right).$$

Figure 6.8 shows both densities for different values of the logistic parameters μ and σ^2 .

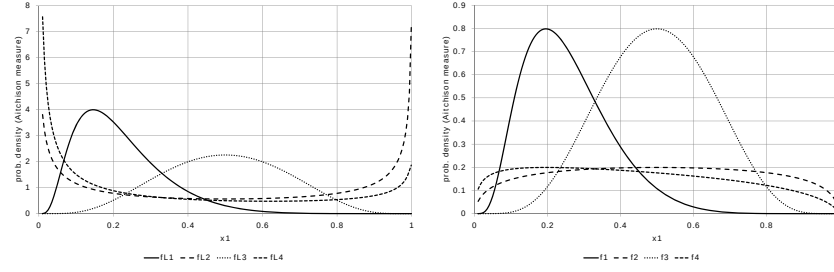


Fig. 6.8. Probability densities of a Normal on $\mathcal{S}^2$. Left panel, with respect to the Lebesgue measure in $[0, 1]$. Right panel, with respect to the Aitchison measure in $[0, 1]$. Parameters: $\mu_1 = -1$, $\sigma_1 = 0.5$; $\mu_2 = 0$, $\sigma_2 = 2$; $\mu_3 = 0$, $\sigma_3 = 0.5$; $\mu_4 = -1$, $\sigma_4^2 = 2$

Exercise 47

Prove Theorem 6.10.

Answer of exercise 47

Theorem 6.10 states that, provided that the variability of a random composition $\mathbf{X} \in \mathcal{S}^D$ exists, the center can be computed as

$$\text{cen}[\mathbf{X}] = \text{ilr}^{-1}(\text{E}[\text{ilr}(\mathbf{X})]) = \text{clr}^{-1}(\text{E}[\text{clr}(\mathbf{X})]) = \mathcal{C} \exp(\text{E}[\ln \mathbf{X}]) .$$

From Definitions 6.8 and 6.9, $\text{cen}[\mathbf{X}]$ is the composition in $\mathcal{S}^D$ which minimizes the expected square Aitchison distance, $\text{E}[\text{d}_a^2(\mathbf{X}, \mathbf{z})]$, with respect to $\mathbf{z} \in \mathcal{S}^D$. As a first step, the Aitchison distance can be expressed in terms of distance between both ilr coordinates and clr coefficients (Properties 4.2 and 4.5). In fact, the variability with respect to a composition $\mathbf{z}$ is

$$\text{var}[\mathbf{X}; \mathbf{z}] = \text{E}[\text{d}_a^2(\mathbf{X}, \mathbf{z})] = \text{E}[\text{d}(\text{ilr}(\mathbf{X}), \text{ilr}(\mathbf{z}))] = \text{E}[\text{d}(\text{clr}(\mathbf{X}), \text{clr}(\mathbf{z}))] ,$$

where all expectations exist due to the hypothesis that variability exists. Hence, the minimization of variability can be re-stated as the minimization of any of the two right-hand members: find the composition $\mathbf{z}$ whose (ilr or clr) transformation minimizes $\text{E}[\text{d}(\text{ilr}(\mathbf{X}), \text{ilr}(\mathbf{z}))]$ or $\text{E}[\text{d}(\text{clr}(\mathbf{X}), \text{clr}(\mathbf{z}))]$, respectively. As both expectations are taken on real vectors and it is known that values are the mean of real vectors, i.e.

$$\begin{aligned} \text{E}[\text{ilr}(\mathbf{X})] &= \underset{\text{ilr}(\mathbf{z}) \in \mathbb{R}^{D-1}}{\text{argmin}} \text{E}[\text{d}(\text{ilr}(\mathbf{X}), \text{ilr}(\mathbf{z}))] , \\ \text{E}[\text{clr}(\mathbf{X})] &= \underset{\text{clr}(\mathbf{z}) \in \mathbb{R}^D}{\text{argmin}} \text{E}[\text{d}(\text{clr}(\mathbf{X}), \text{clr}(\mathbf{z}))] . \end{aligned}$$

Therefore, $\text{E}[\text{ilr}(\mathbf{X})]$ and $\text{E}[\text{clr}(\mathbf{X})]$ are the ilr and clr transformations of $\text{cen}[\mathbf{X}]$, respectively. Taking inverse transformations, it yields the desired result. However, this is only valid for the ilr and clr expressions. The expression using logarithms and expectation requires a detailed development. Starting from the expression of the center in terms of clr, and developing clr-coefficients

$$\begin{aligned} \text{cen}[\mathbf{X}] &= \text{clr}^{-1} \text{E}[\text{clr}(\mathbf{X})] \\ &= \mathcal{C} \{ \exp(\text{E}[\ln \mathbf{X}] - \text{E}[\ln g_m(\mathbf{X})]) \} \\ &= \mathcal{C} \{ \exp(\text{E}[\ln \mathbf{X}]) \cdot \exp(\text{E}[\ln g_m(\mathbf{X})]) \} \\ &= \mathcal{C} \exp(\text{E}[\ln \mathbf{X}]) , \end{aligned}$$

where the multiplicative term $\exp(\text{E}[\ln g_m(\mathbf{X})])$ has been included in the closure operation, given that it is equal for the D components. This completes the proof of Theorem 6.10.

Exercise 48

Prove Theorem 6.19.

Answer of exercise 48

The normal pdf of the random composition $\mathbf{X}$ as a function of the coordinates $\mathbf{x}_0^* \text{ilr}_0(\mathbf{x})$ (Equation 6.5) is

$$f^*(\mathbf{x}_0^*) = \frac{1}{(2\pi)^{(D-1)/2} |\boldsymbol{\Sigma}_0|^{1/2}} \cdot \exp \left[-\frac{1}{2} (\mathbf{x}_0^* - \boldsymbol{\mu}_0) \boldsymbol{\Sigma}_0^{-1} (\mathbf{x}_0^* - \boldsymbol{\mu}_0)^\top \right].$$

To change the argument of this pdf to $\mathbf{x}_1^* = \text{ilr}_1(\mathbf{x})$, $\mathbf{x}_0^*$ has to be expressed as a function of $\mathbf{x}_1^*$. The relationship is

$$\mathbf{x}_0^* = \mathbf{x}_1^* \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top,$$

as implied by $\text{ilr}_i(\mathbf{x}) = \ln(\mathbf{x}) \boldsymbol{\Psi}_i^\top$ for $i = 0, 1$ and their respective inverses:

$$\begin{aligned} \text{ilr}_0(\mathbf{x}) &= \ln(\mathbf{x}) \boldsymbol{\Psi}_0^\top \\ &= \ln(\mathcal{C} \exp(\ln(\mathbf{x}) \boldsymbol{\Psi}_1)) \boldsymbol{\Psi}_0^\top \\ &= (K\mathbf{1} + \ln(\mathbf{x})) \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top, \end{aligned}$$

where the constant term $K\mathbf{1} = [K, K, \dots, K]$, coming from the closure, can be ignored, given that $K\mathbf{1} \boldsymbol{\Psi}_1 = 0$.

The next step is to transform the pdf $f^*(\mathbf{x}_0^*)$ into $f^*(\mathbf{x}_1^*)$. This transformation requires the substitution of $\mathbf{x}_0^* = \mathbf{x}_1^* \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top$ and to introduce a Jacobian factor to account for the change of variable. However, the transformation $\mathbf{x}_0^* = \mathbf{x}_1^* \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top$ is isometric and the Jacobian is 1. Therefore, the new pdf is

$$f^*(\mathbf{x}_1^*) = \frac{1}{(2\pi)^{(D-1)/2} |\boldsymbol{\Sigma}_0|^{1/2}} \cdot \exp \left[-\frac{1}{2} (\mathbf{x}_1^* \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top - \boldsymbol{\mu}_0) \boldsymbol{\Sigma}_0^{-1} (\mathbf{x}_1^* \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top - \boldsymbol{\mu}_0)^\top \right].$$

Taking into account the result stated in Exercise 26,

$$[\boldsymbol{\Psi}_0 \boldsymbol{\Psi}_1^\top]^{-1} = [\boldsymbol{\Psi}_0 \boldsymbol{\Psi}_1^\top]^\top = \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top,$$

and setting $\boldsymbol{\Sigma}_1^{-1} = \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top \boldsymbol{\Sigma}_0^{-1} \boldsymbol{\Psi}_0 \boldsymbol{\Psi}_1^\top$, the term in the exponential of $f^*(\mathbf{x}_1^*)$ (Eq. 6.15) can be rewritten as

$$(\mathbf{x}_1^* \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top - \boldsymbol{\mu}_0) \boldsymbol{\Sigma}_0^{-1} (\mathbf{x}_1^* \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top - \boldsymbol{\mu}_0)^\top = (\mathbf{x}_1^* - \boldsymbol{\mu}_0 \boldsymbol{\Psi}_0 \boldsymbol{\Psi}_1^\top) \boldsymbol{\Sigma}_1^{-1} (\mathbf{x}_1^* - \boldsymbol{\mu}_0 \boldsymbol{\Psi}_0 \boldsymbol{\Psi}_1^\top)^\top.$$

Moreover, setting $\boldsymbol{\mu}_1 = \boldsymbol{\mu}_0 \boldsymbol{\Psi}_0 \boldsymbol{\Psi}_1^\top$, $f^*(\mathbf{x}_1^*)$ is

$$f^*(\mathbf{x}_1^*) = \frac{1}{(2\pi)^{(D-1)/2} |\boldsymbol{\Sigma}_0|^{1/2}} \cdot \exp \left[-\frac{1}{2} (\mathbf{x}_1^* - \boldsymbol{\mu}_1) \boldsymbol{\Sigma}_1^{-1} (\mathbf{x}_1^* - \boldsymbol{\mu}_1)^\top \right]. \quad (6.15)$$

Proving that $\boldsymbol{\Sigma}_1 = \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top \boldsymbol{\Sigma}_0 \boldsymbol{\Psi}_0 \boldsymbol{\Psi}_1^\top$ and $|\boldsymbol{\Sigma}_0| = |\boldsymbol{\Sigma}_1|$, the statement of the theorem holds. The first condition is obtained as

$$\boldsymbol{\Sigma}_1 = [\boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top \boldsymbol{\Sigma}_0^{-1} \boldsymbol{\Psi}_0 \boldsymbol{\Psi}_1^\top]^{-1} = \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top \boldsymbol{\Sigma}_0 \boldsymbol{\Psi}_0 \boldsymbol{\Psi}_1^\top ,$$

Furthermore,

$$|\boldsymbol{\Sigma}_1| = \left| \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top \right| |\boldsymbol{\Sigma}_0| \left| \boldsymbol{\Psi}_0 \boldsymbol{\Psi}_1^\top \right| = |\boldsymbol{\Sigma}_0| ,$$

due to $\left| \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_0^\top \right| \left| \boldsymbol{\Psi}_0 \boldsymbol{\Psi}_1^\top \right| = 1$. Then, Equation (6.15) is the pdf of a multivariate normal with mean $\boldsymbol{\mu}_1$ and covariance matrix $\boldsymbol{\Sigma}_1$, thus proving the statement of Theorem 6.19.

Exercise 49

Using Theorem 6.23, find a way to transform a standard normal distribution (i.e., with ilr mean vector $\boldsymbol{\mu} = \mathbf{0}$ and ilr variance $\boldsymbol{\Sigma} = \mathbf{I}_{D-1}$) to another normal distribution on the simplex with arbitrary ilr mean and variance. Assume that this arbitrary variance matrix is of the form $\boldsymbol{\Sigma} = \mathbf{L} \cdot \mathbf{L}^\top$.

Answer of exercise 49

Real random vectors with a $(D-1)$ -multivariate normal distribution can be obtained using a linear transformation of a $\mathcal{N}(\boldsymbol{\mu} = \mathbf{0}, \boldsymbol{\Sigma} = \mathbf{I}_{D-1})$ distributed random vector. This technique can be applied directly to the coordinates of a normal on the simplex. The key point is the factorization of the covariance matrix of the desired random vector as proposed, $\boldsymbol{\Sigma} = \mathbf{L} \cdot \mathbf{L}^\top$. The notation $\mathbf{L}$ for the factors comes from the standard way of obtaining the factorization using the Cholesky decomposition, where $\mathbf{L}$ is a lower triangular matrix.

Let $\mathbf{X}^*$ be a $(D-1)$ -dimensional real random vector which distribution is $\mathcal{N}(\mathbf{0}, \mathbf{I}_{D-1})$ and assume that $\mathbf{X}^*$ contains the ilr coordinates of a random composition $\mathbf{X}$ in the simplex $\mathcal{S}^D$. Define a new real random vector $\mathbf{Y}_0^* = \mathbf{X}^* \mathbf{L}^\top$, using notation of row vectors. Since a linear transformation of a normal random vector is again normal, the distribution of $\mathbf{Y}_0^*$ will be specified with its mean and covariance matrix. The fact that the mean is $E[\mathbf{X}_0^*] = \mathbf{0}$ is easily checked, because linear combinations of zero mean random variables have also zero mean. To get the covariance matrix, proceed as follows:

$$\boldsymbol{\Sigma}_{Y_0} = E[(\mathbf{Y}_0^*)^\top \mathbf{Y}_0^*] = E[(\mathbf{X}^* \mathbf{L}^\top)^\top \mathbf{X}^* \mathbf{L}^\top] = \mathbf{L} E[(\mathbf{X}^*)^\top \mathbf{X}^*] \mathbf{L}^\top = \mathbf{L} \mathbf{L}^\top = \boldsymbol{\Sigma},$$

where the covariance matrix $E[(\mathbf{X}^*)^\top \mathbf{X}^*] = \mathbf{I}_{D-1}$. As a conclusion, the covariance matrix of $\mathbf{Y}_0^*$ is $\boldsymbol{\Sigma}$ as required in the statement. However, the mean of $\mathbf{Y}_0^*$ is null, but it can be changed to $\boldsymbol{\mu}$ summing it to $\mathbf{Y}_0^*$, i.e. defining $\mathbf{Y}^* = \mathbf{X}^* \mathbf{L}^\top + \boldsymbol{\mu}$, it yields $\mathbf{Y}^* \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The random composition $\mathbf{Y}$ in $\mathcal{S}^D$, which random ilr-coordinates are $\mathbf{Y}^*$, is normal on the simplex, $\mathbf{Y} \sim \mathcal{N}_S(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Note that both $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$, as parameters of the normal distribution on the simplex, depend on the selected ilr-coordinates (see Section 6.3.1).

Exercise 50

Using the expression obtained in Exercise 49, simulate 100 compositions following a normal distribution on the simplex with geometric center and variation matrix

$$\mathbf{g} = \begin{pmatrix} 0.089 \\ 0.239 \\ 0.672 \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} 0.000 & 0.405 & 2.938 \\ 0.405 & 0.000 & 2.911 \\ 2.938 & 2.911 & 0.000 \end{pmatrix},$$

Note that you should first transform the statistics given to a mean vector and variance matrix of a particular ilr of your choice (see Section 5.5). Note as well that the matrix $\mathbf{L}$ can be obtained with the Cholesky decomposition, after transforming the variation matrix to an ilr-variance matrix. (see Sections A.2 and A.3).

Answer of exercise 50

As a first step, select a suitable ilr transformation for $\mathcal{S}^3$. Let

$$b_1 = \sqrt{\frac{2}{3}} \ln \frac{(x_1 x_2)^{1/2}}{x_3}, \quad b_2 = \frac{1}{\sqrt{2}} \ln \frac{x_1}{x_2},$$

be the chosen ilr-coordinates. To specify the contrast matrix Ψ , it is convenient to identify the sign code of the SBP corresponding to the balances b_1 and b_2 , and then build Ψ using Equation (4.10). The contrast matrix Ψ is

$$\Psi = \begin{pmatrix} 0.4082483 & 0.4082483 & -0.8164966 \\ 0.7071068 & -0.7071068 & 0.0000000 \end{pmatrix},$$

where the first and second rows correspond to the clr of the balancing elements giving rise to the coordinates b_1 and b_2 , respectively.

A second step is to get the parameters of the normal distribution of the coordinates $\mathbf{b} = [b_1, b_2]$ in their standard form, i.e. the mean μ and the covariance matrix Σ corresponding to the given center $\mathbf{g}$ and the variation matrix $\mathbf{T}$. The mean coordinate μ is the ilr of $\mathbf{g}$, $\mu = \text{ilr}(\mathbf{g})$, which yields

$$\mu = \text{ilr}([0.089, 0.239, 0.672]) = \ln(\mathbf{g})\Psi^\top = [-1.247369, -0.6984993].$$

The relationship between the variation matrix $\mathbf{T}$ and the covariance matrix of the ilr-coordinates $\mathbf{b}$ is detailed in Appendix A.3. The covariance matrix is

$$\Sigma = -\frac{1}{2}\Psi\mathbf{T}\Psi^\top = \begin{pmatrix} 1.882166667 & 0.007794229 \\ 0.007794229 & 0.202500000 \end{pmatrix},$$

which is near to a diagonal matrix, thus pointing out almost independence of the two balances b_1 and b_2 . The Cholesky decomposition of Σ provides a lower triangular matrix $\mathbf{L}$ such that $\Sigma = \mathbf{L}\mathbf{L}^\top$,

$$\mathbf{L} = \begin{pmatrix} 1.371920795 & 0.0000000 \\ 0.005681253 & 0.4499641 \end{pmatrix}.$$

It is known that, if $\mathbf{Z}^* \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{D-1})$, then $\mathbf{X}^* = \mathbf{Z}^* \mathbf{L}^\top + \boldsymbol{\mu}$ has the desired distribution (see Exercise 49). Therefore, the simulation consists of: (a) simulate a 100-random-sample of a zero-mean standard bivariate normal distribution $\mathbf{z}_i^*$, $i = 1, 2, \dots, 100$, and (b) carry out the transformation $\mathbf{x}_i^* = \mathbf{z}_i^* \mathbf{L}^\top + \boldsymbol{\mu}$ for each sample element $i = 1, 2, \dots, 100$. This provides a 100-sample of coordinates $\mathbf{x}_i^*$. If a sample of compositions $\mathbf{x}_i$ are required, such that $\mathbf{x}_i^* = \text{ilr}(\mathbf{x}_i)$, they can be computed using the ilr-inverse transformation

$$\mathbf{x}_i = \text{ilr}^{-1}(\mathbf{x}_i^*) = \mathcal{C} \exp[\mathbf{x}_i^* \boldsymbol{\Psi}] \quad , \quad i = 1, 2, \dots, 100 \quad .$$

Figure 6.9 shows a simulated 100-sample, both in coordinates $\mathbf{x}_i^*$ (right panel) and as compositions $\mathbf{x}_i$ (left panel) in a ternary diagram. Two families of ellipses are superimposed to the samples. The blue ones correspond to isodensity curves of the theoretical normal distribution $\mathcal{N}_S(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ (left panel) and $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ for coordinates $\mathbf{b}$ (right panel). Red ellipses are also normal isodensity curves, but the parameters of the distribution have been estimated from the simulated 100-sample. Two features can be observed: the ellipses in the right panel have principal axes almost parallel to the plotting axes, thus showing that the two balances are near to be independent; moreover, the red and blue ellipses are not equal but similar as corresponds to a simulation and subsequent estimation of the parameters.

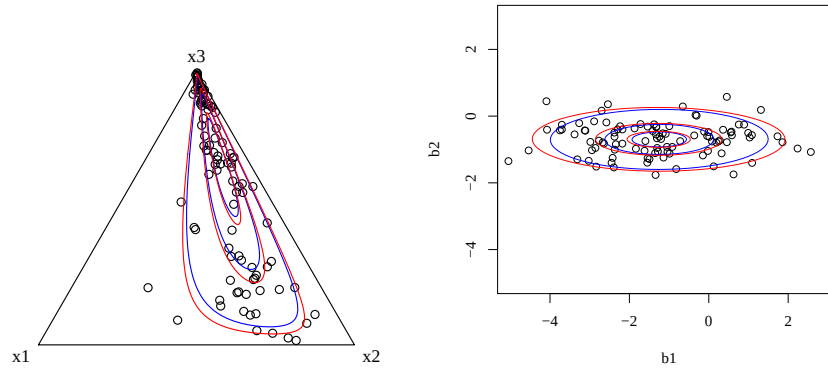


Fig. 6.9. Simulated normal sample of size 100: black circles. Lines are isodensity curves (ellipses) of the theoretical normal distribution (blue) and estimated from the sample (red). The basic ellipse has half-axis equal to the standard deviation on the principal directions; the three curves are scaled by the factors 0.5, 1, and 2.

Exercise 51

Given the definition of the Dirichlet distribution of Section 6.3.2, it seems natural to simulate the Dirichlet composition as the closure of D gamma-distributed values, with the same ν and each with its corresponding α_i . Using this procedure, for $D = 3$, simulate $N = 100$ realizations of a Dirichlet-distributed composition with $\alpha = (1.3, 1.7, 2.0)$. Note that ν can be any arbitrary number. Plot the resulting simulations in a ternary diagram. Compare the results with Figure 6.2.

Answer of exercise 51

The first step is to draw $D = 3$ independent samples of size $N = 100$ from gamma-distributed variables with shape parameters $\alpha = (1.3, 1.7, 2.0)$; the common scale parameter can be, e.g. 1. Most statistical software is prepared to carry out this simulation without difficulties. Figure 6.10 shows the QQ-plots of the three simulated samples with shape parameter $(1.3, 1.7, 2.0)$. The QQ-plots show the points of the sample cdf (it is a stepwise function; the upper points of the steps are represented with filled black circles; the lower points with hollow circles), scaled such that the theoretical gamma cdf appears as a straight-line (red line). The QQ-plots allows a quick comparison of the sample distribution and the theoretical one from which the sample has been drawn. Moreover, the plot shows two green curves representing the 0.05-significance limits from the Kolmogorov-Smirnov goodness of fit test: when any point is outside of the region bounded by the green curves, the sample is rejected to be gamma-distributed at the 0.05-significance level. In the case shown in Figure 6.10, the samples cannot be rejected to be gamma distributed with shape parameters $(1.3, 1.7, 2.0)$ and scale parameter 1.

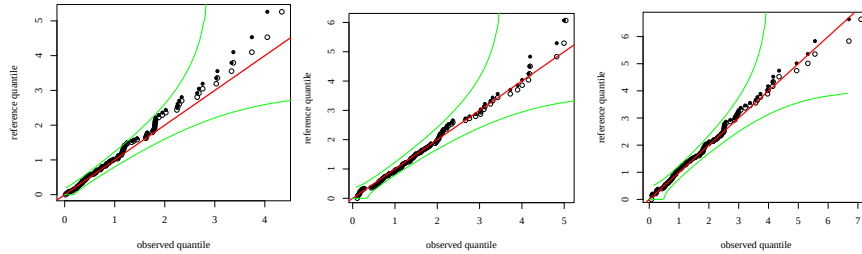


Fig. 6.10. QQ-plots of three simulated gamma samples (shape parameters $(1.3, 1.7, 2.0)$; scale parameter 1), comparing the sample cdf and the gamma-cdf used to simulate (red straight-line). Acceptance region for the Kolmogorov-Smirnov goodness of fit test limited by green curves.

Once the three independent gamma samples have been drawn, closing each trio, coming from the three simulations, produces a simulation of a 3-part

Dirichlet random composition with $\alpha = (1.3, 1.7, 2.0)$. Figure 6.11 shows the simulated sample both in a ternary diagram (left panel) and in ilr-coordinates

$$b_1 = \sqrt{\frac{2}{3}} \ln \frac{(x_1 x_2)^{1/2}}{x_3}, \quad b_2 = \frac{1}{\sqrt{2}} \ln \frac{x_1}{x_2}.$$

Isodensity contours of the theoretical distribution have also been plotted (red curves) as asked by the statement.

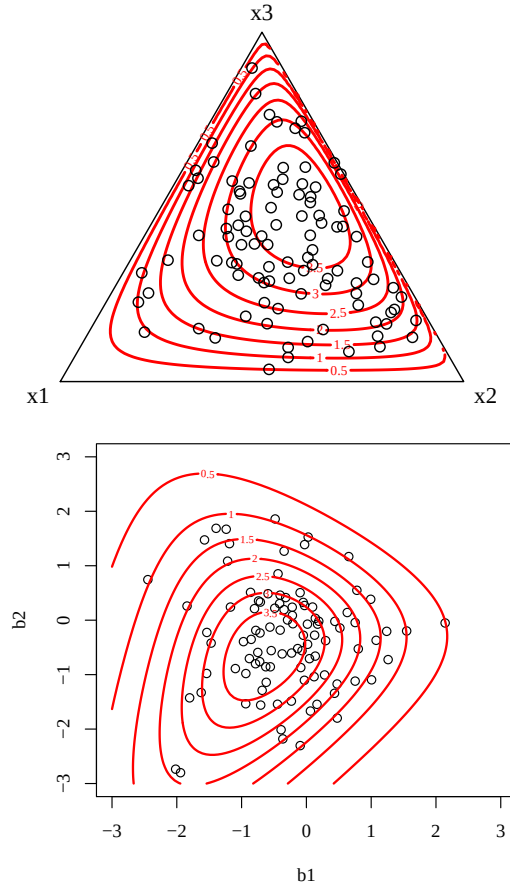


Fig. 6.11. Simulated sample of a 3-part Dirichlet random composition with $\alpha = (1.3, 1.7, 2.0)$.

Exercise 52

Build the subcomposition ($\text{Fe}_2\text{O}_3, \text{FeO}, \text{MnO}$) from the Kilauea data set of Table 5.1, and plot it in a ternary diagram and an ilr scatterplot. Check the normality of this data.

Advanced exercise:

Build the subcompositions $[\text{Fe}_2\text{O}_3, \text{FeO}, \text{MgO}]$ and $[\text{Fe}_2\text{O}_3, \text{FeO}, \text{MnO}, \text{MgO}]$, and test their compositional normality. How can you interpret the result? Which components, or ratios of components, seem to be responsible for a departure from normality?

(Suggestion: Do the advanced exercise with a global test only).

Answer of exercise 52

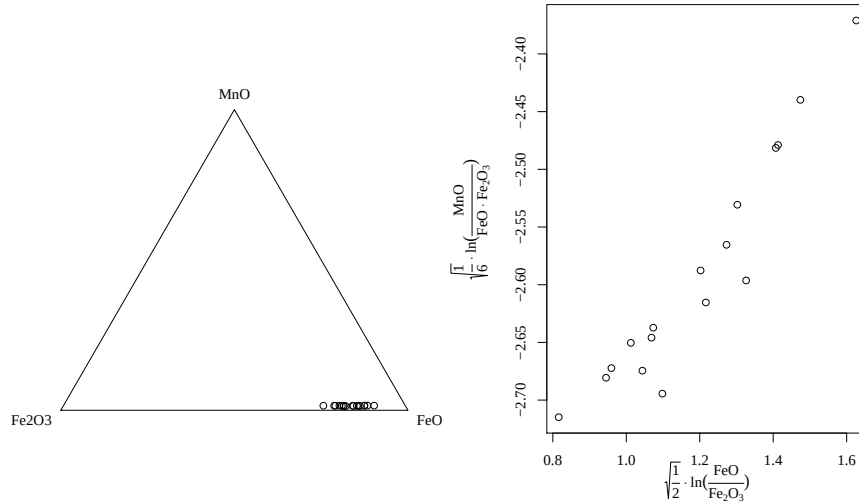
Table 5.1 in the book is reproduced here as Table 7.1 for easy reference. The

Table 7.1. Chemical analysis of rocks from Kilauea Iki lava lake, Hawaii

SiO ₂	TiO ₂	Al ₂ O ₃	Fe ₂ O ₃	FeO	MnO	MgO	CaO	Na ₂ O	K ₂ O	P ₂ O ₅	CO ₂
48.29	2.33	11.48	1.59	10.03	0.18	13.58	9.85	1.90	0.44	0.23	0.01
48.83	2.47	12.38	2.15	9.41	0.17	11.08	10.64	2.02	0.47	0.24	0.00
45.61	1.70	8.33	2.12	10.02	0.17	23.06	6.98	1.33	0.32	0.16	0.00
45.50	1.54	8.17	1.60	10.44	0.17	23.87	6.79	1.28	0.31	0.15	0.00
49.27	3.30	12.10	1.77	9.89	0.17	10.46	9.65	2.25	0.65	0.30	0.00
46.53	1.99	9.49	2.16	9.79	0.18	19.28	8.18	1.54	0.38	0.18	0.11
48.12	2.34	11.43	2.26	9.46	0.18	13.65	9.87	1.89	0.46	0.22	0.04
47.93	2.32	11.18	2.46	9.36	0.18	14.33	9.64	1.86	0.45	0.21	0.02
46.96	2.01	9.90	2.13	9.72	0.18	18.31	8.58	1.58	0.37	0.19	0.00
49.16	2.73	12.54	1.83	10.02	0.18	10.05	10.55	2.09	0.56	0.26	0.00
48.41	2.47	11.80	2.81	8.91	0.18	12.52	10.18	1.93	0.48	0.23	0.00
47.90	2.24	11.17	2.41	9.36	0.18	14.64	9.58	1.82	0.41	0.21	0.01
48.45	2.35	11.64	1.04	10.37	0.18	13.23	10.13	1.89	0.45	0.23	0.00
48.98	2.48	12.05	1.39	10.17	0.18	11.18	10.83	1.73	0.80	0.24	0.01
48.74	2.44	11.60	1.38	10.18	0.18	12.35	10.45	1.67	0.79	0.23	0.01
49.61	3.03	12.91	1.60	9.68	0.17	8.84	10.96	2.24	0.55	0.27	0.01
49.20	2.50	12.32	1.26	10.13	0.18	10.51	11.05	2.02	0.48	0.23	0.01

subcomposition chosen is represented in Figure 7, both in a ternary diagram and as a scatterplot of the two ilr scores. The SVD scores are given in Table 7.2. They are needed to perform the following tests for joint normality:

- Kolmogorov-Smirnoff test of standard normality on each of the two marginal scores, i.e. with zero mean and unit standard deviation;



- Kolmogorov-Smirnoff test of uniformity on the argument of the $\mathbf{U}$ scores within the interval $(0, 2\pi)$;
- Kolmogorov-Smirnoff test that the (square) module of $\mathbf{U}$ rows follow a $\chi^2(2)$ distribution.

Recall that the $\mathbf{U}$ scores are normalized scores obtained from the SVD as described in Section 7.2.

The significance of each of these tests (Table 7.2) is lower than 5%, thus suggesting that the data do not follow a normal distribution. However, one has to take into account that these tests should actually be applied to large samples, or to cases where the mean and the variance of the data are known. If this is not the case, other tests are better suited. In particular, the energy test implemented in R (packages “compositions” and “energy”) delivers the statistic $E = 0.8148$, with an associated p -value = 0.1171, which is high enough to accept joint normality.

Advanced exercise.

Using the same energy test, the normality of subcompositions

$$[\text{Fe}_2\text{O}_3, \text{FeO}, \text{MgO}] \quad \text{and} \quad [\text{Fe}_2\text{O}_3, \text{FeO}, \text{MnO}, \text{MgO}]$$

have, respectively, p -values of 0.2222 and 0.004004. Thus, the energy test suggests non-normality to be concentrated in the ratio MgO/MnO .

Table 7.2. Singular value decomposition of the subcomposition ($\text{Fe}_2\text{O}_3, \text{FeO}, \text{MnO}$) of the Kilauea data set, and transformations needed to test joint normality, as well as the corresponding statistics and p -values.

	u_{i1}	u_{i2}	$\tan^{-1}(u_{i2}, u_{i1})$	$u_{i2}^2 + u_{i1}^2$
1	-0.1327	-0.0873	-2.5595	0.0252
2	0.1789	0.1532	0.7083	0.0555
3	0.1348	0.5145	1.3145	0.2829
4	-0.1277	0.5384	1.8037	0.3062
5	-0.0134	0.2964	1.6161	0.0880
6	0.1427	0.0081	0.0568	0.0204
7	0.1990	-0.1598	-0.6766	0.0651
8	0.2772	-0.1567	-0.5143	0.1014
9	0.1343	-0.0453	-0.3254	0.0201
10	-0.0115	0.0164	2.1802	0.0004
11	0.4165	-0.3449	-0.6916	0.2924
12	0.2596	-0.1727	-0.5869	0.0972
13	-0.5141	-0.2209	-2.7358	0.3131
14	-0.2552	-0.1100	-2.7345	0.0772
15	-0.2619	-0.1098	-2.7444	0.0806
16	-0.0892	0.0904	2.3496	0.0161
17	-0.3375	-0.2100	-2.5850	0.1580
D	0.3385	0.3651	0.6380	0.8551
p -value	0.0306	0.01543	3e-07	1e-14

Exercise 53

Build the subcomposition ($\text{Fe}_2\text{O}_3, \text{FeO}, \text{MgO}$) from the Kilauea data of Table 5.1. Split it into two groups (at your will, or better, at random) and perform the different tests of Chapter 7 on the centres and covariance structures.

Answer of exercise 53

The data (reproduced in Table 7.1) was split randomly in two groups: $A = \{2, 7, 8, 9, 11, 12, 17\}$ and $B = \{1, 3, 4, 5, 6, 10, 13, 14, 15, 16\}$. Taking a basis defined by the matrix

$$\Psi = \begin{pmatrix} -\sqrt{\frac{1}{2}} & +\sqrt{\frac{1}{2}} & 0 \\ -\sqrt{\frac{1}{6}} & -\sqrt{\frac{1}{6}} & +\sqrt{\frac{4}{6}} \end{pmatrix},$$

the ilr means and covariance matrices of each group are

$$\hat{\mu}_A = \begin{pmatrix} 1.0462714 \\ 0.8855813 \end{pmatrix}, \quad \hat{\mu}_B = \begin{pmatrix} 1.293366 \\ 1.003183 \end{pmatrix},$$

$$\hat{\Sigma}_A = \begin{pmatrix} 0.0365 & 0.0047 \\ 0.0047 & 0.0151 \end{pmatrix}, \quad \hat{\Sigma}_B = \begin{pmatrix} 0.0242 & -0.0001 \\ -0.0001 & 0.0625 \end{pmatrix}.$$

The pooled and the combined covariance matrices are

$$\hat{\Sigma}_p = \frac{7}{17} \hat{\Sigma}_A + \frac{10}{17} \hat{\Sigma}_B = \begin{pmatrix} 0.0293 & 0.0019 \\ 0.0019 & 0.0429 \end{pmatrix},$$

$$\hat{\Sigma}_c = \begin{pmatrix} 0.0415 & 0.0084 \\ 0.0084 & 0.0436 \end{pmatrix}.$$

Remember that all these variances are computed using the denominator N , i.e. no bias correction was applied. The corresponding log-determinants of all these covariance matrices are

$$\ln(\hat{\Sigma}_A) = -7.55, \quad \ln(\hat{\Sigma}_B) = -6.49, \quad \ln(\hat{\Sigma}_p) = -6.68, \quad \ln(\hat{\Sigma}_c) = -6.36.$$

Testing the first two hypotheses of Section 7.3, the following results are obtained,

$$\begin{aligned} Q_{1vs4} &= 9.72098, & \Delta\nu &= 5, & p\text{-value} &= 0.083, \\ Q_{2vs4} &= 4.18556, & \Delta\nu &= 3, & p\text{-value} &= 0.242, \end{aligned}$$

which lead to accept equality of means and variances (the simplest hypothesis) at a 5% confidence. With 10% confidence equality of variances can be accepted, but means might differ (hypothesis 2).

Exercise 54

Consider the subcomposition (MnO, MgO, CaO) from the Kilauea data of Table 5.1. Split it into two groups depending on whether CO₂ was zero or not, and perform the different tests of normality. If no strong evidence against normality is found, test the equality of centres and covariance structures.

Answer of exercise 54

Splitting the sample gives 9 observations with nonzero CO₂ (group *A*) and 8 observations with zero CO₂ (group *B*), with respective compositional means and ilr-covariance matrices

$$\hat{\mathbf{g}}_A = \begin{pmatrix} 0.008 \\ 0.558 \\ 0.434 \end{pmatrix}, \quad \hat{\mathbf{g}}_B = \begin{pmatrix} 0.007 \\ 0.611 \\ 0.382 \end{pmatrix},$$

$$\hat{\Sigma}_A = \begin{pmatrix} 0.0201 & -0.0227 \\ -0.0227 & 0.0257 \end{pmatrix}, \quad \hat{\Sigma}_B = \begin{pmatrix} 0.0569 & -0.0628 \\ -0.0628 & 0.0705 \end{pmatrix}.$$

The ilr basis used for this computation was a balance of MgO vs. MnO in the first coordinate, and a balance of CaO vs. the other two in the second coordinate. Proceeding to a SVD of each of the two supopulations, each centered with its own geometric center, the resulting $\mathbf{V}$ matrices were respectively

$$\mathbf{V}_A = \begin{pmatrix} 0.815 & 0.043 \\ -0.445 & 0.684 \\ -0.370 & -0.728 \end{pmatrix}, \quad \mathbf{V}_B = \begin{pmatrix} 0.812 & 0.082 \\ -0.477 & 0.663 \\ -0.336 & -0.744 \end{pmatrix}.$$

On these coordinates, the corresponding normalized $\mathbf{U}$ scores (see Section 7.2) were applied the battery of tests of Section 7.1, which gave the following p -values

test	group <i>A</i>	group <i>B</i>
marginal u_1	0.0006	0.0012
marginal u_2	0.2749	0.2153
angle	0.0063	0.0015
radius	9e-06	5e-06
average	0.0704	0.0545

suggesting that global normality is in both groups barely acceptable, because of the first coordinate. Nevertheless, given the limitations of these tests, we consider the average p -value large enough to assume joint normality and test now equality of means and variances. For this we require the pooled and the combined variances,

$$\hat{\Sigma}_p = \frac{9}{17} \hat{\Sigma}_A + \frac{8}{17} \hat{\Sigma}_B = \begin{pmatrix} 0.0374 & -0.0416 \\ -0.0416 & 0.0468 \end{pmatrix}, \quad \hat{\Sigma}_c = \begin{pmatrix} 0.0376 & -0.0420 \\ -0.0420 & 0.0475 \end{pmatrix}$$

and the logarithms of the determinants of the 4 matrices:

$$\ln(\hat{\Sigma}_A) = -13.55, \quad \ln(\hat{\Sigma}_B) = -9.70, \quad \ln(\hat{\Sigma}_p) = -10.80, \quad \ln(\hat{\Sigma}_c) = -10.82$$

Thus, following the lattice of hypotheses of Section 7.3, from the simplest to the most complex, the statistics, differences of degrees of freedom, and p -values, are:

$$\begin{aligned} Q_{1vs4} &= 15.6732, & \Delta\nu &= 5, & p\text{-value} &= 0.0078, \\ Q_{2vs4} &= 15.9974, & \Delta\nu &= 3, & p\text{-value} &= 0.0011, \end{aligned}$$

which suggest that equality of variances is not admissible. That brings us to test the third hypothesis (equal means and different variances), which seems also reasonable given the statistics of each group. Iterating 200 times the sequence proposed in Section 7.3 point 3, a joint mean and respective variances (in ilr) are obtained:

$$\hat{\mu}_h = \begin{pmatrix} 3.022 \\ 1.542 \end{pmatrix}, \quad \hat{\Sigma}_{Ah} = \begin{pmatrix} 0.0212 & -0.0240 \\ -0.0216 & 0.0245 \end{pmatrix}, \quad \hat{\Sigma}_{Bh} = \begin{pmatrix} 0.1581 & -0.1849 \\ 0.0384 & -0.0516 \end{pmatrix}$$

This gives rise to a statistic

$$Q_{3vs4} = 16.82, \quad \nu = 2, \quad p\text{-value} = 0.00022,$$

showing that no mean or variance can be considered equal between both groups.

Exercise 55

Compute and plot a 95% confidence region for the ilr transformed mean of the data from table 2.1 in $\mathbb{R}^2$. Suggestion: use appendix A.2.

Answer of exercise 55

To draw this ellipse, we choose to work with coordinates obtained with the sign table

	x_1	x_2	x_3
x_1^*	-1	+1	0
x_2^*	-1	-1	+1

On the corresponding coordinates, the estimated ilr mean vector and covariance matrix are

$$\hat{\boldsymbol{\mu}} = \begin{pmatrix} -0.0088 \\ -0.2603 \end{pmatrix}, \quad \hat{\Sigma} = \begin{pmatrix} 1.0021 & -0.0524 \\ -0.0524 & 0.5744 \end{pmatrix}, \quad |\hat{\Sigma}| = 0.5729$$

To build a confidence region for $\boldsymbol{\mu}$ around $\hat{\boldsymbol{\mu}}$, Eq. (7.8) states that a radius is needed,

$$\kappa = \frac{D-1}{n-D+1} c$$

with c the 95% quantile of the Fisher $\mathcal{F}$ distribution with $(D-1, n-D+1)$ degrees of freedom. In this case, with $n = 20$ and $D = 3$, we obtain $c = 3.5546$ and $\kappa = 0.3950$. Using then Eq. (A.3), the desired region can be written as

$$\begin{aligned} \mu_1 &= -0.0088 - \sqrt{\frac{0.5729}{0.5744}} \sqrt{0.3950} \cos \theta + \frac{0.0524}{\sqrt{0.5744}} \sqrt{0.3950} \sin \theta \\ \mu_2 &= -0.2603 - \sqrt{0.5744} \sqrt{0.3950} \sin \theta, \end{aligned}$$

parametrized by an angle θ within the interval $(0, 2\pi)$. This is evaluated in Table 7.3 each 30° .

Table 7.3. Values of a 95% confidence ellipse on the mean of the ilr-transformed data of Table 2.1.

$180\frac{\theta}{\pi}$	0°	30°	60°	90°	120°	150°
μ_1	-0.636	-0.524	-0.273	0.048	0.355	0.563
μ_2	-0.260	-0.498	-0.673	-0.737	-0.673	-0.498
$180\frac{\theta}{\pi}$	180°	210°	240°	270°	300°	330°
μ_1	0.619	0.506	0.255	-0.066	-0.372	-0.581
μ_2	-0.260	-0.022	0.152	0.216	0.152	-0.022

Exercise 56

Transform the confidence region of exercise 55 back into the ternary diagram using the ilr^{-1} .

Answer of exercise 56

The ilr transformation used in the preceding exercise was based on the matrix

$$\Psi = \begin{pmatrix} -\sqrt{\frac{1}{2}} & +\sqrt{\frac{1}{2}} & 0 \\ -\sqrt{\frac{1}{6}} & -\sqrt{\frac{1}{6}} & +\sqrt{\frac{4}{6}} \end{pmatrix}.$$

Applying the inverse ilr transformation with this matrix to each pair (μ_1, μ_2) of the preceding exercise, i.e. for the example values of Table 7.3, produces the results in Table 7.4.

Table 7.4. Values of a 95% confidence ellipse on the geometric center of the dataset of Table 2.1.

$180\frac{\theta}{\pi}$	0°	30°	60°	90°	120°	150°
g_1	0.535	0.540	0.490	0.401	0.311	0.248
g_2	0.217	0.257	0.333	0.430	0.514	0.551
g_3	0.248	0.203	0.177	0.169	0.175	0.201
$180\frac{\theta}{\pi}$	180°	210°	240°	270°	300°	330°
g_1	0.221	0.225	0.258	0.317	0.397	0.480
g_2	0.530	0.461	0.370	0.289	0.235	0.211
g_3	0.249	0.314	0.372	0.394	0.368	0.310

Exercise 57

Compute and plot a 90% probability region for the ilr transformed data of table 2.1 in $\mathbb{R}^2$, together with the sample. Express the ellipse back in compositions and represent it with the data in a ternary diagram. Use the χ^2 distribution, as if mean and variance were actually known.

Answer of exercise 57

Using the same ilr basis of the preceding two exercises, we obtain the same means and variances, and the equation of the ellipse is also the same, just by using a radius κ defined after a χ^2 distribution with 2 degrees of freedom. This gives $\kappa = \chi_{0.90}^2(2) = 4.6052$, and a parametric equation

$$\begin{aligned}\mu_1 &= -0.0088 - \sqrt{\frac{0.5729}{0.5744}}\sqrt{4.6052}\cos\theta + \frac{0.0524}{\sqrt{0.5744}}\sqrt{4.6052}\sin\theta; \\ \mu_2 &= -0.2603 - \sqrt{0.5744}\sqrt{4.6052}\sin\theta.\end{aligned}$$

Figure 7.1 shows the resulting diagrams in ilr-space and as compositions. This diagram also represents twelve points over the ellipse, following the parametric equation with θ each 30° .

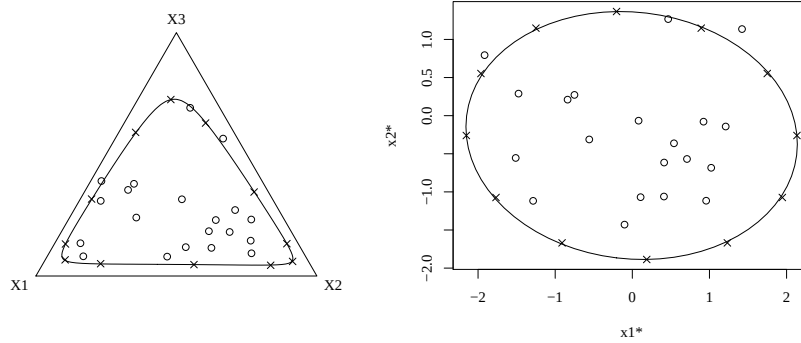


Fig. 7.1. Ilr and ternary diagram representation of the data of Table 2.1, together with a 90% probability region based on the χ^2 distribution. The crosses represent twelve points along this ellipse.

Exercise 58

Simulate two subpopulations, of 50 observations each, of a 3-part random composition. In a basis built with the balances

$$x_1^* = \sqrt{\frac{1}{2}} \ln \frac{x_2}{x_1 x_3}, \quad x_2^* = \sqrt{\frac{1}{2}} \ln \frac{x_3}{x_1}$$

these two subpopulations have the same variability and different means:

$$\boldsymbol{\mu}_A = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad \boldsymbol{\mu}_B = \begin{pmatrix} +1 \\ 0 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 4 \end{pmatrix}.$$

Using the simulated data, test that compositional normality fits each subpopulation. Apply the chain of tests of equal mean and variance, and verify that the hypothesis of equal variances and different means would not be rejected. Plot the two subsets in a ternary diagram (with different colors) and add their respective confidence regions, using the pooled variance.

Answer of exercise 58

After the expressions in appendix A.2, the Cholesky decomposition of $\boldsymbol{\Sigma}$ is

$$\mathbf{U} \cdot \mathbf{U}^\top = \begin{pmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{pmatrix} \begin{pmatrix} u_{11} & 0 \\ u_{12} & u_{22} \end{pmatrix} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{12} & \Sigma_{22} \end{pmatrix} = \boldsymbol{\Sigma},$$

with elements

$$\begin{aligned} u_{22} &= \sqrt{\Sigma_{22}} = \sqrt{4} = 2, \\ u_{12} &= \frac{\Sigma_{12}}{\sqrt{\Sigma_{22}}} = \frac{-1/2}{\sqrt{4}} = -\frac{1}{4}, \\ u_{11} &= \sqrt{\frac{\Sigma_{11}\Sigma_{22} - \Sigma_{12}^2}{\Sigma_{22}}} = \sqrt{\frac{4 - \frac{1}{2^2}}{4}} = \sqrt{\frac{15}{16}}. \end{aligned}$$

Suggestion: check that $\mathbf{U} \cdot \mathbf{U}^\top = \boldsymbol{\Sigma}$ before proceeding.

After exercise 49, a normal simulation with the desired mean and variance matrix can be obtained with the expression $\mathbf{x} = \boldsymbol{\mu} + \mathbf{z} \cdot \mathbf{U}^\top$, from a vector $\mathbf{z}$ corresponding to two standard uncorrelated normal variables. Applying this 50 times with the mean $(-1, 0)$ and 50 times with the mean $(+1, 0)$, the simulated data set shown in Figure 7.2 is obtained. Figure 7.2 shows also confidence regions for each mean, constructed using once the respective estimated means and variance-covariance matrices,

$$\begin{aligned} \boldsymbol{\mu}_A &= \begin{pmatrix} -1.0209 \\ -0.0259 \end{pmatrix}, \quad \boldsymbol{\mu}_B = \begin{pmatrix} 0.9631 \\ 0.0255 \end{pmatrix}, \\ \boldsymbol{\Sigma}_A &= \begin{pmatrix} 0.8327 & -0.7319 \\ -0.7319 & 4.4703 \end{pmatrix}, \quad \boldsymbol{\Sigma}_B = \begin{pmatrix} 1.2184 & -0.3909 \\ -0.3909 & 3.0871 \end{pmatrix}, \end{aligned}$$

and once the pooled variance-covariance matrix,

$$\boldsymbol{\Sigma}_B = \begin{pmatrix} 1.0256 & -0.5614 \\ -0.5614 & 3.7787 \end{pmatrix}.$$

Note that these values are affected by the sampling randomness, different from the theoretical one given in the exercise and from those obtained by the reader. They must nevertheless be approximately equal. We can thus proceed to assess how reliable is the hypothesis of equal means and equal variances, and the hypothesis of equal variances and different means (the *true* ones), for which the log-determinants of all variances are needed,

$$\ln(\hat{\Sigma}_A) = 1.1590, \quad \ln(\hat{\Sigma}_B) = 1.2833, \quad \ln(\hat{\Sigma}_p) = 1.2698, \quad \ln(\hat{\Sigma}_c) = 1.9890.$$

With these, the following statistics and p -values for the several hypotheses are obtained:

$$\begin{aligned} Q_{1vs4} &= 76.7827, & \Delta\nu &= 5, & p\text{-value} &= 4e-15, \\ Q_{2vs4} &= 4.8632, & \Delta\nu &= 3, & p\text{-value} &= 0.1821, \end{aligned}$$

showing that means cannot be considered equal, but variance should be taken as such.

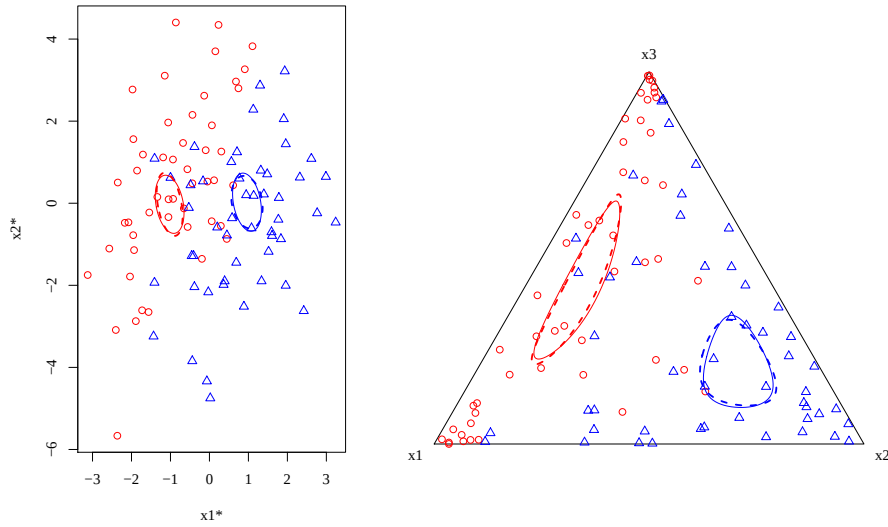


Fig. 7.2. Ilr and ternary diagram representation of simulated data, distinguishing between two groups. Confidence regions for the two means are included, assuming different variances for each group, and a common variance (the pooled variance).

Exercise 59

Consider the data in Table 8.7. They are sand-silt-clay compositions from an Arctic lake taken at different depths (adapted from Coakley and Rust (1968) and reproduced in Aitchison (1986)). The goal is to check whether there is some trend in the composition related to depth. Particularly, using the standard hypothesis testing in regression, check the constant and the straight line models

$$\hat{\mathbf{x}}(\mathbf{t}) = \beta_0, \quad \hat{\mathbf{x}}(\mathbf{t}) = \beta_0 \oplus (t \odot \beta_1),$$

being $t = \text{depth}$. Plot both models, the fitted model and the residuals, in coordinates and the ternary diagram. Try also with depth logarithmically transformed.

Table 8.6. (Table 8.7) Sand, silt, clay composition of sediment samples at different water depths in an Arctic lake.

sample no.	sand	silt	clay	depth (m)	sample no.	sand	silt	clay	depth (m)
1	77.5	19.5	3.0	10.4	21	9.5	53.5	37.0	47.1
2	71.9	24.9	3.2	11.7	22	17.1	48.0	34.9	48.4
3	50.7	36.1	13.2	12.8	23	10.5	55.4	34.1	49.4
4	52.2	40.9	6.9	13.0	24	4.8	54.7	40.5	49.5
5	70.0	26.5	3.5	15.7	25	2.6	45.2	52.2	59.2
6	66.5	32.2	1.3	16.3	26	11.4	52.7	35.9	60.1
7	43.1	55.3	1.6	18.0	27	6.7	46.9	46.4	61.7
8	53.4	36.8	9.8	18.7	28	6.9	49.7	43.4	62.4
9	15.5	54.4	30.1	20.7	29	4.0	44.9	51.1	69.3
10	31.7	41.5	26.8	22.1	30	7.4	51.6	41.0	73.6
11	65.7	27.8	6.5	22.4	31	4.8	49.5	45.7	74.4
12	70.4	29.0	0.6	24.4	32	4.5	48.5	47.0	78.5
13	17.4	53.6	29.0	25.8	33	6.6	52.1	41.3	82.9
14	10.6	69.8	19.6	32.5	34	6.7	47.3	46.0	87.7
15	38.2	43.1	18.7	33.6	35	7.4	45.6	47.0	88.1
16	10.8	52.7	36.5	36.8	36	6.0	48.9	45.1	90.4
17	18.4	50.7	30.9	37.8	37	6.3	53.8	39.9	90.6
18	4.6	47.4	48.0	36.9	38	2.5	48.0	49.5	97.7
19	15.6	50.4	34.0	42.2	39	2.0	47.8	50.2	103.7
20	31.9	45.1	23.0	47.0					

Answer of exercise 59

To represent the composition in one ilr basis use, for instance, the balance x_1^* of silt to sand and the balance x_2^* of clay to (sand, silt). Any other ilr basis will work analogously. On the transformed scores, any software allowing for a regression fit can be used. Then, in the above coordinates,

- the constant model $\hat{\mathbf{x}}(t) = \beta_0$ is $\hat{x}_1^*(t) = 0.8152$ and $\hat{x}_2^*(t) = -0.1660$,
- the model $\hat{\mathbf{x}}(\mathbf{t}) = \beta_0 \oplus (t \odot \beta_1)$, with a linear dependence on depth t , is

$$\begin{aligned}\hat{x}_1^*(t) &= -0.45938 + 0.02653 \, t \\ \hat{x}_2^*(t) &= -1.86726 + 0.03542 \, t;\end{aligned}$$

both slopes are positive, hence both coordinates increase with depth, i.e. silt tends to dominate over sand as clay does over the other two (the deeper the sediment, the finer it tends to be); the significance of the coefficients is reported in Table 8.7 (all slopes have p -values smaller than 0.0005)

		Estimate	Std. Error	norm. Est.	p -value
x_1^*	(Intercept)	-0.459	0.167	-2.744	0.009
	t	0.027	0.003	8.790	0.000
x_2^*	(Intercept)	-1.867	0.263	-7.094	0.000
	t	0.035	0.005	7.462	0.000

Table 8.7. Significance of the coefficients of the model $\hat{\mathbf{x}}(\mathbf{t}) = \beta_0 \oplus (t \odot \beta_1)$.

- the model $\hat{\mathbf{x}}(\mathbf{t}) = \beta_0 \oplus (\ln(t) \odot \beta_1)$, with a linear dependence on the logarithm of depth t , is

$$\begin{aligned}\hat{x}_1^*(t) &= -3.459 + 1.164 \, \ln(t) \\ \hat{x}_2^*(t) &= -5.912 + 1.565 \, \ln(t);\end{aligned}$$

both slopes are again positive, supporting the same interpretation that deeper sediments tend to be finer; Table 8.8 shows that all slopes are significant (p -values smaller than 0.0005).

		Estimate	Std. Error	norm. Est.	p -value
x_1	(Intercept)	-3.459	0.394	-8.773	0.000
	$\ln(t)$	1.164	0.106	11.020	0.000
x_2	(Intercept)	-5.912	0.643	-9.197	0.000
	$\ln(t)$	1.565	0.172	9.087	0.000

Table 8.8. ANOVA table for the model $\hat{\mathbf{x}}(\mathbf{t}) = \beta_0 \oplus (\ln(t) \odot \beta_1)$.

These three models are represented in the three columns of Figure 8.4. The first two rows of figures show classical scatterplots of one coordinate vs. the explanatory variable. The second two plots show a “phase diagram”, i.e. the

two coordinates (or the composition), the explanatory variable not being explicit. In each diagram, circles are data, the model is a red line (or a dot) and the residuals are shown as dashed grey lines joining observed with predicted values. From the significance of the two models with an explanatory variable and from the diagrams we can conclude that the first model is not a good choice.

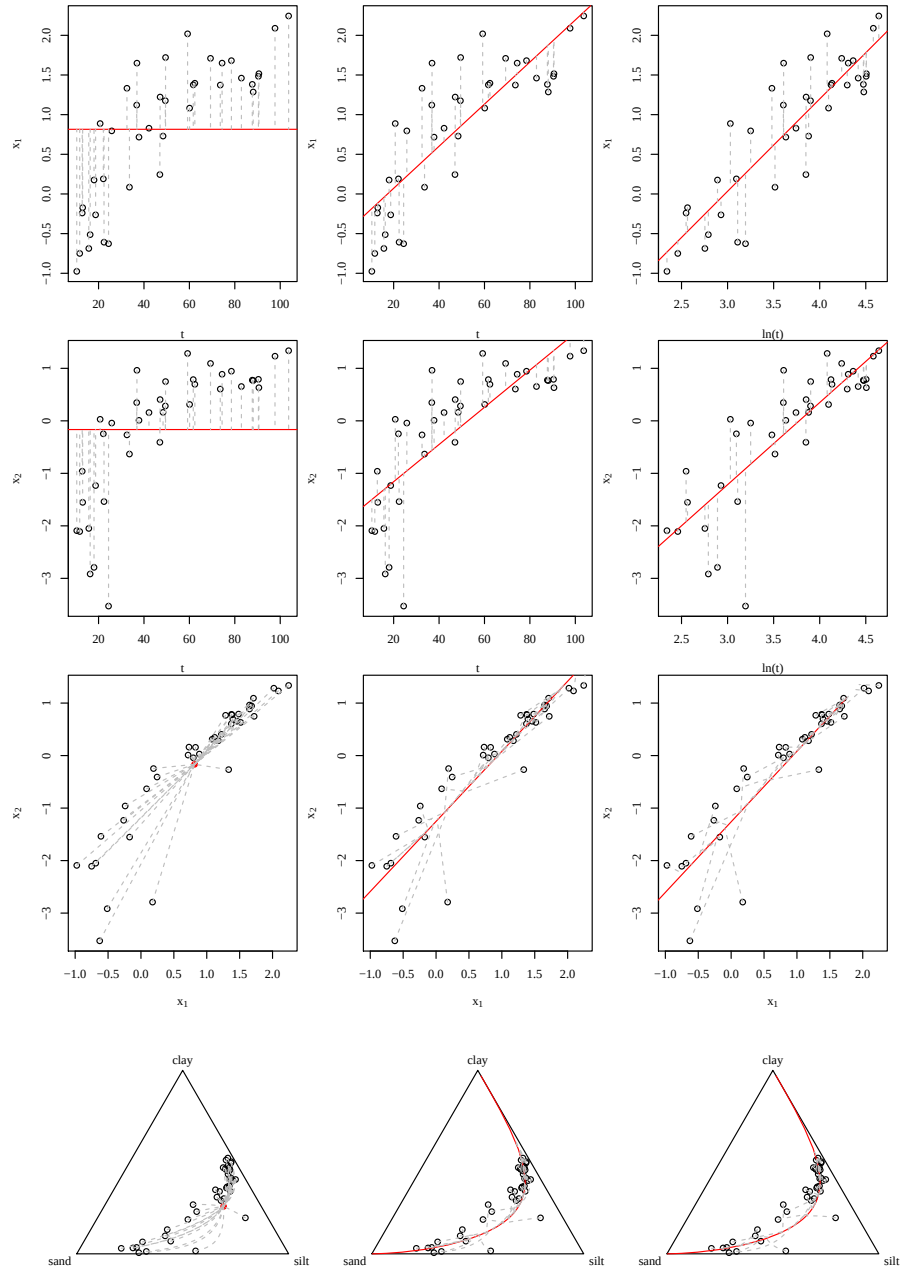


Fig. 8.4. Models fitted in exercise 59. Left column of plots: constant model; middle column: model with depth t as explanatory variable; right column: model with depth $\ln t$ as explanatory variable. First two rows of plots: coordinate values (circles), coordinate residuals (dashed grey lines), and model (red line) as a function of t or $\ln t$. Third and fourth row of plots (phase diagrams) in coordinates and in ternary diagrams (color-line conventions as previous rows).

Exercise 60

Following Exercise 59, one can equivalently check whether the sediment composition brings any information on the depth at which that sample was taken. Using the data from Table 8.7, fit a linear model to explain depth as a function of the composition. Analyze the residuals as usual, as they may be considered real values.

To display the model, you can follow these steps. Split the range of observed depths in several segments of the same length (four to six will be enough in CoDaPack), and give each sample a number corresponding to its depth category. Plot the compositional data in a ternary diagram, using colors for each depth interval. Draw a line on the simplex, from the center of the data set along the gradient of the fitted model.

Answer of exercise 60

This exercise builds upon some results of the preceding one. Using the same ilr basis to represent the sand-silt-clay composition, given the slight curvature shown in several of the first panels of Figure 8.4, it seems reasonable to consider a log-transformed depth (the interested reader can check that the results with a non-transformed depth are marginally worse). Hence, in the following equation, the dependent variable y is log-depth and the explanatory composition is expressed in the balances x_1^* of silt to sand and x_2^* of clay to (sand, silt),

$$y = \hat{y} + \epsilon, \quad \hat{y} = \beta_0 + \beta_1^* x_1^* + \beta_2^* x_2^*.$$

To fit this model, we use function “lm” from a standard R distribution, but equivalent results can be obtained with any multivariate statistics software. The first model is $\hat{y} = 3.2491 + 0.5368 x_1^* + 0.0936 x_2^*$. However, it is always convenient to check that the hypothesis of the regression are satisfied, which can be done with a look at the four diagrams of Figure 8.5, showing that the residuals do not have an apparent non-linear dependence (Residuals vs Fitted), that they are normally distributed (Normal Q-Q) and of homogeneous variance (Scale-Location), but that two leverage points or outliers might be distorting the model (Residuals vs Leverage, samples 7 and 12). Removing these samples and producing the diagnostic plots again (not shown), it is decided that sample 6 can be a leverage point as well. Repeating the analysis for the data set without these three data points, the coefficients in Table 8.9, with their significance, are obtained. Thus, the final model can be written as

$$\hat{y} = 3.6324 + 0.0611 x_1^* + 0.5541 x_2^*.$$

Figure 8.6 compares both models, showing how influential can a leverage point be. Each of the two lines shown represents a straight line on the simplex passing through the center of the data set $\hat{\mathbf{g}} = [0.1780, 0.5636, 0.2584]$ and with leading vector $\text{ilr}^{-1}[0.0611, 0.5541]$ (solid line, no leverage points), respectively $\text{ilr}^{-1}[0.5368, 0.0936]$ (dashed line, all data). The first model could even be simplified by removing the influence of x_2^* (p -value 0.3843, not shown here),

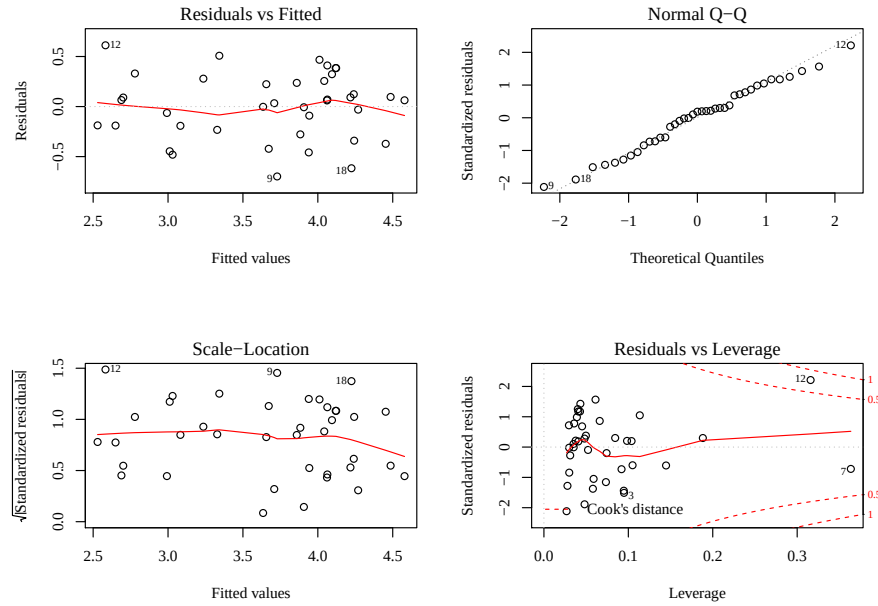


Fig. 8.5. Diagnostic plots of the model of exercise 60.

	Estimate	Std. Error	norm. Est.	<i>p</i> -value
(Intercept)	3.6324	0.2198	16.5266	0.0000
x_1^*	0.0611	0.2529	0.2414	0.8107
x_2^*	0.5541	0.2217	2.4998	0.0176

Table 8.9. Coefficients and their significance for model $y = \beta_0 + \beta_1^* x_1^* + \beta_2^* x_2^*$.

while in the second model it is the influence of x_1^* (p -value 0.8107) which is not significant! As in non-compositional case studies, it is of an extreme importance to remove leverage points in a regression model before doing any test. Note that the curves drawn do not go along the data. This is not an error. Each of these curves shows the gradient of *log-depth*, that is, the direction on the simplex along which *log-depth* is forced to vary the most according to the model. That is not necessarily the direction on the simplex where the *composition* shows largest variations (the first principal component, which is shown as a dashed black line for the sake of comparison).

A last consideration is about the appropriateness of using standard regression to predict depth from the sand-silt-clay composition. In standard regression, the response measurements are assumed to be affected by random

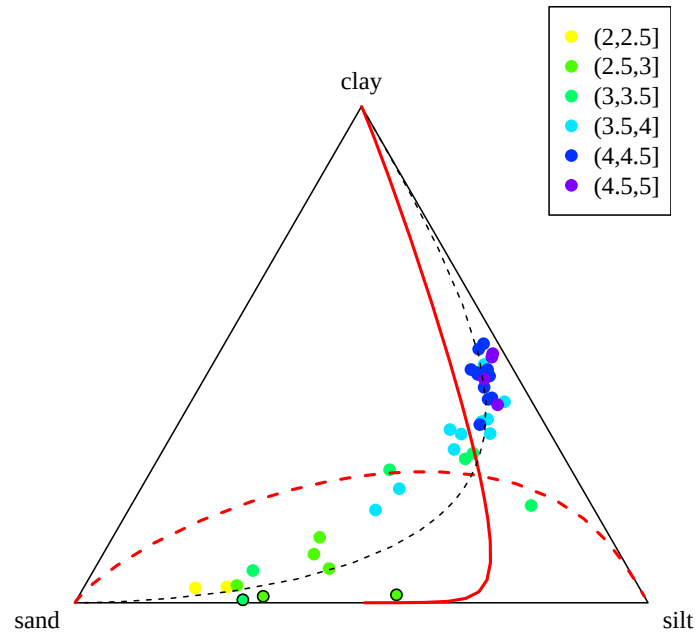


Fig. 8.6. Fitted models (red lines) and data (colored dots) from exercise 60; as shown in the legend, data points are colored using equally spaced intervals of the predicted variable, log-depth. The dashed red line is the model fitted using the complete data set, the solid red line is the model after removal of three outliers/leverage points (these points are circled), and the dashed black line is the first principal component of the sand-silt-clay composition.

deviations, while explanatory variables are considered free of measurement errors which are not taken into account. In the present case, depth or log-depth (response variables) are likely to be less affected by measurement errors than the composition (explanatory variables), and regression techniques considering both types of deviations may produce better models.

Exercise 61

Following Example 8.3, we want to choose a basis that allows for the maximal reduction on the number of coordinates necessary for prediction. For this goal,

- write down the matrix Ψ of 3 rows and 4 columns, that was used to calculate the balances of Table 8.4;
- take the vector of ilr slope coefficients $\beta^* = [-53.132, 26.715, 6.138]$ and express them as clr slope coefficients, that is, calculate $\beta^* \Psi = \beta_{clr}^*$;
- considering the signs and magnitudes of the resulting clr coefficients, choose a hierarchical partition isolating first the quasi-zero components from the other components and then splitting the nonzero components between positive and negative components; within each group (quasi-zero, positive, negative), you can choose how to split the components; write the associated contrast matrix Ψ' ;
- calculate the coordinates with respect to this basis;
- fit the model again using the new set of coordinates as explanatory variables; test the significance of each logratio;
- recast the clr-coefficients to the new set of balances, with $\beta_{clr}^* \cdot (\Psi')^\top$; compare with the slopes obtained in the preceding point.

Given these results (in particular, the several p -values), can any component be considered to be superfluous to predict the year?

Advanced exercise:

Consider the possibility to restrict the dependence of year to one single coordinate. Which coordinate should that be? Build the corresponding model, and calculate the predicted times $\hat{y}(\mathbf{x})$. Calculate the SSE (Equation 8.6) using these predictions and the predictions provided by the general model (without simplifications, the one with the three slopes of the main exercise). Denote these two SSEs as s_0^2 and s_g^2 . Using the generalized likelihood ratio test of Equation (7.6), test the hypothesis that the two models are equivalent, that is, obtain the logratio

$$Q(\mathbf{x}, y) = 2 \ln \frac{s_0^2}{s_g^2}.$$

Obtain the p -value of Q according to a χ^2 distribution of 2 degrees of freedom, the number of restrictions introduced passing from a model with three predictor coordinates to one coordinate. This can be done, for example, in excel with the function CHIINV or in R with function `pchisq(..., lower.tail=FALSE)`. Given this p -value, is there any reason to reject the model with one single predictor coordinate?

Answer of exercise 61

The logcontrast matrix used to obtain the results of Table 8.4 is

$$\Psi = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} & 0 \\ 1/\sqrt{12} & 1/\sqrt{12} & 1/\sqrt{12} & -3/\sqrt{12} \end{pmatrix},$$

considering the parts “Catholic”, “Evangelist”, “Other” and “No religion” in this order. The product $\beta^* \Psi = \beta_{clr}^*$ yields

$$[-24.8920, 50.2479, -20.0406, -5.3153].$$

With these results, we decide to use a hierarchical partition to split “No religion” from the other first, then “Evangelists” from the other two and finally a balance between “Catholic” and “Other”. This yields the logcontrast matrix:

$$\Psi' = \begin{pmatrix} 1/\sqrt{12} & 1/\sqrt{12} & 1/\sqrt{12} & -3/\sqrt{12} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} & 0 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} & 0 \end{pmatrix},$$

Fitting a model to the data transformed by the ilr linked to this basis (table

Table 8.10. Coordinates of the proportions of religions in Brazil obtained using the log-contrast Ψ' (see Exercise 61 for explanation).

year	x'_1	x'_2	x'_3
1940	2.13	0.99	3.38
1950	1.79	0.76	3.37
1960	2.15	0.72	3.21
1970	1.97	0.50	3.20
1980	1.65	0.25	3.25
1991	0.50	0.01	3.13
2000	0.59	-0.13	2.37

8.10), to predict the year as a function of the religious composition of society, i.e. $\hat{y} - 1940 = \beta_0 + \beta_1^* x'_1 + \beta_2^* x'_2 + \beta_3^* x'_3$ (remember to subtract the starting year to the data to obtain more stable estimates), generates the following estimates (and p -values) for the global model,

	Estimate	Std. Error	norm. Est.	p -value
(Intercept)	57.7165	13.0422	4.4254	0.0214
x'_1	6.1376	3.2924	1.8642	0.1592
x'_2	-59.3709	6.0891	-9.7504	0.0023
x'_3	-3.4305	4.5231	-0.7584	0.5033

These estimates are exactly the same as those obtained multiplying the clr estimate

$$\beta_{clr}^* = [-24.8920, 50.2479, -20.0406, -5.3153]$$

with the new logcontrast basis,

$$\beta_{clr}^* \cdot (\Psi')^\top = [6.1376, -59.3709, -3.4305].$$

It highlights the fact that the regression model does not depend on which basis is used, because the obtained coefficients can be transformed back and forth between the clr and the several ilr representations.

With regard to the possibility to simplify the model, it is now apparent that the last coordinate is superfluous (the p -value is larger than 0.05), hence the logratio between “Catholic” and “Other” to predict the year is not needed. Removing that ilr variable and fitting the model again yields

	Estimate	Std. Error	norm. Est.	p -value
(Intercept)	48.1266	3.0223	15.9238	0.0001
x'_1	5.9317	3.1020	1.9122	0.1284
x'_2	-61.2216	5.2742	-11.6078	0.0003

which allows to suppress the first coordinate. In this approach of variable elimination, one must proceed step-by-step because each of the p -values depends on which other variables are considered (compare the p -values for x'_1 and x'_2 in the two preceding tables). The simplified model contains one single predictive logratio, producing the following estimates and p -values:

	Estimate	Std. Error	norm. Est.	p -value
(Intercept)	53.2681	1.7078	31.1917	0.0000
x'_2	-52.1993	2.9163	-17.8992	0.0000

Thus, a very simplified model for predicting the year as a function of religious proportions can be written as

$$year = 1940 + 53.2681 - 52.1993 \frac{1}{\sqrt{6}} \ln \frac{C \cdot O}{E^2} = 1993.3 - 21.3 \ln \frac{C \cdot O}{E^2}.$$

The advanced exercise proposes a different way to arrive to this simplified model, by jumping from the first to the last model. For this we need to compare the residuals obtained with the two models (table 8.11). Computing the sum of squares of the residuals,

$$\begin{aligned} s_g^2 &= (-0.53)^2 + (-1.91)^2 + 2.75^2 + 0.75^2 + (-1.96)^2 + 1.63^2 + (-0.74)^2 \\ &= 19.14 \\ s_0^2 &= (-1.70)^2 + (-3.48)^2 + 4.25^2 + 2.77^2 + (-0.30)^2 + (-1.66)^2 + 0.11^2 \\ &= 43.65, \end{aligned}$$

yields $Q = 1.6493$ which, for two degrees of freedom, gives a p -value of 0.4. The conclusion of the test is that the global and the simplified models explain the available data comparably well. Applying the principle of parsimony, the

Table 8.11. Predictions and residuals obtained with the starting model and with the simplified model

original data	global model		simplified model	
	fitted values	residuals	fitted values	residuals
0	0.53	-0.53	1.70	-1.70
10	11.91	-1.91	13.48	-3.48
20	17.25	2.75	15.75	4.25
30	29.25	0.75	27.23	2.77
40	41.96	-1.96	40.30	-0.30
51	49.37	1.63	52.66	-1.66
60	60.74	-0.74	59.89	0.11

simplified one can be kept. Readers should be aware that this omnibus test is asymptotic, i.e. valid only when the number of rows N of the data set is very large in comparison with the number of parameters of the model P . This condition is by no means satisfied in this case study. The exercise should be taken as a learning example. Actually, statistical software, able to deal with these regression models, includes also the classical ANOVA table (Table 8.12), which, besides the several sums of squares and numbers of parameters, includes a (non-asymptotic) Fisher test, more convenient for comparing linear models than the asymptotic omnibus χ^2 test. Note, nevertheless, that using any of the two tests we would prefer the simplified model, because the only relevant aspect of a p -values is that it is above 0.05.

Table 8.12.

	$N - P$	s^2	degrees of freedom	Sum of Squares	F	p -value
global model	3	19.14				
simplified model	5	43.65	-2	-24.52	1.92	0.2903

Exercise 62

Following Example 8.4, compute the center (Definition 5.1) of the nutrition data set separately for the Eastern and Western countries. Calculate the perturbation difference between them. Compare the two centers and the perturbation with $\hat{\beta}_1$ and $\hat{\beta}_2$ as obtained in Example 8.4 (note that you will have to apply the inverse clr transformation to obtain these estimates first).

Answer of exercise 62

Table 8.13 reports the required means and the perturbation between them, as well as their clr transformations. Comparing these values (particularly columns 3 and 6) with the results of Example 8.4 (the clr coefficients given in the second table of page 162), it is clear that the parameter $\hat{\beta}_1 = \hat{\mathbf{g}}_E$ corresponds to the estimated mean for the first group (in this case the Eastern countries), while the second parameter vector $\hat{\beta}_2 = \hat{\mathbf{g}}_W \ominus \hat{\mathbf{g}}_E$ is the perturbation between the two groups. This is a general result of ANOVA: the first vector of coefficients usually corresponds to the average response for one of the groups (typically the first one) taken as a sort of *baseline*, while the following vectors show *increments* from that baseline to the other groups, and are always *differences* of means (in the case of compositional data, perturbations between the centers of each group with the mean of the first group).

Table 8.13. Means by groups (Eastern vs. Western countries), their clr and perturbation from East to West

	$\hat{\mathbf{g}}_E$	$\hat{\mathbf{g}}_W$	$\text{clr}(\hat{\mathbf{g}}_E)$	$\text{clr}(\hat{\mathbf{g}}_W)$	$\hat{\mathbf{g}}_W \ominus \hat{\mathbf{g}}_E$	$\text{clr}(\hat{\mathbf{g}}_W \ominus \hat{\mathbf{g}}_E)$
red meat	0.092	0.132	0.402	0.493	0.105	0.091
white meat	0.082	0.089	0.283	0.094	0.079	-0.190
eggs	0.023	0.040	-0.975	-0.696	0.127	0.279
milk	0.144	0.229	0.850	1.043	0.116	0.192
fish	0.015	0.060	-1.427	-0.303	0.295	1.124
cereals	0.522	0.321	2.136	1.380	0.045	-0.756
starch	0.040	0.053	-0.441	-0.427	0.097	0.014
nuts	0.038	0.027	-0.487	-1.079	0.053	-0.592
vegetables	0.044	0.049	-0.341	-0.504	0.082	-0.163

Exercise 63

Repeat Example 8.4 and check for differences between Northern and Southern countries.

Answer of exercise 63

Following the ideas of the preceding exercise, we first generate the coefficients of the model, which will be the center of nutrition patterns for the northern countries as $\hat{\beta}_1$, and the difference between the two centers as $\hat{\beta}_2$ (Table 8.14). In order to test which differences are significant, it is necessary to choose

Table 8.14. Means by groups (Eastern vs. Western countries), their clr and perturbation from East to West

	$\text{clr}(\hat{\mathbf{g}}_N)$	$\text{clr}(\hat{\mathbf{g}}_S)$	$\text{clr}(\hat{\mathbf{g}}_S \ominus \hat{\mathbf{g}}_N)$	$\hat{\beta}_1$	$\hat{\beta}_2$
redmeat	0.477	0.431	-0.046	0.523	-0.046
whitemeat	0.369	-0.206	-0.575	0.944	-0.575
eggs	-0.613	-1.123	-0.510	-0.103	-0.510
milk	1.136	0.685	-0.450	1.586	-0.450
fish	-0.631	-0.843	-0.212	-0.420	-0.212
cereals	1.452	2.007	0.555	0.897	0.555
starch	-0.265	-0.729	-0.464	0.199	-0.464
nuts	-1.299	-0.095	1.204	-2.503	1.204
vegetables	-0.624	-0.127	0.497	-1.122	0.497

an ilr basis which coefficients can be interpreted and have good chances to identify groups of variables that behave in the same way. An option is to choose a basis that groups variables with similar clr coefficient $\hat{\beta}_2$. Hence the partition matrix (Table 8.15) is used to define the basis. We then represent

Table 8.15. Sign code for the proposed SBP.

	RM	WM	E	M	F	C	S	N	V
v_1	0	0	0	-1	0	0	1	0	0
v_2	0	0	0	0	0	-1	0	0	1
v_3	0	0	-1	1	0	0	1	0	0
v_4	0	-1	1	1	0	0	1	0	0
v_5	-1	0	0	0	1	0	0	0	0
v_6	1	-1	-1	-1	1	0	-1	0	0
v_7	0	0	0	0	0	1	0	-1	1
v_8	-1	-1	-1	-1	-1	1	-1	1	1

the composition in the ilr basis associated to this partition matrix, and fit an ANOVA model to each of the coordinates. The results are reported in

Table 8.16, which show that almost all coefficients can be considered to be null. Only differences in v_7 and v_8 are significant, i.e. the diet of southern and northern countries shows no differences within the subcomposition [RM, WM, E, M, F, S] (protein sources plus starch), or within the subcomposition [C, V] (cereals and vegetables): their differences are in the balance between these two subcompositions, and in the proportion of nuts (N). Looking at the signs of v_8 , this balance tends to increase the relative importance of the vegetal nutrition against the protein sources (incl. starch). Considering that this is the largest coefficient in Table 8.16 (and the most significant one), one might conclude that the Mediterranean diet represents a larger influence of cereals and vegetables (including fruits) with respect to starch and protein sources, not a big surprise.

Table 8.16. Estimated coefficients and p -values of the ANOVA model where several coordinates are explained as a function of the N-S difference between European countries.

	Estimate	Std. Error	norm. Est.	p -value
v_1	-0.0095	0.1869	-0.0509	0.9598
v_2	-0.0409	0.1567	-0.2608	0.7966
v_3	0.0435	0.1115	0.3905	0.6997
v_4	0.0871	0.1583	0.5498	0.5878
v_5	-0.1172	0.2967	-0.3951	0.6964
v_6	0.4286	0.2452	1.7481	0.0938
v_7	-0.5534	0.1518	-3.6453	0.0014
v_8	1.5958	0.2813	5.6727	0.0000

These features can be easily visualised in a balance dendrogram (Section 5.5) using the last SBP, and considering Northern (red) and Southern (blue) countries as separated populations. Figure 8.7 shows such a balance dendrogram. According with p -value shown in Table 8.16, balance v_8 have quite different mean for the Northern and Southern countries. This is visualised, in Figure 8.7, as the separation between the red and blue vertical lines of the upper partition in the dendrogram which corresponds to v_8 . This is confirmed by the corresponding box-plots. Also, for Northern-Southern countries, the mean values of the balance v_7 , (cereals, vegetables) over (nuts), seem quite different as the central boxes do not overlap; the p -value 0.0014 confirms this impression. The balance v_6 (red meat, fish) over (white meat, milk, eggs, starch) shows also some difference between the two sets of countries, but the central part of the boxes overlap and, in fact, the p -value is 0.0938, a poor significance level. Other balances show similar means by group of countries, with clearly overlapping box-plots and, accordingly, the p -values in Table 8.16 are clearly non-significant. The compositional dendrogram is useful for a visual comparison between variances of each balance within the two groups of countries.

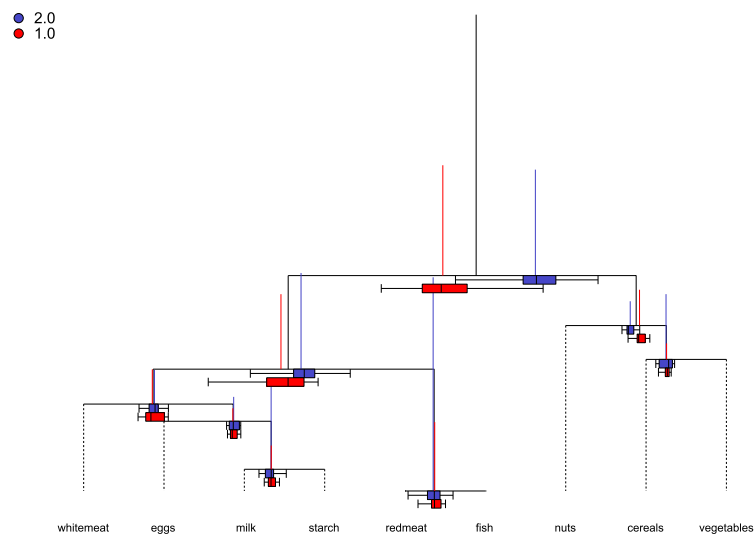


Fig. 8.7. Compositional dendrogram of Nutrition data set (Example 8.4) considering Northern (red) and Southern (blue) countries, using the SBP proposed in Table 8.15.

Some of them are very similar (e.g. for balance v_6), while other ones differ substantially (e.g. v_2 comparing cereals and vegetables).

Exercise 64

In the webpage (www.wiley.com/go/glahn/practical), you can find a file named “Hydrochem.csv”, containing an extensive data set of the geochemical composition of water samples from several rivers and tributaries of the Llobregat river, the most important river in the Barcelona province (NE Spain). This data was studied in detail by Otero et al. (2005) and Tolosana-Delgado et al. (2005), and placed, with the authors consent, in the public domain within the R package “**compositions**” (Boogaart and Tolosana-Delgado 2008). Table 8.8 provides a random sample for illustration.

Table 8.17. Main anion composition of some water samples from four different rivers in Barcelona province (NE Spain).

river	Cl	SO ₄	HCO ₃	river	Cl	SO ₄	HCO ₃
A	197.43	857.99	348.39	U	16.54	71.88	182.20
A	312.37	487.83	377.13	U	27.29	93.35	197.97
A	15.49	239.93	146.00	U	26.00	96.81	176.96
A	118.09	445.63	341.50	U	29.15	76.87	188.60
A	352.84	341.68	557.50	U	37.14	94.72	179.60
A	309.78	371.71	538.50	U	22.86	84.46	244.80
A	432.24	357.35	393.70	U	33.29	116.76	180.10
L	142.80	120.34	210.30	U	9.57	42.96	197.31
L	305.74	199.97	222.45	U	7.79	25.75	171.29
L	309.67	164.40	206.32	U	6.07	36.85	174.20
L	325.76	151.63	201.90	U	108.14	96.16	180.45
L	256.18	145.33	189.20	U	24.79	109.86	209.70
L	242.42	196.08	187.10	C	15.22	83.35	177.40
L	373.26	166.62	249.70	C	265.84	116.69	188.70
L	382.45	222.31	219.96	C	385.13	118.58	191.70
L	228.30	181.83	368.40	C	634.93	164.80	232.56
L	14.02	55.52	245.90	C	519.88	397.32	220.10
L	445.39	455.62	286.67	C	844.45	154.68	175.10
L	300.05	469.89	287.40	C	10.22	83.98	180.44
L	1133.39	581.08	613.60	C	194.83	228.07	293.60
L	652.03	517.47	410.78				

Fit an ANOVA model to these three-part compositions. Draw the data set using colors to distinguish between the four rivers. Plot the means of the four groups, as estimated by the ANOVA fit.

Advanced exercise:

Extract the variance-covariance matrix of each group. Draw *confidence* regions for each of their means, as explained in Sections 7.4 and A.2. Obtain the pooled variance-covariance matrix (see section 7.3), and draw its ellipse three times, around the centers of the three groups. Compare the different ellipses and

the different matrices. Is the assumption of equal variances between the three groups reasonable?

Answer of exercise 64

The whole data set in Hydrochem.csv has 485 rows and 19 parts, including several descriptive variables, two categorical variables (Location and River) and 14 chemical components. From those, the present problem proposes to consider the subcomposition (Cl, SO₄, HCO₃), which is often used in Hydro-geochemical studies within a Piper diagram. The problem can be solved by any software able to work with multivariate ANOVA. After applying an *ilr* (here the balance of SO₄ vs Cl as x_1^* , and the balance of HCO₃ vs the other two as x_2^*), the results of Table 8.18 (first 4 rows, first 2 columns), are obtained. The application of the inverse *ilr* to each of these rows produces the ANOVA coefficients represented as compositions (first 4 rows, last 3 columns). The lower half of the table is the predicted means (expressed in *ilr* coordinates on the first 2 columns, and as a composition in the last 3 columns). These predictions are obtained as explained in the preceding exercises. The mean for the first group (ANOVA) is equal to the intercept. The means for the other groups are obtained adding the vector (or perturbing the compositions) of the intercept with the vector (or the composition) corresponding to the target group: for instance, the Lower Llobregat mean in coordinates, $\hat{\mathbf{g}}_3 = [-0.2487, 0.2684]$, is obtained as the intercept plus the row marked as “†LowerLLobregat”, $[0.5773, 0.1054] + [-0.8260, 0.16310]$. Figure 8.8 reports the means calculated this way as colored dots. The advanced part of the

Table 8.18. Results of an ANOVA for the three-part composition, to be distinguished among the four subrivers of the Llobregat river basin, using the whole data Hydrochem.csv

	x_1^*	x_2^*	Cl	SO ₄	HCO ₃
(Intercept)	0.5773	0.1054	0.2011	0.4549	0.3441
†Cardener	-0.6713	0.2402	0.4501	0.1742	0.3757
†Lower LLobregat	-0.8260	0.1631	0.5021	0.1561	0.3418
†Upper LLobregat	0.1914	0.9031	0.1733	0.2271	0.5996
Anoia	0.5773	0.1054	0.2011	0.4549	0.3441
Cardener	-0.0941	0.3456	0.3027	0.2650	0.4324
Lower LLobregat	-0.2487	0.2684	0.3486	0.2452	0.4062
Upper LLobregat	0.7687	1.0085	0.1011	0.2999	0.5989

exercise requires the calculation of the variance for each group as well as the pooled variance (see Section 7.3). These are reported in a compact way in Table 8.19. Figure 8.9 shows a balance dendrogram using the balances x_1^* and x_2^* . The groups of data-points by rivers are shown separately. The dendrogram in black corresponds to the pooled sample. The dendrogram allows a visual

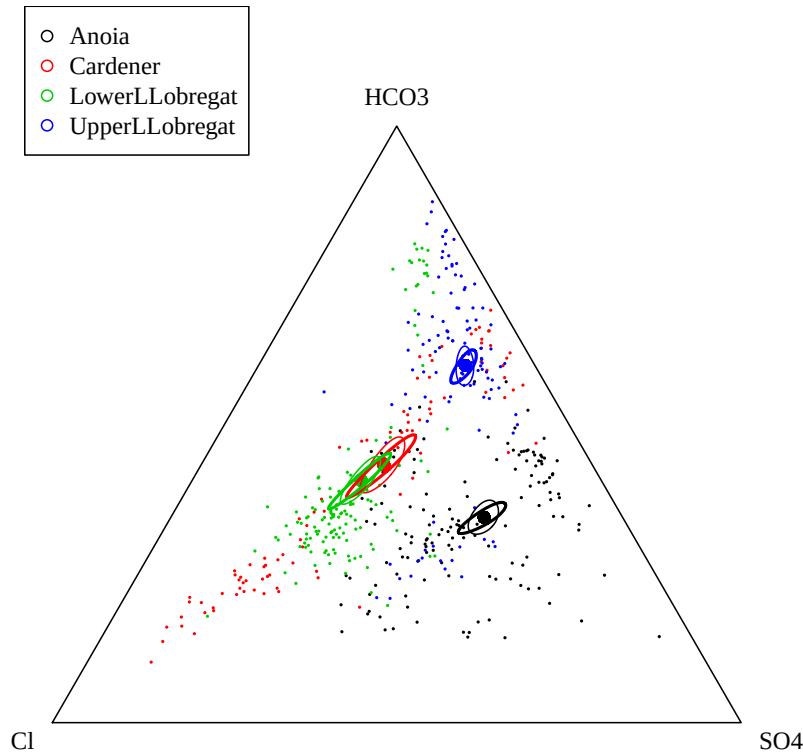


Fig. 8.8. Predicted means (big dots) and data (small points) from exercise 64, by rivers. Thick line ellipses show confidence regions computed using for each group its own variance, while thin line ellipses show the equivalent ellipses constructed using the pooled within-group variance.

comparison of the balance means and variances. The main features of the dendrogram using this coordinates are: balance x_1^* seems to distinguish the means of Cardener and Lower Llobregat from those of Anoia and Upper Llobregat although there is a possible difference in variances between Cardener and Lower Llobregat; on the other hand, x_2^* only shows a possible significant difference of Upper Llobregat mean from means of other rivers. Whether these differences in mean or variances are significant, require a statistical analysis beyond this exploratory and visual analysis.

To plot the ellipses, the radius obtained from Eq. (7.8) in Section 7.4 is used, where $D = 3$ in all cases; for each group a different n is used, depending

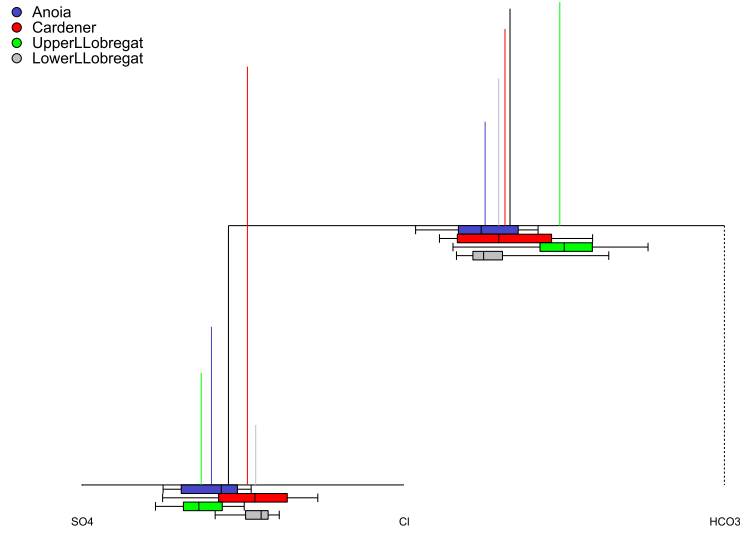


Fig. 8.9. Balance dendrogram for different river samples using the balances described in the text. Dendrograms and boxplots in color correspond to those described in the legend. The black dendrogram is computed from the pooled sample.

on the number of observations in each group: $n = 143$ in Anioia, $n = 95$ in Cardener, $n = 135$ in LowerLlobregat and $n = 112$ in UpperLlobregat. The obtained radii are then $\kappa = 0.2083$ (Anioia), $\kappa = 0.2580$ (Cardener), $\kappa = 0.2147$ (LowerLlobregat) and $\kappa = 0.2366$ (UpperLlobregat). The resulting ellipses are reported in Figure 8.8. Though this is not a formal test of equal variance among the four groups, we may conclude that the four variances are acceptably similar, with the possible exception of Cardener river, and that the hypothesis of equal variance matrix is not blatantly violated (recall that variances should be compared in *orders of magnitude*).

Table 8.19. Group clr-variances and within-group pooled clr-variance for the data Hydrochem.csv.

group		Cl	SO ₄	HCO ₃
Anoia	Cl	0.301	-0.131	-0.170
	SO ₄	-0.131	0.109	0.022
	HCO ₃	-0.170	0.022	0.147
Cardener	Cl	0.851	-0.376	-0.474
	SO ₄	-0.376	0.182	0.195
	HCO ₃	-0.474	0.195	0.280
LowerLLobregat	Cl	0.201	-0.012	-0.190
	SO ₄	-0.012	0.031	-0.019
	HCO ₃	-0.190	-0.019	0.209
UpperLLobregat	Cl	0.345	-0.040	-0.305
	SO ₄	-0.040	0.052	-0.012
	HCO ₃	-0.305	-0.012	0.317
(pooled)	Cl	0.391	-0.125	-0.266
	SO ₄	-0.125	0.088	0.037
	HCO ₃	-0.266	0.037	0.230

Exercise 65

Find a reclassification between Northern and Southern countries from Example 8.5 using only animal products: red meat, white meat, eggs, milk, and fish.

Answer of exercise 65

A linear discriminant analysis of the northern vs the southern countries requires the pooled covariance matrix and the mean for each of the groups. To obtain them, an ilr transformation is first applied. For the selected 5-part composition, we choose the basis linked to the partition

	RM	WM	E	M	F
v_1	-1	1	0	0	0
v_2	-1	-1	1	0	0
v_3	-1	-1	-1	1	0
v_4	-1	-1	-1	-1	1

The group means $\boldsymbol{\mu}_N^*$, $\boldsymbol{\mu}_S^*$ and the pooled covariance $\boldsymbol{\Sigma}$ for the transformed data are reported in Table 8.20. The number of countries in each group are $n_N = 16$ and $n_S = 9$, which give estimates for the prior probabilities of $p_N = 0.64$ and $p_S = 0.36$. With these, all functions in page 164 can be computed. The discriminating vector is

$$\mathbf{v}_{SN} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_S^* - \boldsymbol{\mu}_N^*) = [-3.8301, -3.5594, -3.4767, 0.3237],$$

the intercept function $A_{SN} = -1.5117$, and the discriminant function

$$d_{SN}(\mathbf{x}^*) = -1.5117 + [-3.8301, -3.5594, -3.4767, 0.3237] \cdot \mathbf{x}^*.$$

Applying this function to the ilr transformed subcomposition provides the

Table 8.20. Group means and pooled covariance for the ilr-transformed nutrition subcomposition, using only food classes of animal origin.

group	v_1	v_2	v_3	v_4
N	-0.0764	-0.8457	0.9163	-0.8708
S	-0.4507	-1.0088	0.8530	-0.7064
	v_1	v_2	v_3	v_4
v_1	0.1565	-0.0037	-0.0684	-0.0797
v_2	-0.0037	0.0609	-0.0024	0.0966
v_3	-0.0684	-0.0024	0.0971	0.0113
v_4	-0.0797	0.0966	0.0113	0.7493

discriminating scores for each country, which can then be converted to the probability of belonging to the southern group as

$$p_S = \frac{\exp(d_{SN})}{1 + \exp(d_{SN})},$$

because the discriminating function can be seen as the logratio transformation of the vector of two probabilities $[p_S, p_N]$. Table 8.21 reports the values for this function and for the probability of belonging to the southern group, as well as the attributed group: in a natural way, if a country $p_S > 0.5$ then it is considered to be in the southern group. These results are also displayed in figure 8.10, which shows that perhaps a reduction of the threshold of attribution of a country to the southern block would produce a more balanced misclassification, e.g. by putting the threshold below Italy, we would classify the Czech Republic, the former USSR and Belgium as southern countries, but classify correctly many more of the southern countries.

Table 8.21. Discriminant scores, probability of belonging to a southern country and true and estimated groups.

country	d_{SN}	p_S	estimated group	true group
Al	4.32	0.99	S	S
Au	-3.12	0.04	N	N
Be	-0.16	0.46	N	N
Bu	0.97	0.72	S	S
Cz	-0.35	0.41	N	N
Dk	-2.10	0.11	N	N
Gw	-0.98	0.27	N	N
Fi	-2.86	0.05	N	N
Fr	1.45	0.81	S	S
Gr	-0.40	0.40	N	S
Hn	-3.29	0.04	N	N
Ei	-1.70	0.15	N	N
It	-0.67	0.34	N	S
Nt	-2.88	0.05	N	N
Nw	-1.61	0.17	N	N
Pl	-3.20	0.04	N	N
Po	3.04	0.95	S	S
Ro	-1.04	0.26	N	S
Es	-0.23	0.44	N	S
Sw	-2.28	0.09	N	N
Sz	-0.93	0.28	N	N
UK	0.51	0.62	S	N
USR	-0.46	0.39	N	N
Ge	-1.41	0.20	N	N
Yu	-1.94	0.13	N	S

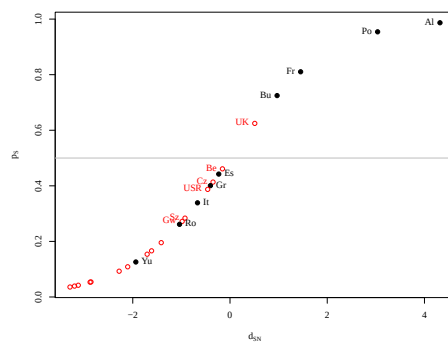


Fig. 8.10. Probability of belonging to a southern country vs the discriminant scores, with colour/symbol after true group (red circles=North, black dots=South). Some country abbreviations are reported for reference.

Exercise 66

A typical way of assessing the goodness of an LDA is to reclassify the sample assigning to each observation the class with its highest predicted probability. Among other fundamental problems, this conventional procedure can be criticized because it reduces a rich information on assignment probabilities to just one label: misclassifying an observation of group A into group B when the predicted probabilities of these two group were $(p_A, p_B) = (0.48, 0.52)$ seems less grievous than if the probabilities were $(p_A, p_B) = (0.08, 0.92)$. Given that these probabilities are indeed compositions themselves, they can be summarized in a compositionally compliant way instead of just by counting how often each category was given a larger probability. Following Example 8.5, have a look at the probabilities of assignment of Table 8.6. Fill in the classical reclassification table

true group	reclassified group		total
	East	West	
East			
West			

An alternative compositional view to this table would be to obtain the geometric center of the vectors of probabilities within the two groups defined by the true country and arrange the resulting numbers in a square table. Upscale these means to sum up to the number of observations within each class. Compare the two reclassification tables.

Answer of exercise 66

For the example in Section 5.8 the classical reclassification table (Table 8.22) shows that LDA had a misclassification rate of $100 \cdot 3/25 = 12\%$. The al-

Table 8.22. Classical reclassification table for the nutrition data of Section 5.8.

true group	reclassified group		total
	East	West	
East	6	3	9
West	0	16	16

ternative table required in the exercise cannot be computed, because there are rounded zeros in the estimated probabilities. Instead, we resort to apply the proposed alternative to the preceding exercise. The reclassification table (Table 8.23) gives a misclassification rate of $100 \cdot (5 + 1)/25 = 24\%$. The two centers of the probability vectors $\mathbf{p} = [1 - p_S, p_S]$ reported in table 8.21 are $\mathbf{g}(\mathbf{p}|N) = [0.8425, 0.1575]$ and $\mathbf{g}(\mathbf{p}|S) = [0.3519, 0.6481]$. Upscaling these vectors by $n_N = 16$ and $n_S = 9$ gives Table 8.24. As can be seen, the two tables are reasonably similar. Though the proposed compositional alternative

Table 8.23. Compositional reclassification table for the nutrition data of Section 5.8.

true group	reclassified group		total
	North	South	
North	15	1	16
South	5	4	9

Table 8.24. Re-scaled probabilities.

true group	rescaled probability		total
	North	South	
North	13.5	2.5	16
South	3.2	5.8	9

does not depend on the threshold probability placed to select which countries are considered in which block (an issue discussed in the preceding example), experience in its usefulness, sense and meaning is still needed.

Exercise 67

Between three groups, three borders can be drawn (A with B , B with C , A with C). Show that these three boundaries intersect in one single point (a triple junction). Find the equation of that point. Now assume that the discriminating composition has three components: note that in this case, the boundaries could be drawn as segments from the triple junction along the directions of some vectors $\mathbf{v}_{jk}^\perp$ orthogonal to $\mathbf{v}_{jk}$. Obtain the equations of these boundaries.

Answer of exercise 67

Note that there are two typos in the book, in the equations of page 164 about linear discriminant analysis. The discriminant function between groups j and k , i.e. the log-ratio of evidence in group j against evidence in group k $\ln(p_j(\mathbf{x})/p_k(\mathbf{x}))$, is

$$d_{jk}(\mathbf{x}) = A_{jk} + \mathbf{v}_{ij}^\top \cdot \mathbf{x}$$

with leading vector

$$\mathbf{v}_{jk} = \Sigma^{-1}(\boldsymbol{\mu}_j^* - \boldsymbol{\mu}_k^*)$$

and intercept

$$A_{ij} = \ln \frac{p_j^0}{p_k^0} - \frac{1}{2}(\boldsymbol{\mu}_j^* - \boldsymbol{\mu}_k^*)^\top \cdot \Sigma \cdot (\boldsymbol{\mu}_j^* + \boldsymbol{\mu}_k^*).$$

The boundaries are then affine hyperplanes orthogonal to $\mathbf{v}_{jk}$ passing through the point

$$\mathbf{x}_{jk}^0 = \frac{1}{2}(\boldsymbol{\mu}_j^* + \boldsymbol{\mu}_k^*) - \frac{2A_{jk} + \mathbf{v}_{jk}^\top \cdot (\boldsymbol{\mu}_j^* + \boldsymbol{\mu}_k^*)}{\mathbf{v}_{jk}^\top \cdot (\boldsymbol{\mu}_j^* - \boldsymbol{\mu}_k^*)}(\boldsymbol{\mu}_j^* - \boldsymbol{\mu}_k^*),$$

i.e. the boundary is the set of points such that $\mathbf{v}_{jk}^\top \cdot (\mathbf{x} - \mathbf{x}_{jk}^0) = 0$. Note that these boundaries are not necessarily orthogonal to the vector of differences between means $\boldsymbol{\mu}_j^* - \boldsymbol{\mu}_k^*$, because Σ^{-1} acts as a deformation of the space.

To proceed with the required proof, note there is a geometric intuitive *demonstration*, which relates to the properties of triangles. This is *a priori* valid only for the case $p_A^0 = p_B^0 = p_C^0$ and $\Sigma = \mathbf{I}$ (i.e. the case where the boundaries are indeed orthogonal to $\boldsymbol{\mu}_j^* - \boldsymbol{\mu}_k^*$), but which is still helpful. Remember that the three orthogonal bisectors (*mediatriz* in Spanish, *Mittelsenkrechte* in German) of the three sides of a triangle cross at a point (the circumcenter). Given that for $p_A^0 = p_B^0 = p_C^0$ and $\Sigma = \mathbf{I}$ the boundaries between each pair of groups correspond to the orthogonal bisector of the segment joining the two group means, these boundaries must cross at the circumcenter of the three means. A more formal proof can be obtained using the three discriminant functions $d_{AB}(\mathbf{x}) = \ln(p_A/p_B)$, $d_{AC}(\mathbf{x}) = \ln(p_A/p_C)$ and $d_{BC}(\mathbf{x}) = \ln(p_B/p_C)$. The fact that any of them can be obtained as a linear combination of the other two, e.g. $d_{AC}(\mathbf{x}) = d_{AB}(\mathbf{x}) + d_{BC}(\mathbf{x})$, proofs that if a point satisfies

that $d_{AB}(\mathbf{x}) = d_{BC}(\mathbf{x}) = 0$ (i.e. it belongs to the boundary between groups $A - B$ and also to the boundary between groups $B - C$) then it must also satisfy $d_{AC}(\mathbf{x})$ (i.e. it belongs to the boundary $A - C$). Hence, there will always be a set of points that satisfies all three equations, i.e. where $p_A = p_B = p_C$. If $\mathbf{x} \in \mathbb{R}^2$, then this set must have one single point.

The location of this point is best found realizing that it must satisfy $\mathbf{v}_{AB}^\top \cdot (\mathbf{x} - \mathbf{x}_{AB}^0) = 0$ and $\mathbf{v}_{BC}^\top \cdot (\mathbf{x} - \mathbf{x}_{BC}^0) = 0$ (or the equivalent for $A - C$, but two equations are sufficient). Because we are in a 2D problem, these equations are sufficient to find the point if we write them as a system of equations

$$\begin{aligned}\mathbf{v}_{AB}^\top \mathbf{x} &= \mathbf{v}_{AB}^\top \mathbf{x}_{AB}^0, \\ \mathbf{v}_{BC}^\top \mathbf{x} &= \mathbf{v}_{BC}^\top \mathbf{x}_{BC}^0,\end{aligned}$$

or in matrix form

$$\underbrace{\begin{pmatrix} \mathbf{v}_{AB}^\top \\ \mathbf{v}_{BC}^\top \end{pmatrix}}_{\mathbf{V}} \cdot \mathbf{x} = \begin{pmatrix} \mathbf{v}_{AB}^\top \mathbf{x}_{AB}^0 \\ \mathbf{v}_{BC}^\top \mathbf{x}_{BC}^0 \end{pmatrix}.$$

The matrix $\mathbf{V}$ is square, and it will be invertible as well if $\mathbf{v}_{AB}$ and $\mathbf{v}_{BC}$ are not parallel, i.e. if the three means are not aligned. Thus, the solution will be

$$\mathbf{x}_{ABC}^0 = \mathbf{V}^{-1} \cdot \begin{pmatrix} \mathbf{v}_{AB}^\top \mathbf{x}_{AB}^0 \\ \mathbf{v}_{BC}^\top \mathbf{x}_{BC}^0 \end{pmatrix}.$$

The equation of the boundary $A - B$ is then a line passing through $\mathbf{x}_{ABC}^0$ along the vector orthogonal to $\mathbf{v}_{AB} = [v_1^{AB}, v_2^{AB}]$ i.e. along the vector $\mathbf{v}_{AB}^\perp = [-v_2^{AB}, v_1^{AB}]$,

$$\mathbf{y}(\alpha) = \mathbf{x}_{ABC}^0 + \alpha \cdot \mathbf{v}_{AB}^\perp.$$

Analogous equations can be written for any of the other three boundaries.

Exercise 68

Using the data set from Exercise 64, obtain the discriminant functions between rivers A, U, and L (remove C for this exercise). This may be easily done by computing the ilr coordinates in CoDaPack and exporting them to your favorite statistical software. Draw the data in the ilr plane and in a ternary diagram, using colors to distinguish between rivers. Add the group centers and the boundaries between groups. If you use R, LDA is available with function “lda” in the package “MASS”.

Answer of exercise 68

We use the ilr representation based on the partition shown in Table 8.25. Representing the data set in coordinates, the group counts, means and pooled

Table 8.25. Partition table for Exercise 68

	SO ₄	HCO ₃	Cl
v_1	-1	+1	0
v_2	-1	-1	+1

Table 8.26. Statistics for constructing an LDA, for the dataset “Hydrochem.csv” without the Cardener river.

group	counts	mean	
		x_1^*	x_2^*
Anoia	143	-0.1974	-0.5526
LowLl	135	0.3568	0.0812
UppLl	112	0.4890	-1.1700
variance			
		x_1^*	x_2^*
		0.1428	-0.1307
		x_2^*	0.4165

variance can be computed (Table 8.26). Apply then the process shown in the solution of the preceding exercise. To find the triple point, choose to work with the two boundaries with respect to Upper Llobregat. The two mean differences are $\boldsymbol{\mu}_A^* - \boldsymbol{\mu}_U^* = [-0.6864, 0.61739]$ and $\boldsymbol{\mu}_L^* - \boldsymbol{\mu}_U^* = [-0.1322, 1.2511]$ and the two leading vectors are $\mathbf{v}_{AU} = [-4.8428, -0.0378]$ and $\mathbf{v}_{LU} = [2.5605, 3.8074]$. The two constants are $A_{AU} = 0.91793$ and $A_{LU} = 1.1765$, and the two crossing points of the boundaries with the segments joining the means are $\mathbf{x}_{AU}^* = [0.1966, -0.9070]$ and $\mathbf{x}_{LU}^* = [0.4327, -0.6370]$. Finally, the system of equations is

$$\underbrace{\begin{pmatrix} 0.0378 & -4.8427 \\ -3.8074 & 2.5605 \end{pmatrix}}_{\mathbf{V}} \cdot \mathbf{x}_{ABC}^0 = \begin{pmatrix} 4.3998 \\ -3.2786 \end{pmatrix},$$

which delivers the triple point $\mathbf{x}_{ABC}^0 = [0.2514, -0.9066]$. This point and the boundary segments along $\mathbf{v}_{AU}^\perp$, $\mathbf{v}_{LU}^\perp$ and $\mathbf{v}_{AL}^\perp = \mathbf{v}_{AU}^\perp - \mathbf{v}_{LU}^\perp = [3.8452, -7.4032]$ are shown in Figure 8.11.

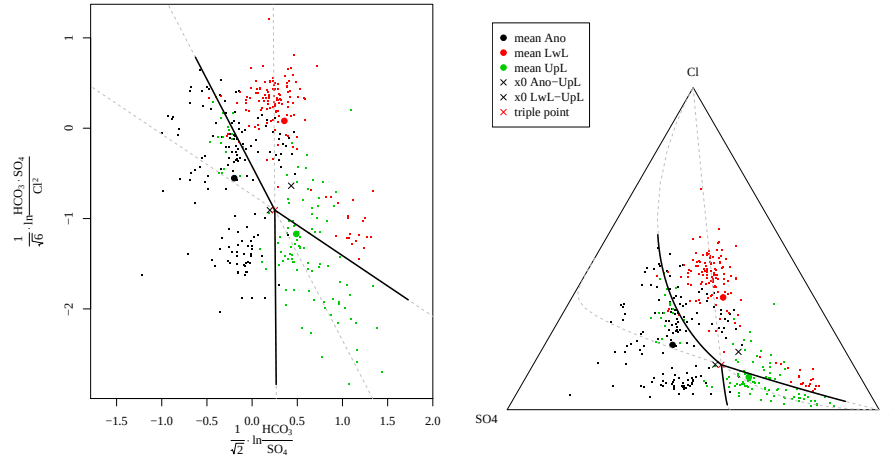


Fig. 8.11. Graphical representation of the group centers, boundaries and middle points of the discriminant analysis

Exercise 69

Select two arbitrary three-part compositions, $\mathbf{x}(0)$, $\mathbf{x}(t_1)$, and consider the linear process from $\mathbf{x}(0)$ to $\mathbf{x}(t_1)$. Determine the direction of the process normalized to one and the time, t_1 , necessary to arrive from $\mathbf{x}(0)$ to $\mathbf{x}(t_1)$. Plot the process (a) in a ternary diagram, (b) in balance coordinates, and (c) as evolution in time of the parts normalized to a constant.

Answer of exercise 69

Let $\mathbf{x}(0) = \mathcal{C}[0.9, 0.1, 0.2]$ and $\mathbf{x}(t_1) = \mathcal{C}[0.05, 0.7, 0.95]$ be the initial and final 3-part compositions. They have been chosen in a non-closed form to insist on the fact that closure does not play any role in the operations in the simplex and the logratio representations. Also, $t_1 = 10$ has been chosen arbitrarily. The direction of a compositional line passing through $\mathbf{x}(0)$ and $\mathbf{x}(t_1)$ is the perturbation difference

$$\mathbf{x}(t_1) \ominus \mathbf{x}(0) = \mathcal{C} \left[\frac{0.05}{0.9}, \frac{0.7}{0.1}, \frac{0.95}{0.2} \right] = [0.004705882, 0.592941176, 0.402352941] .$$

A line segment in the simplex with this direction, starting at $\mathbf{x}(0)$ and ending at $\mathbf{x}(t_1)$, is

$$\mathbf{x}(t) = \mathbf{x}(0) + \frac{t}{t_1} \odot (\mathbf{x}(t_1) \ominus \mathbf{x}(0)) \quad , \quad 0 \leq t \leq t_1 .$$

The balances

$$b_1 = \sqrt{\frac{2}{3}} \ln \frac{(x_1 x_2)^{1/2}}{x_3} \quad , \quad b_2 = \frac{1}{\sqrt{2}} \ln \frac{x_1}{x_2} ,$$

have been selected for representation in ilr-coordinates. Using these coordinates, the line is

$$b_1(t) = b_1(0) + \frac{t}{t_1} \cdot (b_1(t_1) - b_1(0)) \quad , \quad b_2(t) = b_2(0) + t \cdot (b_2(t_1) - b_2(0)) ,$$

where $b_1(t), b_2(t)$ are the coordinates of $\mathbf{x}(t)$. Particularly,

$$[b_1(0), b_2(0)] = [0.3311, 1.5537] \quad , \quad [b_1(10), b_2(10)] = [-1.3267, -1.8661] .$$

Note that, $[b_1(t_i), b_2(t_i)]$, $i = 1, 2, \dots, 100$ are first obtained; then, applying ilr^{-1} , the process of proportions, $\mathbf{x}(t)$, is computed. This procedure is numerically more convenient than the equivalent of directly evaluating the model in proportions. The three figures required in the statement of the Exercise are shown in Figure 9.19. The upper left panel shows the linear evolution of the coordinates $b_1(t), b_2(t)$ (blue and red lines, respectively). The upper right panel shows the changes of proportions $x_1(t), x_2(t), x_3(t)$ (blue, red, and green lines, respectively). The lower figure shows the trajectory of $\mathbf{x}(t)$ in a ternary diagram. In this case, time is implicit, but from the initial point, $\mathbf{x}(0) = \mathcal{C}[0.9, 0.1, 0.2]$, we can deduce, that it is the extreme of the trajectory that is closer to the vertex x_1 .

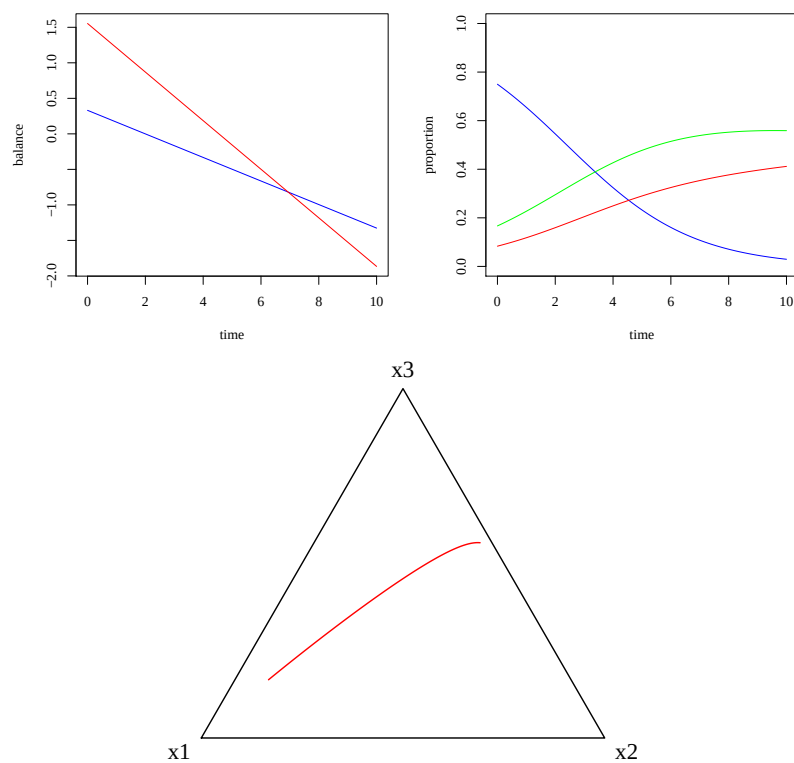


Fig. 9.19. Three representations of the same linear process as a function of time. Upper left: in coordinates; upper right: in proportions; lower figure: trajectory in a ternary diagram.

Exercise 70

Consider the linear process starting with $\mathbf{x}(0) = [82.7, 16.5, 0.8]$ and rates of decay $\lambda_1 = -3$, $\lambda_2 = -2$ and $\lambda_3 = -1$. Draw the equivalents of Figures 9.1 and 9.2, for example, you can take time steps between 0 and 5 at intervals of 0.05 time units, and calculate $\mathbf{x}(t)$ for each one of the resulting 101 time moments. Use the same basis as in Example 9.1. Compare the resulting diagrams with Figures 9.1 and 9.2. What can you conclude with regard to the signs of each individual λ_i ? Repeat for $\boldsymbol{\lambda} = [-1, 0, +1]$, if you deem it necessary.

Answer of exercise 70

A process of exponential decay of masses or abundances follows the model

$$\mathbf{z}(t) = \mathbf{z}(0) \cdot \exp(\boldsymbol{\lambda}t), \quad t > 0,$$

where $\boldsymbol{\lambda} = [-3, -2, -1]$ has all components negative. Applying closure, i.e. $\mathbf{x}(t) = \mathcal{C}\mathbf{z}(t)$, and using operations in the simplex the model is

$$\mathbf{x}(t) = \mathbf{x}(0) \oplus (t \odot \exp(\boldsymbol{\lambda})), \quad t > 0.$$

Taking the time sampling interval as 0.05, from $t = 0$ to $t = 5$, 100 points are obtained. The ilr basis chosen corresponds to the balances

$$b_1 = \sqrt{\frac{2}{3}} \ln \frac{(x_1 x_2)^{1/2}}{x_3}, \quad b_2 = \frac{1}{\sqrt{2}} \ln \frac{x_1}{x_2}.$$

These balance-coordinates are evaluated at the time sampling points, and the evolution of proportions is obtained as ilr^{-1} of the balances.

Figure 9.20 shows the results. The upper left panel shows the linear evolution in time of the coordinates $b_1(t)$, $b_2(t)$ (blue and red lines, respectively). The upper right panel shows the changes of proportions $x_1(t)$, $x_2(t)$, $x_3(t)$ (blue, red, and green lines, respectively). The lower left figure shows the trajectory of $\mathbf{x}(t)$ in a ternary diagram and the lower right panel is this trajectory in the coordinate space, i.e. b_1 against b_2 (a circle points out the initial value at $t = 0$).

Comparing with Example 9.1, the rates of exponential change have been changed from $\boldsymbol{\lambda} = [1, 2, 3]$, an abundance growth, to $\boldsymbol{\lambda} = [-3, -2, -1]$, so that abundances decay in time. However, the obtained figures are the same as those shown in Figures 9.1 and 9.2. The key point is that the direction of the compositional line is exactly the same in both processes. This can be checked taking into account that

$$\exp(-4) \cdot \exp[1, 2, 3] = \exp[1 - 4, 2 - 4, 3 - 4] = \exp[-3, -2, -1],$$

and

$$\mathcal{C} \exp[1, 2, 3] = \mathcal{C} \exp[-3, -2, -1] = \mathcal{C} \exp[-1, 0, 1].$$

Therefore, from the compositional point of view, proportions (not abundances) follow exactly the same process. This remarks that proportions do not inform

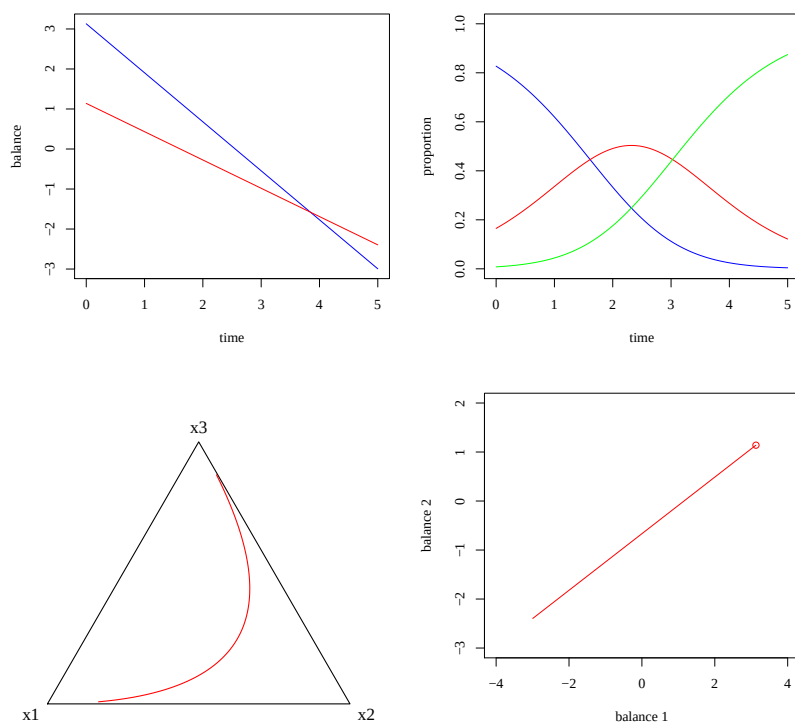


Fig. 9.20. Evolution of a linear process as a function of time. Upper left: in coordinates; upper right: in proportions; lower left: trajectory in a ternary diagram; lower right: trajectory in the coordinate space.

about the decay or growth of abundances, only proportions are informative; they can be the same, despite the fact that masses or abundances decay or grow. Therefore, it is not necessary to repeat the exercise for $\lambda = [-1, 0, 1]$; again, the direction of the line is equal to the two cases in Example 9.1.

Exercise 71

Chose $\mathbf{x}(0)$ and $\mathbf{p}$ in $\mathcal{S}^3$. Consider the process $\mathbf{x}(t) = \mathbf{x}(0) \oplus t \odot \mathbf{p}$ with $0 \leq t \leq 1$. Assume that the values of the process at $t = j/49$, $j = 1, 2, \dots, 50$ are perturbed by observation errors, $\mathbf{y}(t)$, distributed as a normal on the simplex $\mathcal{N}_s(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with $\boldsymbol{\mu} = \mathcal{C}[1, 1, 1]$ and $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}_3$ ($\mathbf{I}_3$ unit (3×3) -matrix). Observation errors are assumed independent of t and $\mathbf{x}(t)$. Plot $\mathbf{x}(t)$ and $\mathbf{z}(t) = \mathbf{x}(t) \oplus \mathbf{y}(t)$ in a ternary diagram and a balance-coordinate plot. Try with different values of σ^2 .

Answer of exercise 71

The following parameters have been chosen:

$$\mathbf{x}(0) = [0.9, 0.07, 0.03] \quad , \quad \mathbf{p} = [0.05, 0.7, 0.25] \quad .$$

The balance-coordinates used are

$$b_1 = \sqrt{\frac{2}{3}} \ln \frac{(x_1 x_2)^{1/2}}{x_3} \quad , \quad b_2 = \frac{1}{\sqrt{2}} \ln \frac{x_1}{x_2} \quad .$$

In these ilr-coordinates, the error free process is

$$\text{ilr}(\mathbf{x}(t)) = [b_1(t), b_2(t)] = [b_1(0), b_2(0)] + t \odot \mathbf{p} \quad , \quad 0 \leq t \leq 1 \quad .$$

This model can be sampled at time points $t = j\delta$, for $j = 0, 1, \dots, 50$, and $\delta = 1/49$, to obtain 50 values of $[b_1(t), b_2(t)]$, which are on a straight line in the coordinate space. In order to introduce deviations distributed as a normal in the simplex, this deviations can be simulated in coordinates.

The center of the deviations is $\boldsymbol{\mu} = \mathcal{C}[1, 1, 1]$ which corresponds to null balances $\text{ilr}(\boldsymbol{\mu}) = [0, 0]$. The variance-covariance matrix $\boldsymbol{\Sigma}$ depends on the chosen coordinates (see Section 6.3.1). As the process is defined in $\mathcal{S}^3$, the coordinates are 2-dimensional and the covariance matrix $\boldsymbol{\Sigma}$ is a $(2, 2)$ -matrix. Note that there is a mistake in the statement of the problem, as this matrix is said to be a $(3, 3)$ -matrix. Here, two values of σ^2 have been chosen to develop this example: $\sigma_1^2 = 0.3^2$, $\sigma_2^2 = 1.4^2$, and the matrix is then $\boldsymbol{\Sigma}_j = \sigma_j^2 \mathbf{I}_2$, $j = 1, 2$. Two independent bivariate normally distributed samples, $\boldsymbol{\epsilon}_i$, of size 50 were sampled from the corresponding bivariate normal distribution. They were added to $[b_1(t), b_2(t)]$ to get a perturbed sample, expressed in balance-coordinates, i.e. $\text{ilr}(\mathbf{y}(t_i)) = [b_1(t_i), b_2(t_i)] + \boldsymbol{\epsilon}_i$. Note that the covariance matrix $\boldsymbol{\Sigma}_j$, $j = 1, 2$, is in both cases diagonal; thus, the vector $\boldsymbol{\epsilon}_i$ could be simulated from univariate normal distributions, one for each component.

Using ilr^{-1} on $[b_1(t), b_2(t)]$ and their perturbed version $\text{ilr}(\mathbf{y}(t_i))$, the process in proportions $\mathbf{x}(t_i)$ and its perturbations $\mathbf{y}(t_i)$ are readily obtained. The following set of Figures shows the results. Left panels correspond to perturbations with $\sigma_1^2 = 0.3^2$ and right panels to $\sigma_2^2 = 1.4^2$. Figure 9.21 shows the change in time of the two balances (lines for non-perturbed process and circles for the respective deviated process; b_1 in blue, b_2 in red). It can be observed

the substantial increase of variability of the deviated process in the right panel looking at the vertical scale.

Figure 9.22 shows the trajectory of the process in coordinates (plot b_1 against b_2). The non-perturbed process in blue and the perturbed processes in red. Note that the left and right blue lines are identical, but the scale is changed due to the change of variance of deviations.

Figure 9.23 is the representation of the trajectories in ternary diagrams. These plots have been centered so that something can be seen. When these plots are not centered, everything is close to the left-bottom corner of the triangle. In this figures the deviated samples (red) appear very dispersed from the non-perturbed process (blue). In the right panel something similar to three modes are distinguishable, similar to what happens when the density of a normal in the simplex, with large variance, is plotted in a ternary diagram taking the Lebesgue measure as a reference (See Section 6.3.1). This circumstance makes the ternary diagram inconvenient and unreliable to observe sample distributions.

Figure 9.24 shows the non-perturbed process in proportions (lines, x_1 blue, x_2 red, x_3 green) and the deviated samples (circles with the same colors). After visualizing the previous plots, it is rather surprising the apparent low dispersion in these plots even in the right panel. It should be remarked that this impression is mainly caused by the scale of the previous plots.

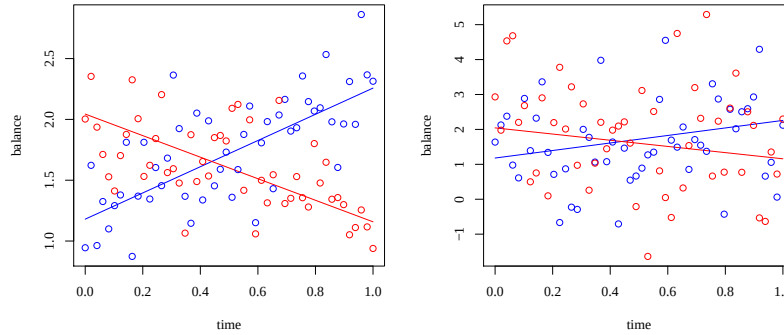


Fig. 9.21. Change in time of two balances.

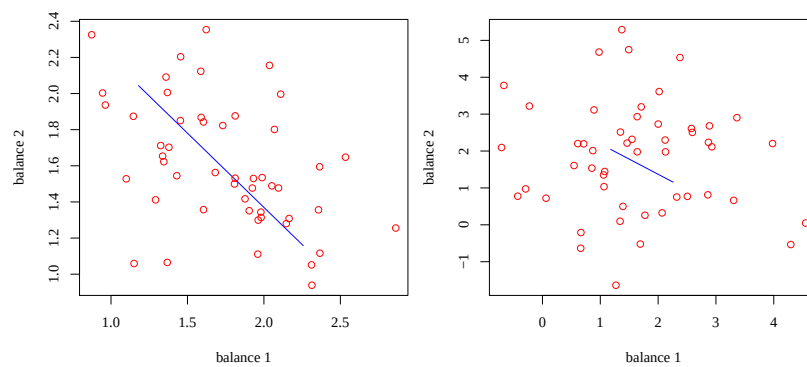


Fig. 9.22. Trajectory of the process in coordinates.

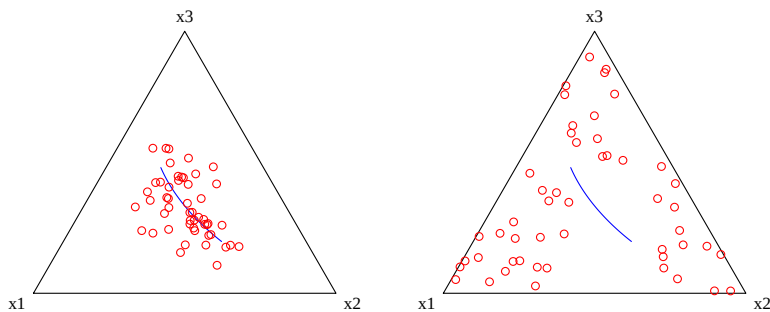


Fig. 9.23. Representation of the trajectories in ternary diagrams.

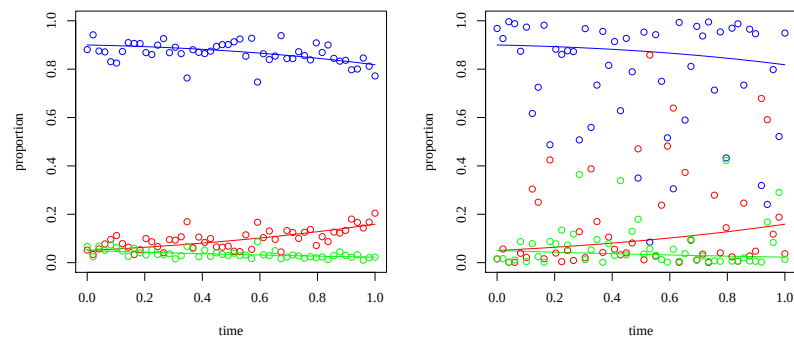


Fig. 9.24. Non-perturbed process in proportions.

Exercise 72

In Example 9.3, set $x_1(0) = 1$, $x_2(0) = 2$, $x_3(0) = 3$, $V = 100$, $Q = 5$. Find the sequential binary partition used in the example. Plot the evolution in time of the coordinates and mass concentrations including the water $x_0(t)$. Plot, in a ternary diagram, the evolution of the subcomposition x_0, x_1, x_2 .

Answer of exercise 72

In the washing process of Example 9.3, the main feature is that the concentration of $\mathbf{x}(t) = [x_1(t), x_2(t), x_3(t)]$, at the outgoing fluid, is constant. However, the example refers only to this subcomposition, and water $x_0(t)$ was not taken into account. Let be $\mathbf{y}(t) = [y_0(t), y_1(t), y_2(t), y_3(t)]$ the composition including water. For simplicity, assume that the units of concentration are g/L ; thus, $V = 100L$ and $Q = 5L/s$. In this case, $y_i(t) = x_i(t)$ for $i = 1, 2, 3$ and $y_0(t) = (1000 - y_1(t) - y_2(t) - y_3(t))g/L$, assuming that the density of water—with or without pollutants—is constant and equal to 1. As this equation does not correspond to an exponential decay, but to a linear relationship between components, the process $\mathbf{y}(t)$ is a settling process, although—because of the negative signs—it could also be called a removal process. To determine the process along a given period of time $0 \leq t$ the equations governing the process are needed. They are

$$y_j(t) = y_j(0) \cdot \exp(-tQ/V), \quad j = 1, 2, 3, \quad y_0(t) = (1000 - y_1(t) - y_2(t) - y_3(t)).$$

This process can be sampled at time intervals δ , resulting in compositions $\mathbf{y}(t_i)$. The balance-coordinates can be obtained using the sequential binary partition (SBP) coded in Table 9.7, where also the number of positive and negative values (n_+ , n_- respectively) are indicated.

Table 9.7. Sequential binary partition for Exercise 55.

coordinate	y_0	y_1	y_2	y_3	n_+	n_-
b_1	-1	+1	+1	+1	3	1
b_2	0	+1	+1	-1	2	1
b_3	0	0	+1	+1	1	1

Note that balances b_2 and b_3 are equal to those defined in the Example 9.3. Remarkably, b_1 can be considered as a (real) index of contamination, given that x_1, x_2 and x_3 are considered pollutants; in fact, it represents the balance of pollutants over clear water:

$$b_1 = \sqrt{\frac{3}{4}} \ln \frac{(y_1 y_2 y_3)^{1/3}}{x_0}.$$

Moreover, all information about the water part y_0 is conveyed by b_1 , as the other balances do not depend on y_0 , but only on the pollutants.

Once the corresponding balances $b_j(t_i)$ are computed for each t_i , they can be plotted as required by the statement. Figure 9.25, left panel, shows the evolution of the balance-coordinates in time. The balances b_2 and b_3 , characterizing the subcomposition of pollutants $[y_1, y_2, y_3]$ (blue and red lines) are constant in time as explained in Example 9.3. The balance b_1 (black line) is approximately linear, although it is actually non-linear as happens in settling processes. The right panel shows the evolution of concentrations in time. Blue, red and green lines (pollutants) decay exponentially with the same rate. In order to draw the curve of concentration of water in the same plot, $y_1(t) - y_1(0)$ has been plotted instead (black line). The proportion of water tends to 1 with time; this is the reason to call this kind of evolution *washing process*.

Figure 9.26 represents the trajectory of the subcomposition $[y_0, y_1, y_2]$ in the

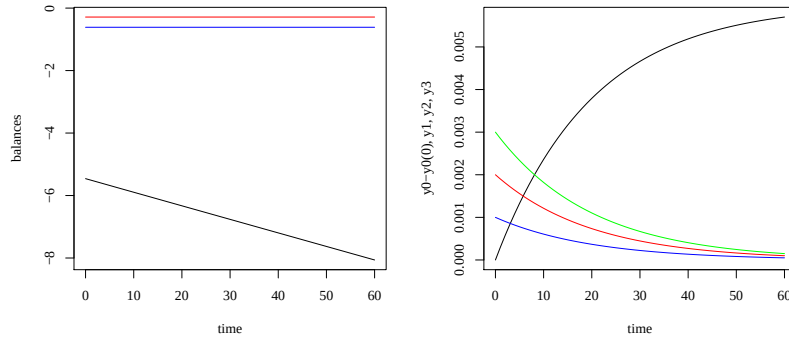


Fig. 9.25. Left: evolution of the balance-coordinates in time. Right: Evolution of concentrations in time.

ternary diagram. The initial subcomposition is marked with a circle. The plot has been centered for easy visualization. If centering is not carried out, the whole line is placed almost in the corner labelled y_0 . The main feature is that, for increasing time, the proportion of water $y_0(t)$ increases.

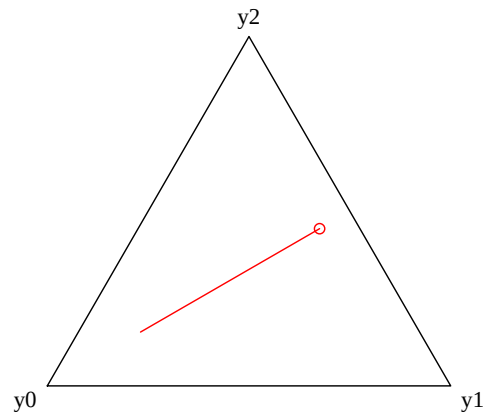


Fig. 9.26. Left: evolution of the balance-coordinates in time. Right: Evolution of concentrations in time.

Exercise 73

In Example 9.2, find the necessary volume m_2 and the composition in container B, $\mathbf{y}$. Find the direction of the simplicial linear process to go from $\mathbf{z}_0$ to $\mathbf{z}_1$.

Answer of exercise 73

In Example 9.2 (obtaining a mixture), container A has volume $m_1 = 93\text{vu}$ (volume units) and the volume proportions of the three components is $\mathbf{x} = [0.3226, 0.5376, 0.1398]$. After pouring the materials into container B, the volume and proportions in container B are $m_1 + m_2$ and $\mathbf{z} = [0.0411, 0.2740, 0.6849]$, respectively. The equations governing the mixture are

$$(m_1 + m_2)\mathbf{z} = m_1\mathbf{x} + m_2\mathbf{y} \quad \text{or} \quad m_1(\mathbf{z}_1 - \mathbf{x}) + m_2\mathbf{z} = m_2\mathbf{y} ;$$

it is clear that, given m_2 , the concentrations $\mathbf{y}$ in container B can be easily found. As the three compositions $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ are closed to 1, the volume m_2 in the right most member of the equation is consistent with the value of m_2 multiplying $\mathbf{z}$. However, m_2 is restricted to values which provide positive values to $\mathbf{y}$. Intuitively, the mixture proportions $\mathbf{z}$ must be on an intermediate point between $\mathbf{x}$ and $\mathbf{y}$: this is the condition that must be fulfilled by m_2 . Analytically, the lower bound of admissible m_2 is obtained equating to 0 the three right-hand equations corresponding to $m_1(\mathbf{z}_1 - \mathbf{x}) + m_2\mathbf{z} = m_2\mathbf{y}$; the lower bound m_2^{low} for m_2 is

$$m_2^{\text{low}} = \max_{i=1,2,3} \left\{ m_1 \frac{x_i - z_i}{z_i} \right\} = 636.9708 .$$

Some values of m_2 , $m_2 > m_2^{\text{low}}$, have been selected and the corresponding $\mathbf{y}$ compositions computed. Table 9 shows these results and their balance coordinates

$$b_1 = \sqrt{\frac{2}{3}} \ln \frac{(x_1 x_2)^{1/2}}{x_3} \quad , \quad b_2 = \frac{1}{\sqrt{2}} \ln \frac{x_1}{x_2} .$$

Figure 9.27 also shows these results both in a (centered) ternary diagram (left panel) and in balance-coordinates (right panel). For reference, the balance coordinates of $\mathbf{x}$ and $\mathbf{z}$ are $[0.8912478, 0.9524035]$ and $[-1.5225250, -0.6478122]$, respectively; they are identified in Figure 9.27 with a red circle for $\mathbf{x}$ and a green circle for $\mathbf{z}$. Also in Figure 9.27, the $\mathbf{y}$ compositions in Table 9 are plotted as blue circles. Note that $\mathbf{z}$ takes an intermediate position between $\mathbf{x}$ (red) and any of the values of $\mathbf{y}$ (blue) along the mixture curve. This fact makes visible the condition fulfilled by the masses m_2 . Finally, observe that the mixture curve appears as a straight-line in the ternary diagram, while it is curved in coordinates, thus pointing out that a mixture process is non-linear with respect to the Aitchison geometry of the simplex.

Table 9.8. A selection of volumes $m_2 > m_2^{low}$ (right column) and the corresponding compositions $\mathbf{y}$ as proportions and their balances b_1, b_2 .

y_1	y_2	y_3	b_1	b_2	m_2
0.0004069307	0.2358945	0.7636986	-3.556689	-0.8307008	643.3405
0.0033414967	0.2386425	0.7580160	-2.686283	-0.8172302	693.3405
0.0058812818	0.2410208	0.7530980	-2.446115	-0.8056154	743.3405
0.0081009288	0.2430993	0.7487998	-2.307213	-0.7954964	793.3405
0.0100573788	0.2449313	0.7450113	-2.211691	-0.7866008	843.3405
0.0117948250	0.2465583	0.7416469	-2.140236	-0.7787189	893.3405
0.0133480910	0.2480128	0.7386391	-2.084012	-0.7716863	943.3405
0.0147449891	0.2493208	0.7359342	-2.038236	-0.7653724	993.3405
0.0160080001	0.2505035	0.7334885	-2.000034	-0.7596722	1043.3405
0.0171554926	0.2515781	0.7312664	-1.967546	-0.7545003	1093.3405
0.0182026219	0.2525586	0.7292388	-1.939503	-0.7497862	1143.3405
0.0191620035	0.2534570	0.7273810	-1.915002	-0.7454717	1193.3405

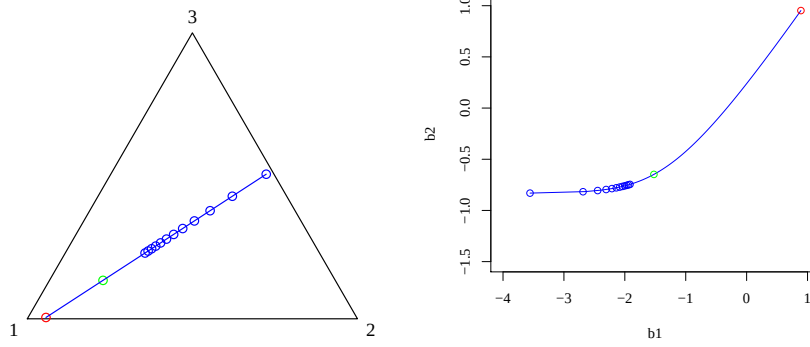


Fig. 9.27. Composition in container A, red circle. Final composition after merging container A and B, green circle. Some possible solutions for the initial composition of container B, blue circles. Blue line, mixture process. Left panel: representation in a centered ternary diagram. Right panel: representation using balance-coordinates.

Exercise 74

A container has a constant volume $V = 100\text{vu}$ (vu = volume units) and initially contains a liquid whose composition is $\mathbf{x}(0) = \mathcal{C}[1, 1, 1]$. A constant flow of $Q = 1\text{vu}$ per second, with volume composition $\mathbf{x} = \mathcal{C}[80, 2, 18]$, is introduced into the container. After a complete mixing, there is an output whose flow equals Q with the volume composition $\mathbf{x}(t)$ at the time t . Model the evolution of the volumes of the three components in the container using ordinary linear differential equations and solve them (Hint: these equations are easily found in textbooks, e.g., Albarède (1995, pp. 345-350)). Are you able to plot the curve for the output composition $\mathbf{x}(t)$ in the simplex without using the solution of the differential equations? Is it a mixture?

Answer of exercise 74

To avoid confusion on the notation, the input flow constant composition is denoted here $\mathbf{y}$, with $\mathbf{y} = [y_1, y_2, y_3] = [80, 2, 18]$, although in the statement it was called $\mathbf{x}$. The closure constant has been taken as $\kappa = 100$, so that y_i is either the concentration in percent and/or the volume of the i -th liquid. Using this closure constant, the initial condition is $\mathbf{x}(0) = [V/3, V/3, V/3]$. The differential equations governing the volumes of the three liquids are

$$V \partial_t x_i(t) = Q y_i - Q x_i(t) \quad , \quad x_i(0) = \frac{V}{3} \quad , \quad i = 1, 2, 3 \quad , \quad (9.31)$$

as proposed in Equation (7.2.1) in Albarède (1995, pp. 346). The reason to state these differential equations can be: in a small increment of time dt , the volume of the i th-liquid entering the container is $Q y_i dt$; this liquid is simultaneously removed from the container as $Q x_i(t) dt$; thus, the volume increment of the i th-liquid in dt is $(Q y_i - Q x_i(t)) dt$. The change of volume of the i th-liquid in the container, in a short interval of time dt , is also $d(V x_i(t)) = V dx_i(t)$. Equation (9.31) is obtained equating both expressions of volume change and taking the limit when $dt \rightarrow 0$. Note that this system of equations in (9.31) is a non-homogeneous linear system of ordinary differential equations, obtained for the evolution of volumes, and it is not a simplicial differential equation.

Equations (9.31) can be separately solved for each liquid species, as the system does not show up any interaction between different species. Each equation is solved in two steps. The first step consists in getting the general solution of the homogeneous equation, i.e. removing the term $Q y_i$ for any initial condition $x_i(0)$. The second step requires to get a particular solution of the complete equation, for an arbitrary initial condition. The sum of the two solutions is the general solution of the complete equations. To proceed, first consider the homogeneous equation and its general solution

$$\partial_t x_i^{(h)}(t) = -\frac{Q}{V} x_i^{(h)}(t) \quad , \quad x_i^{(h)}(t) = c_i \exp \left[-\frac{Q}{V} t \right] \quad ,$$

where the solution is valid for any positive value c_i . In order to find a particular solution of the complete equation, set

$$x_i^{(c)}(t) = \exp\left(-\frac{Q}{V} t\right) + y_i ,$$

which can be checked to be the solution of the complete Equation (9.31). Note that $x_i^{(c)}(0) = 1 + y_i$ for $i = 1, 2, 3$, which are not the required initial conditions. The general solution is then

$$x_i(t) = x_y^{(c)}(t) + c_i \exp\left[-\frac{Q}{V} t\right] ,$$

for an arbitrary value of $c_i > 0$. Equating $x_i(0) = V/3$, the constant is $c_i = V/3 - 1 - y_i$ and the desired solution is

$$x_i(t) = y_i + \left(\frac{V}{3} - y_i\right) \exp\left(-\frac{Q}{V} t\right) . \quad (9.32)$$

Check that, in fact, for $t = 0$, the initial condition is fulfilled, and that for $t \rightarrow +\infty$, $x_i(+\infty) = y_i$, thus confirming the intuitive idea that only for long runs the concentrations in the container approach $\mathbf{y}$.

The evolution in time of the process is shown in Figure 9.28. Left panel

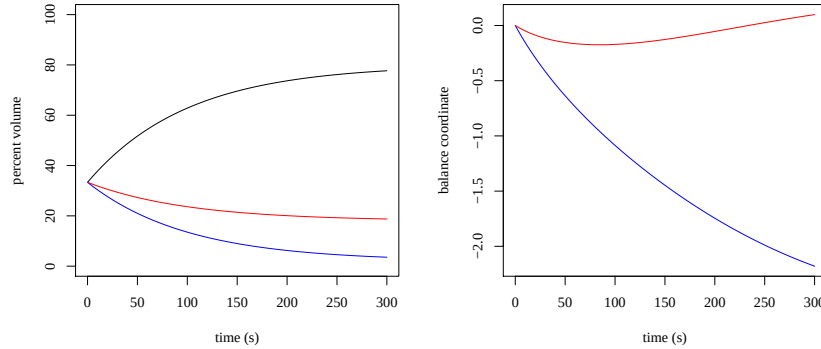


Fig. 9.28. Time evolution of the process, in percent of volume (left panel) and in coordinates (right panel). Left panel: $x_1(t)$ = black line, $x_2(t)$ = blue line, $x_3(t)$ = red line. Right panel: coordinates $b_1(t)$ in blue and $b_2(t)$ in red.

shows the percents in volume of the three components. The curves start at $\mathbf{x}(0) = [100/3, 100/3, 100/3]$ and approach $\mathbf{y} = [80, 2, 18]\text{vu}$ for $t = 300\text{s}$. The value $\mathbf{y}$ is only attained at infinite time t . The process in its coordinates (balances),

$$b_1 = \sqrt{\frac{2}{3}} \ln \frac{(x_1 x_2)^{1/2}}{x_3}, \quad b_2 = \frac{1}{\sqrt{2}} \ln \frac{x_1}{x_2},$$

is shown in the right panel of Figure 9.28. The evolution pattern of $b_1(t)$ and $b_2(t)$ is not as straight-lines, thus demonstrating that the process is not linear in the simplex. Note that the values of both balances is 0 at the initial condition. The trajectories of the process are shown in Figure 9.29; in the left panel, it is represented in a ternary diagram, where it appears as a straight-segment (initial point marked with a circle), thus suggesting that the process is a mixture. Note that the process goes along the segment with an exponentially decreasing velocity; the final point $\mathbf{y}$ is approached for very large t . The right panel shows the same trajectory in coordinates $(b_1(t), b_2(t))$ which is not linear but a hyperbola-shaped curve, thus showing again non-linearity in the simplex.

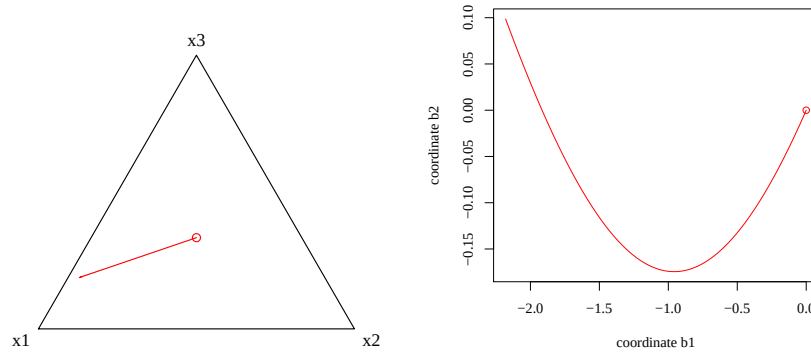


Fig. 9.29. Process trajectories (red lines) represented in a ternary diagram (left panel) and in coordinates (left panel). The initial point, $\mathbf{x}(0) = [100/3, 100/3, 100/3]$ is marked with a circle. the time span is $(0, 300)$ s.

The question about whether the process is a mixture process is answered positively. In fact, at each time t the composition $\mathbf{x}(t)$ is a mixture which depends on the proportion of the volume coming from the input flow with composition $\mathbf{y}$, and the volume accumulated in the initial container $\mathbf{x}(0)$. Denoting these pair of volumes by $\mathbf{v}(t) = [v_{in}(t), v_{con}(t)]$, the process can be described as the mixture

$$\mathbf{x}(t) = v_{in}(t)\mathbf{y} + (V - v_{in}(t))\mathbf{x}(0) \quad , \quad v_{con}(t) = V - v_{in}(t) .$$

This allows to draw both panels in Figure 9.29, without having in hands the solution of the differential equation (9.32), just taking values of $v_{in}(t)$ along the interval $[0, 1]$. However, to plot the curves in Figure 9.28, the function $\mathbf{v}(t) = [v_{in}(t), v_{con}(t)]$ is needed. The solution for $\mathbf{v}(t)$ can be obtained from

the same differential equations (9.31) applied to the composition of volumes $\mathbf{v}(t)$ and suitable initial conditions.

Exercise 75

Check that the simplicial derivative can be computed using Equation (9.8).

Answer of exercise 75

Taking any ilr transformation in the definition of simplicial derivative (Eq. 9.7)

$$\text{ilr}[\partial_{\oplus} \mathbf{x}(t)] = \text{ilr} \left[\lim_{h \rightarrow 0} \frac{1}{h} \odot (\mathbf{x}(t+h) \ominus \mathbf{x}(t)) \right].$$

The limit operation commutes with the ilr under mild conditions of convergence. Using the isometric properties of ilr , it yields

$$\text{ilr}[\partial_{\oplus} \mathbf{x}(t)] = \lim_{h \rightarrow 0} \frac{1}{h} \cdot [\text{ilr}(\mathbf{x}(t+h)) - \text{ilr}(\mathbf{x}(t))] = \partial \text{ilr}(\mathbf{x}(t)).$$

Applying ilr^{-1} the statement holds.

Exercise 76

Consider the settling process $\mathbf{x}(t)$ defined by the equations

$$\begin{aligned} x_1(t) &= a_1 \exp(\lambda_1 t), \\ x_2(t) &= a_2 \exp(\lambda_2 t), \\ x_3(t) &= a_3 + a_1(1 - \exp(\lambda_1 t)) + a_2(1 - \exp(\lambda_2 t)), \end{aligned}$$

with $t \in T$, $T = (0, t_1)$, and assuming that $x_3(0) > 0$. Compute the simplicial derivative of $\mathbf{x}(t)$ for $t \in T$, and pay attention to the behavior of $\partial_{\oplus} \mathbf{x}(t)$ near to a large enough end point t_1 . Plot the process $\mathbf{x}(t)$ and the tangent process at some values of t , both in proportions and in balance coordinates.

Answer of exercise 76

First of all, notice that the total mass of the process is constant. This is not a characteristic of additive transfer processes, it is only a possibility. In order to compute the simplicial derivative, the expression $\partial_{\oplus} \mathbf{x}(t) = \mathcal{C} \exp[\partial \ln(\mathbf{x}(t))]$ is used. The required derivative is

$$\partial_{\oplus} \mathbf{x}(t) = \mathcal{C} \left[\exp(\lambda_1), \exp(\lambda_2), \exp \left(-\frac{a_1 \lambda_1 \exp(\lambda_1 t) + a_2 \lambda_2 \exp(\lambda_2 t)}{a_3 + a_1(1 - \exp(\lambda_1 t)) + a_2(1 - \exp(\lambda_2 t))} \right) \right].$$

Remarkably, for large $t > 0$, the third component of $\partial_{\oplus} \mathbf{x}(t)$ tends to a constant. For instance, if $\lambda_1 < 0$, $\lambda_2 < 0$, for $t \rightarrow +\infty$, the limit value is 1, and the derivative for large t is approximately constant. This fact explains why the coordinates of additive transfer processes appear as bent lines. This can be observed in Figure 9.30. The parameters used are $a_1 = 1$, $a_2 = 2$, $a_3 = 0.3$, $\lambda_1 = -0.5$, $\lambda_2 = -0.2$. Note that $\mathcal{C}[a_1, a_2, a_3]$ is the composition at $t = 0$, and these proportions are going to change along time as can be seen in the left panel (dotted and dashed curves) of Figure 9.30: proportion $x_3(t)$ increases monotonically to 1, while $x_1(t)$ and $x_2(t)$ decrease. Moreover, the evolution of $[x_1(t), x_2(t), x_3(t)]$ has a curved pattern, thus indicating that this process is not a mixing process. In the right panel of Figure 9.30, the two balances have been represented (dotted and dashed curves) confirming the asymptotic behavior as straight lines. The tangent process at $t = 2$ has also been plotted (solid lines), both in proportions and in coordinates. As expected, the tangent process in coordinates is represented by straight lines, although it seems that $t = 2$ is not large enough to make the process and its tangent undistinguishable at the right tail.

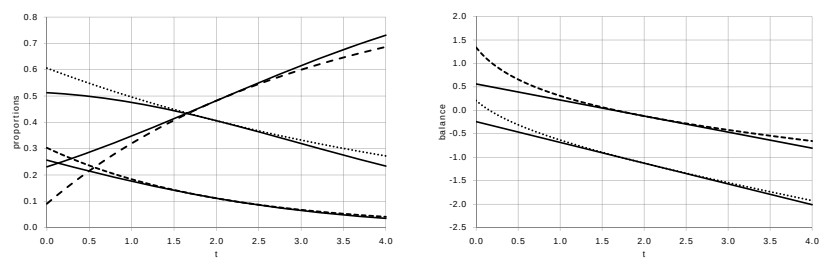


Fig. 9.30. Additive transfer process represented in proportions (left panel) and in balance-coordinates (right panel), as dotted and dashed curves. The tangent process at $t = 2$ is also represented (solid lines).

Exercise 77

Consider the two-dimensional real harmonic oscillator

$$\partial^2 \mathbf{x}^*(t) + \mathbf{A}_0^* \mathbf{x}^*(t) = \mathbf{0}, \quad \mathbf{A}_0^* = \begin{pmatrix} 2.0 & -0.5 \\ -0.5 & 1.0 \end{pmatrix},$$

with initial conditions $\partial \mathbf{x}^*(0) = [0.5, -0.5]$ and $\mathbf{x}^*(0) = [0.1, -0.3]$. Find the simplicial differential equation when $\mathbf{x}^*(t)$ is taken as the coordinate vector of a simplicial process $\mathbf{x}(t)$, using the coordinates given in Equation (9.27). Transform both the coordinate and simplicial equations into first-order differential equations. Solve the first-order differential equation for coordinates and find the corresponding solution in the simplex. Plot the time evolution for $t = 0$ to $t = 10$, as a function of time and in the phase plane $(x_1^*(t), x_2^*(t))$ and as a ternary diagram for $\mathbf{x}(t)$. Hint: the harmonic oscillator has explicit solution, but it is recommended to solve the equations numerically.

Answer of exercise 77

The sign code and the contrast matrix corresponding to the given coordinates are

$$\begin{array}{c|ccc} & x_1 & x_2 & x_3 \\ \hline x_1^* & +1 & +1 & -1 \\ x_2^* & +1 & -1 & 0 \end{array}, \quad \Psi = \begin{pmatrix} 0.408 & 0.408 & -0.816 \\ 0.707 & -0.707 & 0.000 \end{pmatrix}.$$

Taking ilr^{-1} in the differential equation of coordinates, it yields

$$\partial_{\oplus}^2 \mathbf{x}(t) \oplus \mathbf{A}_0 \mathbf{x}(t) = \mathbf{n},$$

where $\mathbf{n} = \mathcal{C}[1, 1, 1]$ is the neutral element in $\mathcal{S}^3$, and

$$\mathbf{A}_0 = \Psi^\top \mathbf{A}_0^* \Psi = \begin{pmatrix} 0.5446582 & -0.1666667 & -0.3779915 \\ -0.1666667 & 1.1220085 & -0.9553418 \\ -0.3779915 & -0.9553418 & 1.3333333 \end{pmatrix}.$$

Reduction of a second order differential equation into a first order one is based on the definition of first order derivatives as new variables, say $\mathbf{z}^*(t) = \partial \mathbf{x}^*(t)$. The equation of the exercise is then written as

$$\begin{aligned} \partial \mathbf{y}^*(t) &= [\partial z_1^*(t), \partial z_2^*(t), \partial x_1^*(t), \partial x_2^*(t)] \\ &= [z_1^*(t), z_2^*(t), x_1^*(t), x_2^*(t)] \cdot \begin{pmatrix} 0 & 0 & -a_{11}^* & -a_{12}^* \\ 0 & 0 & -a_{21}^* & -a_{22}^* \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} = \mathbf{y}^*(t) \mathbf{B}^*, \end{aligned}$$

where a_{ij}^* denote the entries of $\mathbf{A}_0^*$. Taking ilr^{-1} , from $\mathbb{R}^4$ into $\mathcal{S}^5$, in this equation gives an equivalent SLODE. Unfortunately, the SLODE would have a difficult interpretation, as the parts of the first and second simplicial derivatives are mixed in some way. To get a better representation, working in $\mathcal{S}^6$ is

advisable. For this, the three first parts can be taken proportional to $\partial_{\oplus}\mathbf{x}(t)$, and the other three parts to $\mathbf{x}(t)$. This $\mathcal{S}^6$ composition is defined by the two mentioned subcompositions $\partial_{\oplus}\mathbf{x}(t)$, $\mathbf{x}(t)$, but the balance between them is undefined; it can be chosen arbitrarily. An obvious choice is to add to the system of four equations a fifth equation like $\partial w^*(t) = 0$, where $w^*(t)$ is the balance between the two subcompositions. The corresponding SLODE is obtained by taking ilr^{-1} in $\mathcal{S}^6$. The resulting SLODE is $\partial_{\oplus}\mathbf{y}(t) = \mathbf{y}(t) \boxminus \mathbf{B}$, where $\mathbf{B} = \Psi^\top \mathbf{B}^* \Psi$ and $\mathbf{y}(t) = \mathcal{C} \exp(\mathbf{y}^*(t) \Psi)$, and $\mathbf{y}^*(t)$ and $\mathbf{B}^*$ are the versions extended to five components. Most software for solving ordinary differential

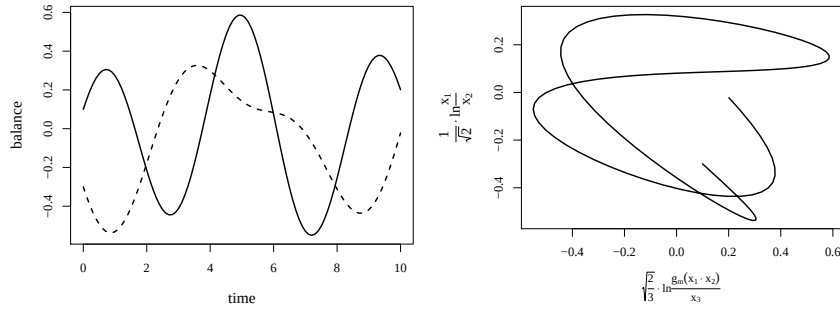


Fig. 9.31. Solution of the harmonic oscillator. Left panel: $x_1^*(t)$ solid line; $x_2^*(t)$ dashed line. Right panel: evolution in the phase plane $x_1^*(t)$ versus $x_2^*(t)$.

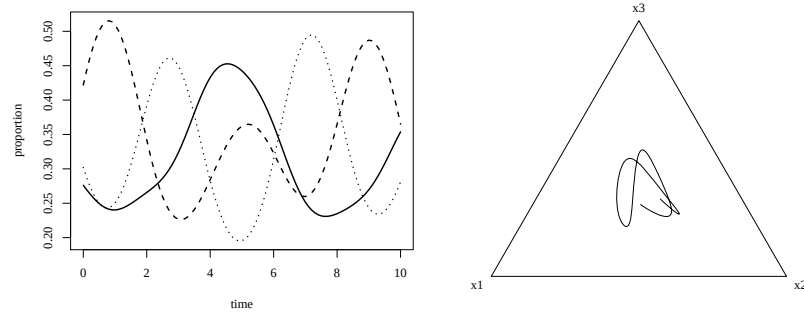


Fig. 9.32. Solution of the harmonic oscillator. Left panel: evolution of $x_1(t)$ (solid line), $x_2(t)$ (dashed line), $x_3(t)$ (dotted line). Right panel: $\mathbf{x}(t)$ in a ternary diagram.

equations is prepared to solve first order differential equations, and it is able to solve the first order equation in coordinates representing the second order harmonic oscillator. Figure 9.31 shows the desired solution in balance-coordinates.

The left panel shows the evolution of the two balances $x_1^*(t)$, $x_2^*(t)$ in time and the right panel the same balances in the phase plane. Figure 9.32 shows the time evolution of the compositional process $\mathbf{x}(t) = [x_1(t), x_2(t), x_3(t)]$ (left panel) and the described orbit in the ternary diagram. Note that the left panels of Figures 9.31 and 9.32 show curves that are similarly shaped; this is due to the fact that the harmonic oscillator moves around the neutral element in the simplex, where the Aitchison geometry and the Euclidean geometry of the coordinates are quite similar. The reader can change the initial values for this oscillator to larger coordinates and observe that the similarity of the curves is progressively lost when oscillations enlarge their amplitudes.

A

Errata

1. Page 25, Caption of Figure 3.1. In panel (a), the symbols of initial compositions must be $*$ and, after perturbation the symbol must be $\circ$.
2. Page 38, in equation Equation (4.9) a_- must be changed to $-a_-$, that is

$$b_k = \sqrt{\frac{rs}{r+s}} \ln \frac{(x_{i_1}x_{i_2}\dots x_{i_r})^{1/r}}{(x_{j_1}x_{j_2}\dots x_{j_r})^{1/s}} = \ln \frac{(x_{i_1}x_{i_2}\dots x_{i_r})^{a_+}}{(x_{j_1}x_{j_2}\dots x_{j_r})^{-a_-}} \quad (4.9)$$

3. Page 70, after Equation (5.2), orthogonality of matrices $\mathbf{U}$ and $\mathbf{V}$ must be $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_r$ and $\mathbf{V}^\top \mathbf{V} = \mathbf{I}_r$.
4. Page 72, first and second lines. The values of $\nu = 0$ and $\nu = 1$ are misplaced. The $\nu = 1$ corresponds to the *covariance biplot* and $\nu = 0$ produces a *form biplot*.
5. Page 78, two last lines of the page mention *links* twice but they should be *rays*, the sentence must be:
 "... which implies that orthogonal rays suggest uncorrelated coordinates and nearly parallel rays suggest well-correlated coordinates;".
6. Page 109, Definition 6.9
 The subscript of functions argmin and min , should be $\mathbf{z} \in \mathcal{S}^D$, in place of $z \in \mathcal{S}^D$.
7. Page 129, Exercise 50
 The vector $\mathbf{g}$ is shown as a column vector. It must be a row vector or, alternatively, it must be transposed.
8. Page 177, Example 9.2, Equation (9.5)
 In the last vector, the bottom entry is $1 - y_2 - y_3 - z_3$, but it must be $1 - y_1 - y_2 - z_3$.
9. Page 204, Exercise 71
 The covariance matrix is defined as $\Sigma = \sigma^2 \mathbf{I}_3$ ($\mathbf{I}_3$ unit (3×3) -matrix). It must be $\Sigma = \sigma^2 \mathbf{I}_2$ ($\mathbf{I}_2$ unit $(2, 2)$ -matrix). Moreover, the mean $\mu = \mathcal{C}[1, 1, 1]$ should be referred to as *center*. But, following notation in Chapter 6, $\mathcal{N}_S(\mu, \Sigma)$ for a normal distribution in the simplex, must be $\mu = [0, 0]$.
 The correct version of Exercise 71 is therefore:
Exercise 71. Chose $\mathbf{x}(0)$ and $\mathbf{p}$ in $\mathcal{S}^3$. Consider the process $\mathbf{x}(t) = \mathbf{x}(0) \oplus t \odot \mathbf{p}$ with $0 \leq t \leq 1$. Assume that the values of the process at $t = j/49$, $j = 1, 2, \dots, 50$ are perturbed by observation errors, $\mathbf{y}(t)$, distributed as a normal on the simplex $\mathcal{N}_s(\mu, \Sigma)$, with $\mu = [\text{ilr}(\mathcal{C}[1, 1, 1])] = [0, 0]$ and $\Sigma = \sigma^2 \mathbf{I}_2$ ($\mathbf{I}_2$ unit (2×2) -matrix). Observation errors are assumed independent of t and $\mathbf{x}(t)$. Plot $\mathbf{x}(t)$ and $\mathbf{z}(t) = \mathbf{x}(t) \oplus \mathbf{y}(t)$ in a ternary diagram and a balance-coordinate plot. Try with different values of σ^2 .
10. Page 226-227. Appendix A.3. Two contrast matrices Ψ_i , $i = 1, 2$ are used, but in the text $i = 0, 1$ appear two times. The typo is amended taking $i = 1, 2$ both times.
11. Page 227, Appendix A.3
 The last equation of the Appendix A.3 must be

$$t_{ij} = \sigma_{ii} + \sigma_{jj} - 2\sigma_{ij}$$